

ME 3211: Aircraft Design Project

Semester: 5

DESIGN OF A SAILPLANE WING

By

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Date of submission	17/11/2024
Due date of submission	17/11/2024

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CHAPTER 01

Aerodynamic Design

1 Aerodynamic design

1.1 Aerofoil Profile Selection

When selecting an optimal aerofoil profile for high-performance gliders, a comprehensive approach was taken. Initially, a literature survey was conducted to gather detailed specifications of commercially available gliders. The data collected include critical parameters such as the length, wingspan, wing area, aspect ratio, weight, cruise speed, maximum speed, maximum glide ratio, and the specific aerofoil cross sections utilized in these gliders. This information provides valuable insights into the diverse range of factors influencing glider performance, serving as a foundational reference for the subsequent stages of the project.

Table 01: Analysis of Existing Gliders

Sail Plane	Length (m)	Wingspan (m)	Wing Area (m ²)	Aspect Ratio	Gross Weight (kg)	Cruise/Max Speed (km/h)	Glide ratio	Aero foil Used
Binder EB29	8	29.3	16.8	51	850	160	69	
LP 49	-	15	13.3	16.9	318	90	33	64(3)-618
Ventus-3	6.63	18	10.84	29.9	600	280 Max		-
JS-3	6.86	18	-	32.8	600	120	55	-
HIRTH Arcus M	8.73	20	15.59	25.7	850	280 Max	50	-
AS 33	-	18	10	32.4	600	-	-	-
HIRTH Duo Discus	8.73	20	16.4	24.4	750	262 Max	47	DFVLR HX 83
HIRTH Quintus	7.83	23	14.7	36	850	-	60	Delft DU 97-127/15
Stemme S10	8.42	23	18.7	28.3	850	259	50	-
Pipistrel Taurus Electro	7.3	14.97	12.33	18.6	550	150	32	-
Jantar Standard	6.71	15	10.66	21.1	385	200	40	NN-8
ETA Glider	9.75	30.9	18.61	51.33	850	108	70	HQR 2 inboard, HQR 1 outboard

Considering this initial data, a survey was done to study high performance aerofoils. The details from this survey are listed below, providing a clear view of the available options. This approach helps us to understand both commercial glider specifications and potential aerofoil profiles, making it easier to choose the best aerofoil.

Existing High-Performance Aerofoils

- Eppler E385 (8.41%)
- WORTMANN FX 63-137 AIRFOIL
- WORTMANN FX 72-MS-150B AIRFOIL
- WORTMANN FX 05-191 AIRFOIL
- WORTMANN FX 049-915 AIRFOIL
- DAE-21 AIRFOIL
- WORTMANN FX 76-MP-120 AIRFOIL

Assuming a Reynolds number of one million, all seven aerofoil profiles were imported into XFLR5 for batch analysis. This analysis enables a thorough evaluation of the aerodynamic performance of each profile under consistent conditions, facilitating an objective comparison and selection of the most suitable aerofoil for the intended glider application.

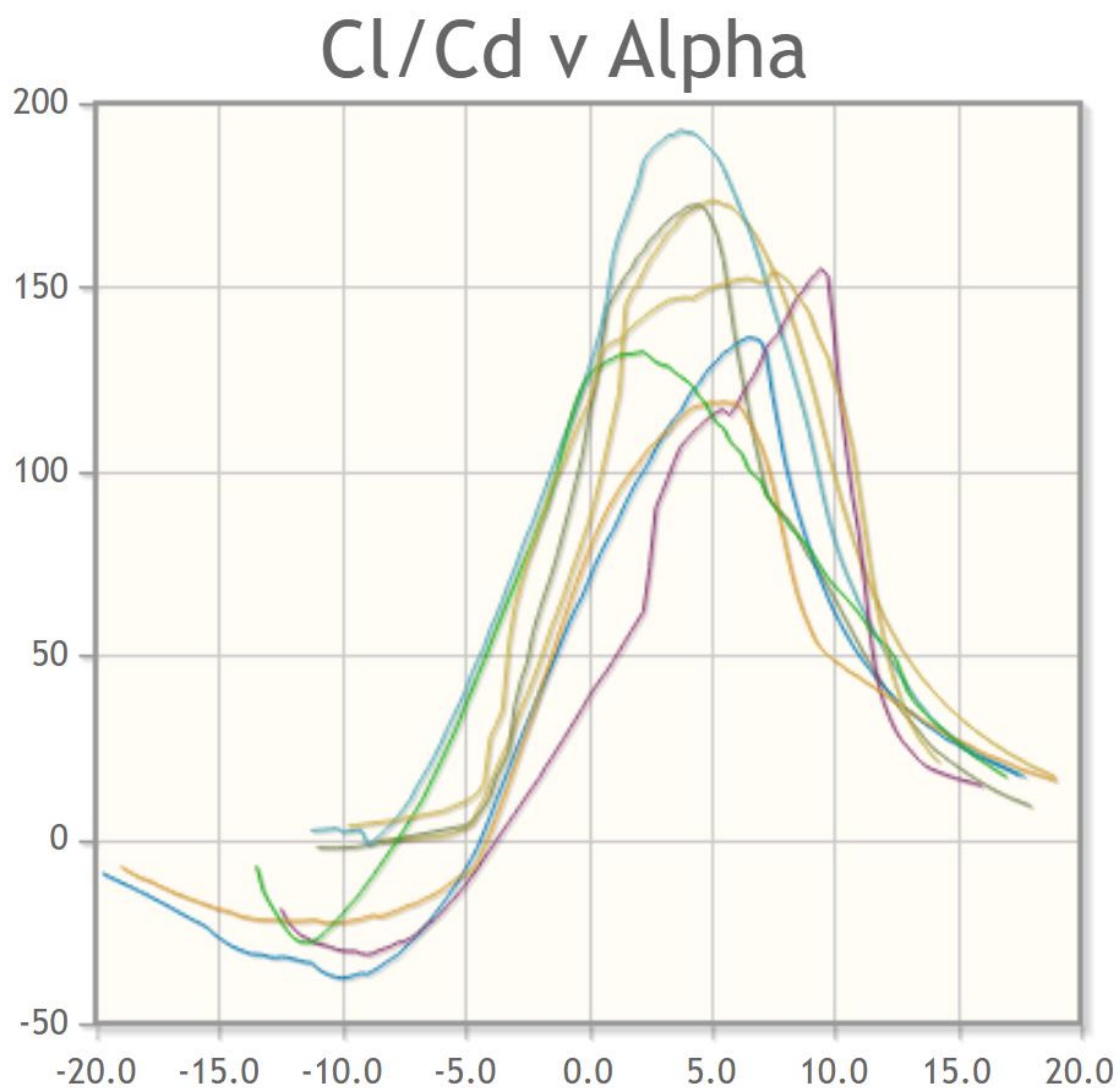
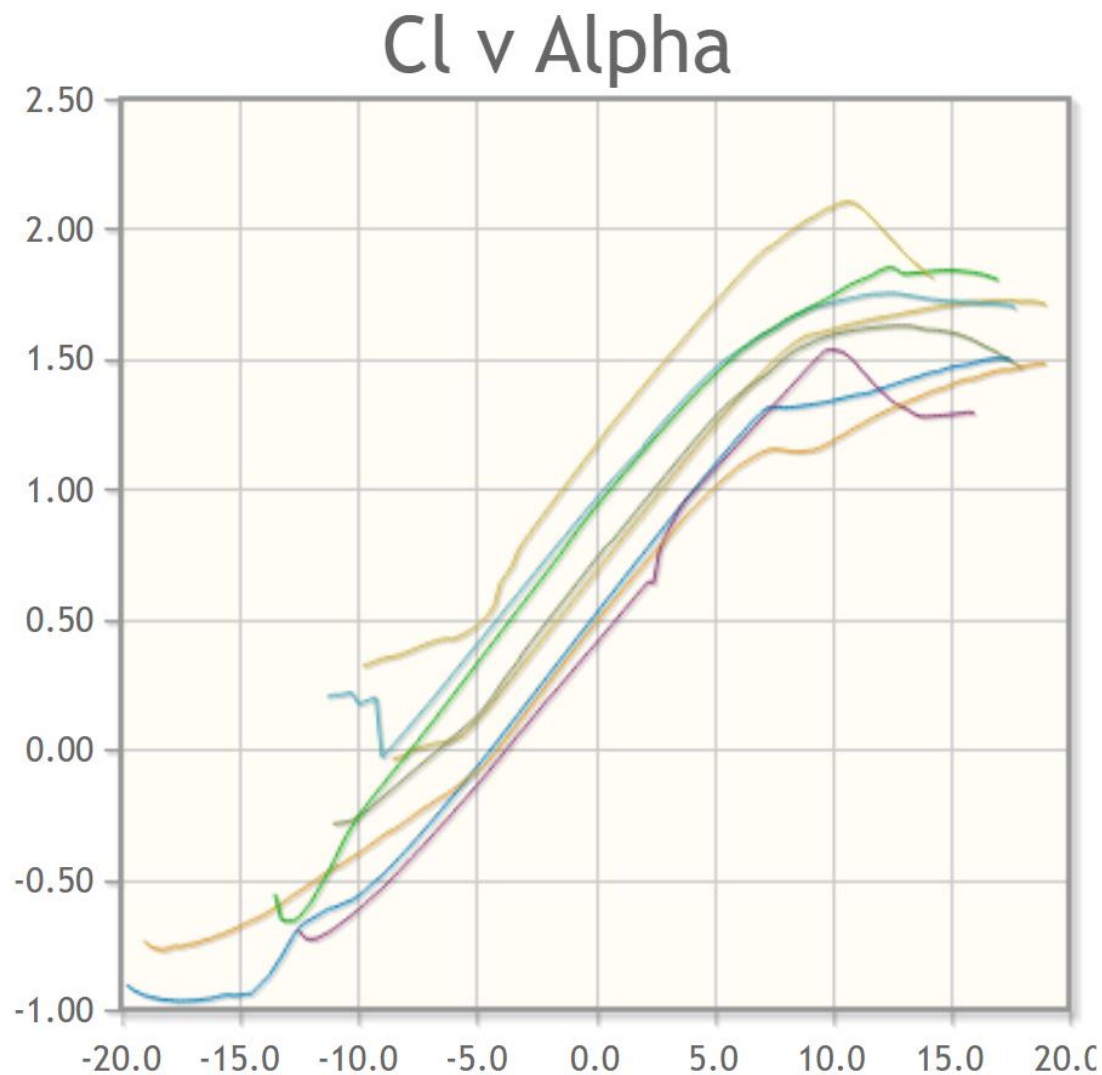


Figure 01: Cl/ Cd values of batch analysis



Aerofoil Performance Analysis

To maximize the range of a glider, it is essential to maximize the lift to drag ratio (Cl/Cd) of the aerofoil. According to the batch analysis of all aerofoils at $Re = 1000000$, the **fx76mp120-il**, **dae21-il**, and **e385-il** aerofoils produced the highest Cl/Cd values at specific angles of attack. Specifically:

- The **fx76mp120-il** aerofoil achieved a maximum Cl/Cd value of 192.6 at an angle of attack (α) of 3.75° .
- The **dae21-il** aerofoil achieved a maximum Cl/Cd value of 173.5 at an angle of attack (α) of 5° .
- The **e385-il** aerofoil achieved a maximum Cl/Cd value of 172.3 at an angle of attack (α) of 4.5° .

These results highlight the superior aerodynamic efficiency of the fx76mp120-il, dae21-il, and e385-il aerofoils, making them strong candidates for optimizing the performance of the glider.

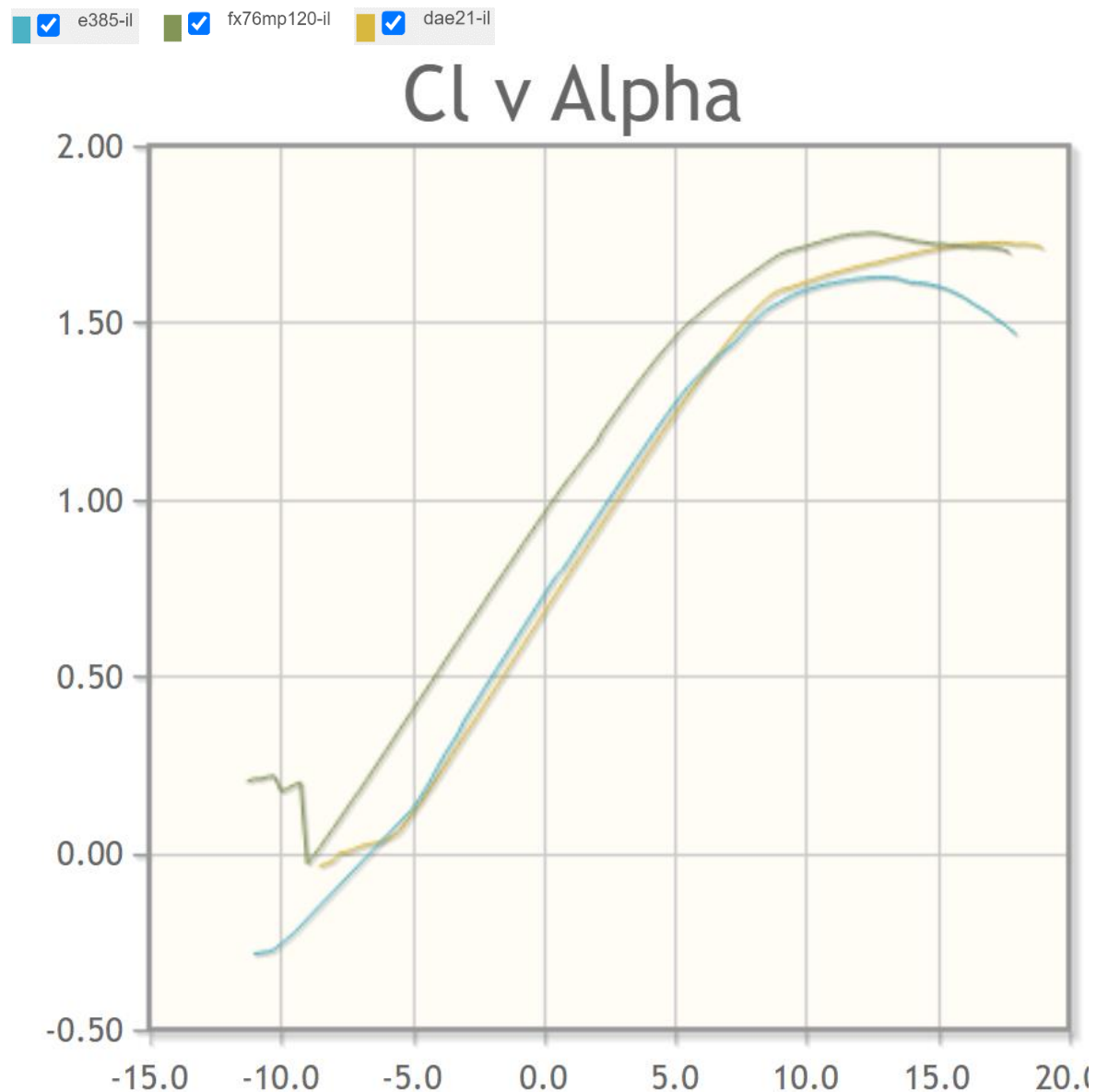


Figure 03: Cl vs alpha values of batch analysis

Aerofoil Selection

Considering both the Cl vs. α and Cl/Cd vs. α graphs, the fx76mp120-il aerofoil demonstrates the best performance among the evaluated profiles. Its superior lift and lift to drag ratio across the range of angles of attack indicate its suitability for achieving optimal aerodynamic efficiency. Therefore, the **WORTMANN FX 76-MP-120** Aerofoil has been selected for the wing, ensuring enhanced performance and range for the glider.

Characteristics of WORTMANN FX 76-MP-120 Aerofoil

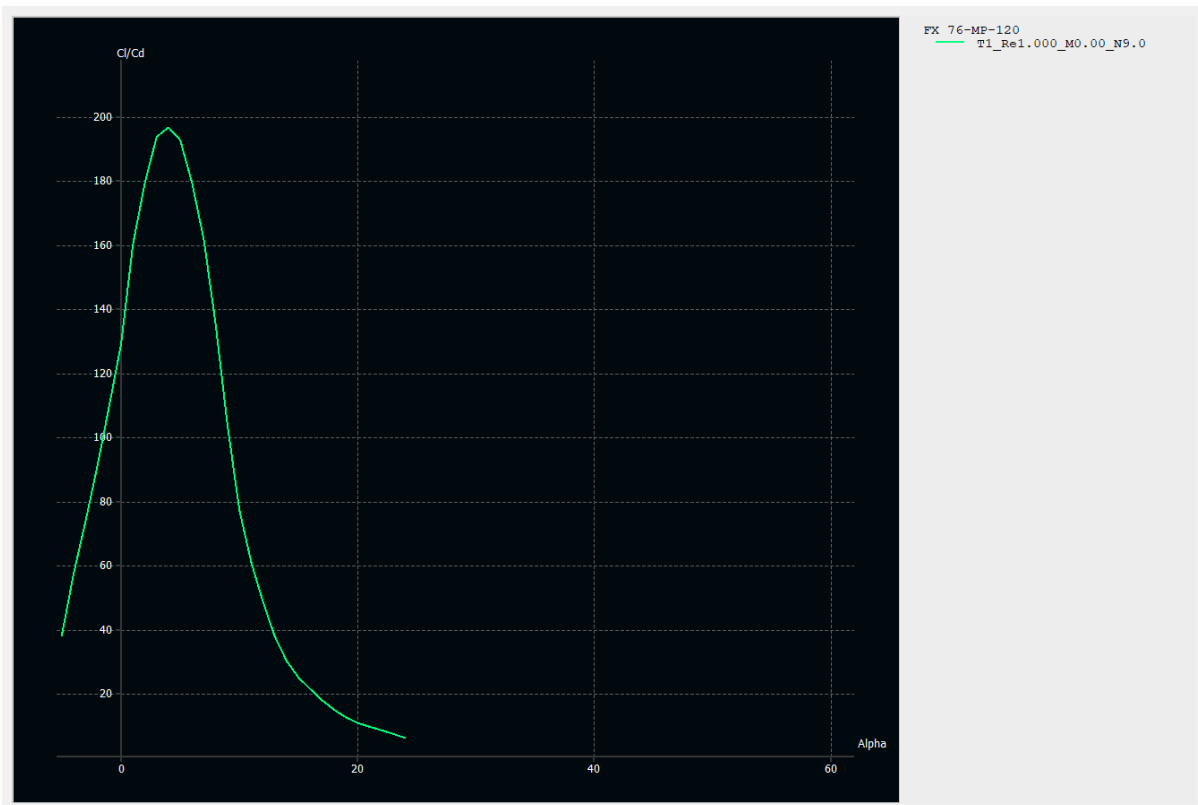


Figure 04: Cl/ Cd graph of WORTMANN FX 76-MP-120 aerofoil

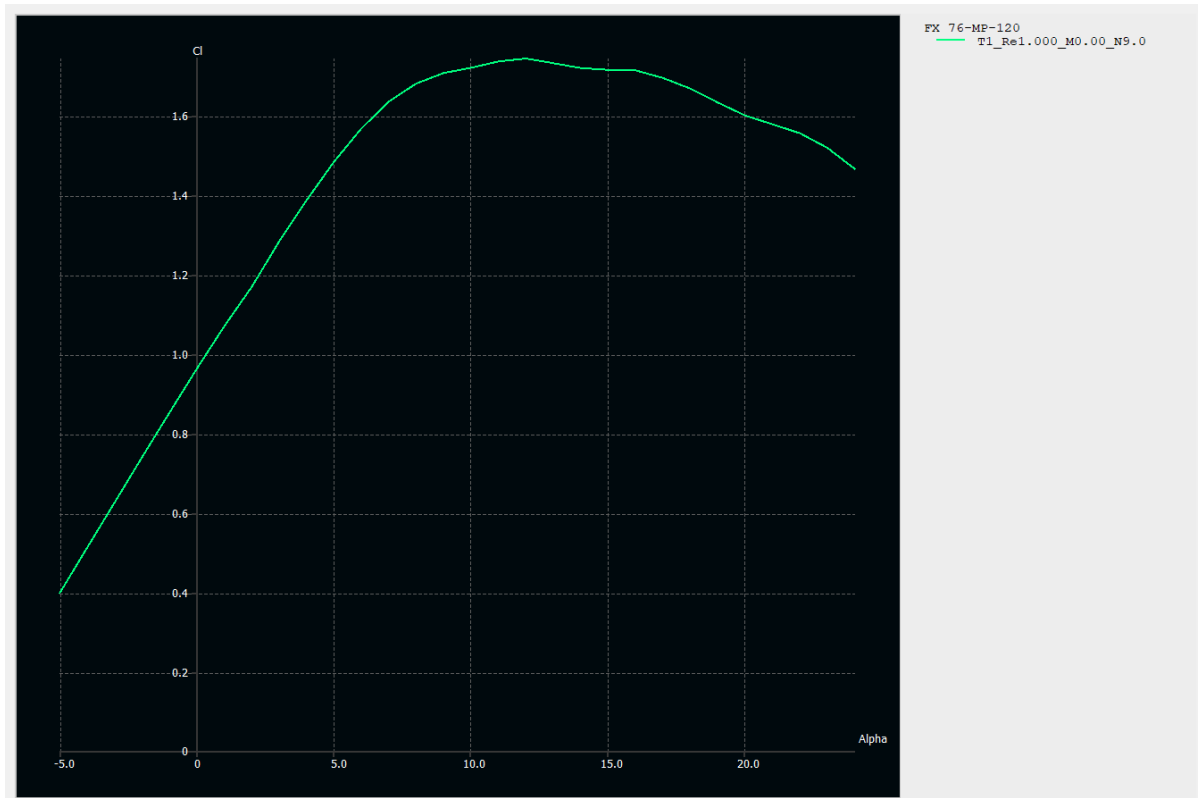


Figure 05: Cl vs alpha graph of WORTMANN FX 76-MP-120 aerofoil

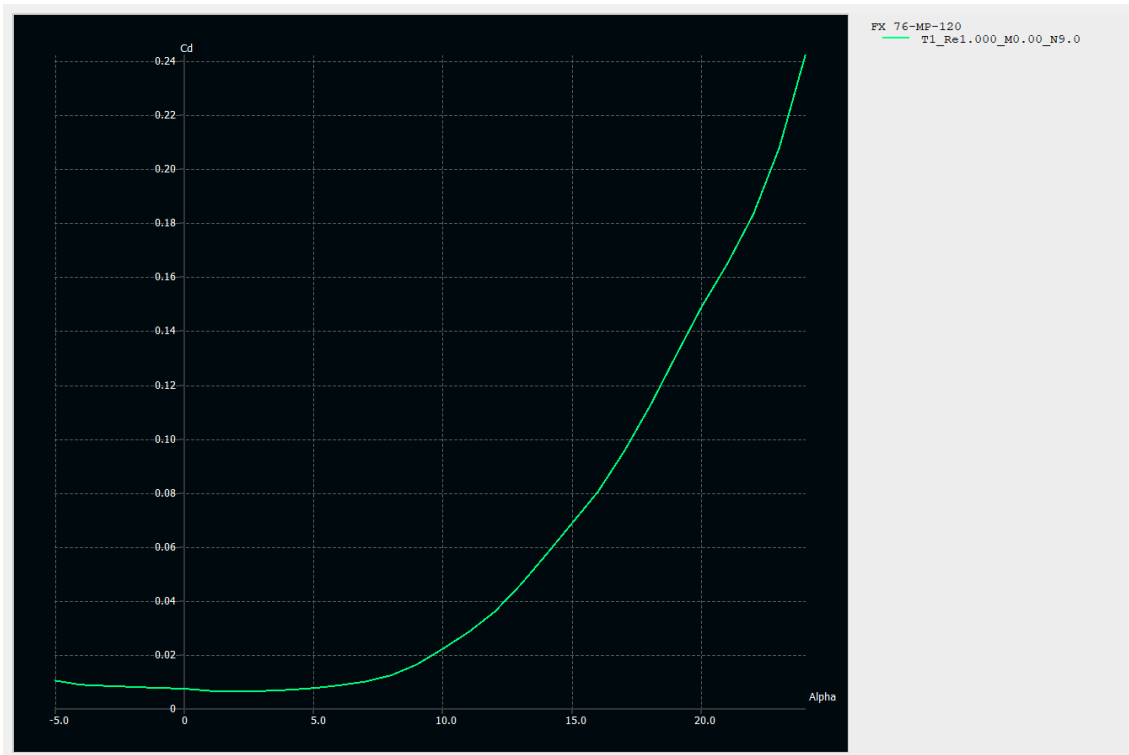


Figure 06: C_d vs α graph of WORTMANN FX 76-MP-120 aerofoil

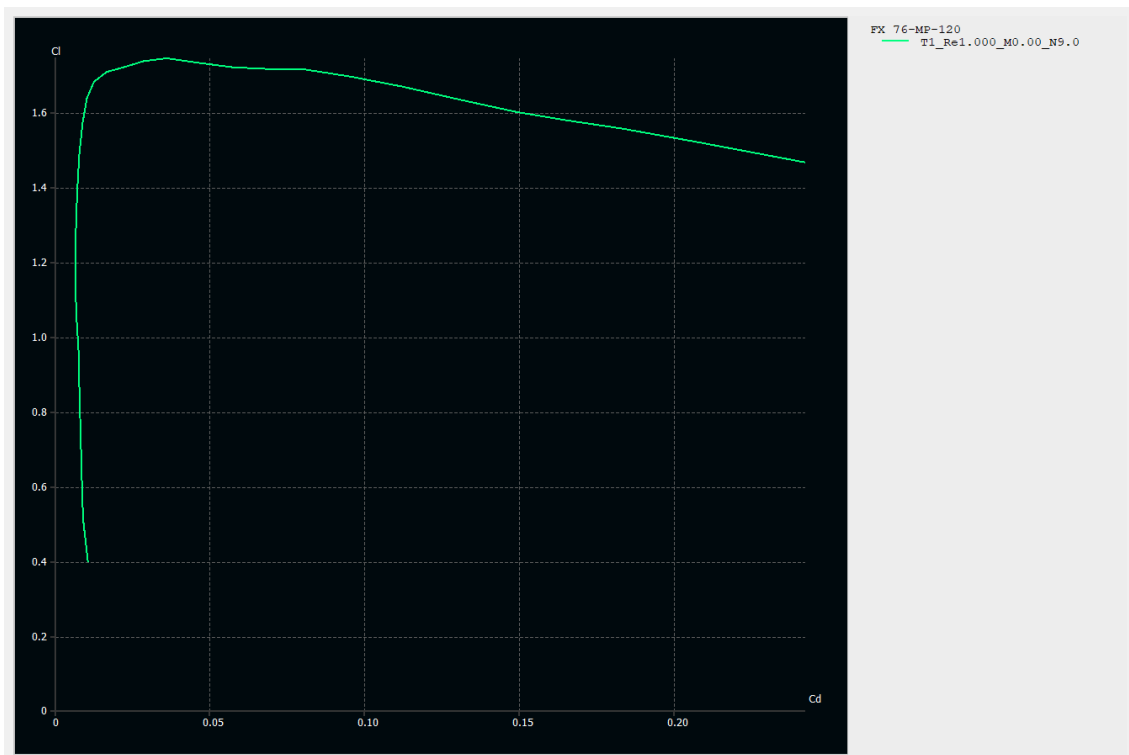


Figure 07: C_l vs C_d graph of WORTMANN FX 76-MP-120 aerofoil

1.2 Aspect Ratio Selection

The aspect ratio, defined as the ratio of the wingspan to the mean chord length, affects the distribution of lift and the magnitude of induced drag. Higher aspect ratios generally lead to more efficient lift production and lower induced drag.

Upon selecting the aerofoil, the polar data file, which includes all the C_l , C_d and C_m versus α data, was imported into MATLAB as a text file. This data was generated using AirfoilTool for a Reynolds number of 1 million with an N_{crit} of 9. The imported data was stored in MATLAB arrays as matrices for further analysis.

In the context of an aircraft, the finite wing effect is significant and must be considered for accurate performance predictions. Unlike an idealized infinite wing, a real aircraft wing has tips where the airflow can spill over from the high pressure underside to the low pressure topside. This spillover creates wingtip vortices, which are spirals of rotating air that trail behind the wing.

These wingtip vortices induce a downward component of velocity in the airflow around the wing, known as downwash. The downwash reduces the effective angle of attack experienced by the wing, which leads to a reduction in lift compared to an infinite wing. This induced drag, a byproduct of lift generation, also increases due to the energy lost in the formation of these vortices.

This effect can be incorporated into calculations using the following equation: [7]

$$a = \frac{a_0}{1 + 57.3 a_0 / (\pi \epsilon_1 AR)}$$

The slope of the finite wing lift curve is dependent on the span efficiency factor e and the aspect ratio. For a wing with an elliptical lift distribution, e equals 1.

The efficiency of a finite wing is characterized by the span efficiency factor, which accounts for deviations from the ideal elliptical lift distribution. In practical terms, optimizing the wing's aspect ratio can mitigate the adverse effects of finite wing aerodynamics. For a wing with an

elliptical lift distribution, e equals to 1. However, achieving an elliptical distribution is impractical in our case. Therefore, we assume a span efficiency factor of 0.9.

To determine the optimal aspect ratio for the wing, a range of aspect ratios from 12 to 30 was considered. By plotting the finite lift curves (C_l vs α) for a range of aspect ratios using MATLAB, we can analyze and determine the most efficient configuration for our specific application, balancing aerodynamic performance with practical considerations.

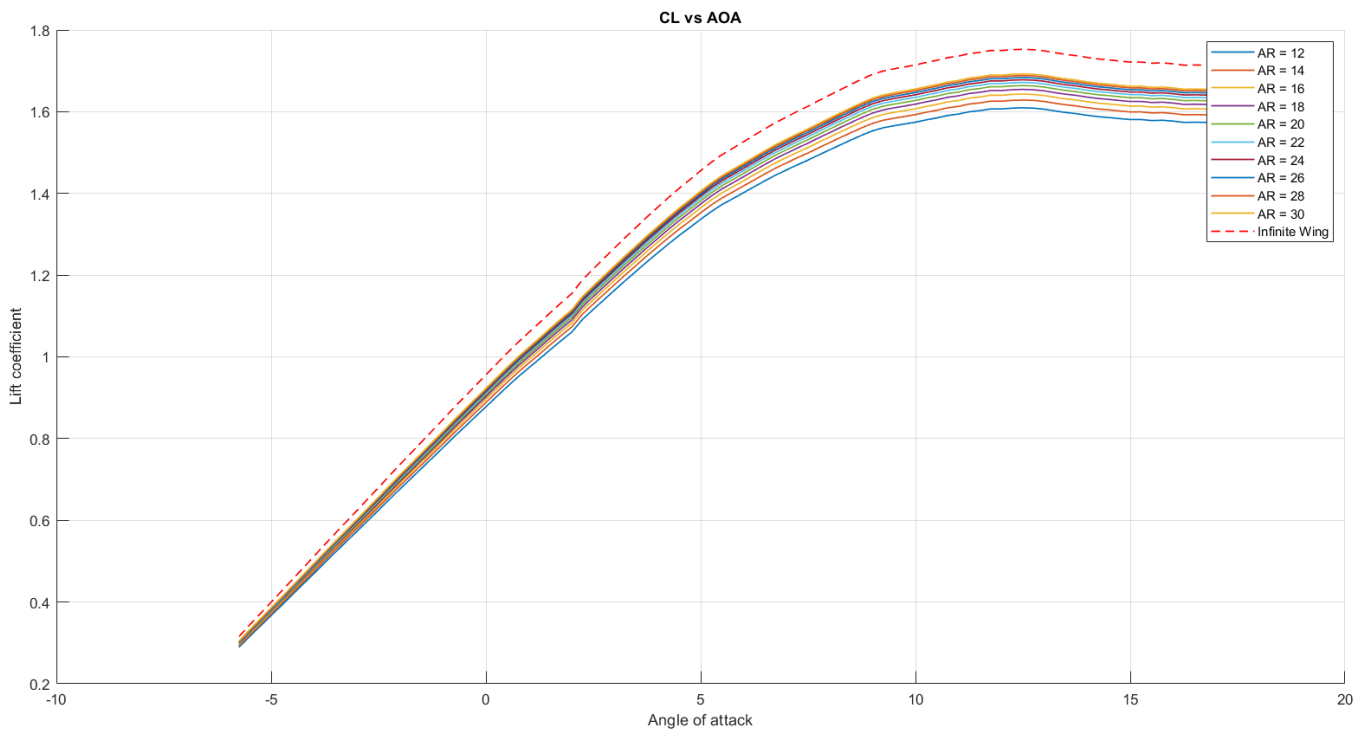


Figure 08: C_l vs α graph for a range of aspect ratios graph

Oswald Efficiency Factor

The Oswald efficiency factor, denoted as e_0 , is a parameter used in aerodynamics to quantify the efficiency of a wing in producing lift while minimizing drag. It accounts for the distribution of lift across the span of the wing and the induced drag that results from wingtip vortices. A higher Oswald efficiency factor indicates a more efficient wing with lower induced drag. The efficiency factor is influenced by the aspect ratio (AR) and taper ratio of the wing.

The taper ratio, which is the ratio of the tip chord to the root chord, also impacts the lift distribution and aerodynamic efficiency. Wings with an optimal taper ratio can achieve a more elliptical lift distribution, enhancing the Oswald efficiency factor and reducing induced drag.

$$e = \frac{1}{\left(1 + 0.12 \cdot M^2\right) \left(1 + \frac{0.142 + f(\lambda) \cdot A \cdot (10 \cdot t/c)^{0.33}}{(\cos \varphi_{25})^2} + \frac{0.1 \cdot (3N_e + 1)}{(4 + A)^{0.8}}\right)}$$

with:

$$f(\lambda) = 0.005 \cdot (1 + 1.5 \cdot (\lambda - 0.6)^2).$$

Here,[8]

M = Mach number

t/c = Airfoil thickness ratio

λ = Taper ratio

A = Aspect ratio

Φ = Sweep angle at 25% of chord line

N_e = Number of engines over the wing

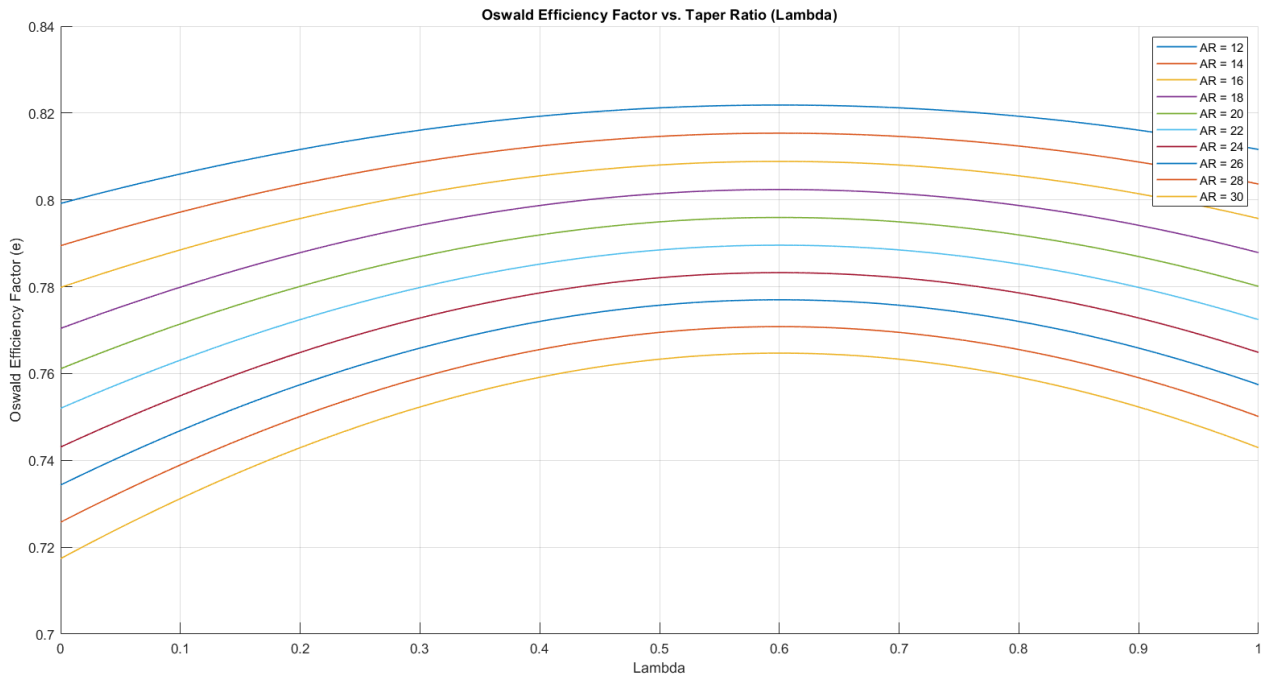


Figure 09: Oswald Efficiency factor vs taper ratio for range of aspect ratios graph

Utilizing this empirical equation, curves for the Oswald efficiency factor vs taper ratio were plotted for a range of aspect ratios. In the calculation process, the optimal gliding velocity cannot be directly input. Instead, this velocity is determined after completing all relevant calculations. Once the optimal gliding velocity is established, the corresponding Mach number can be calculated. Subsequently, the program should be run again with this updated information. The sweep angle and number of engines over the wing were taken as zero.

After analyzing both the C_l comparison and Oswald efficiency factor comparison graphs, an aspect ratio value of 26 was selected for the wing.

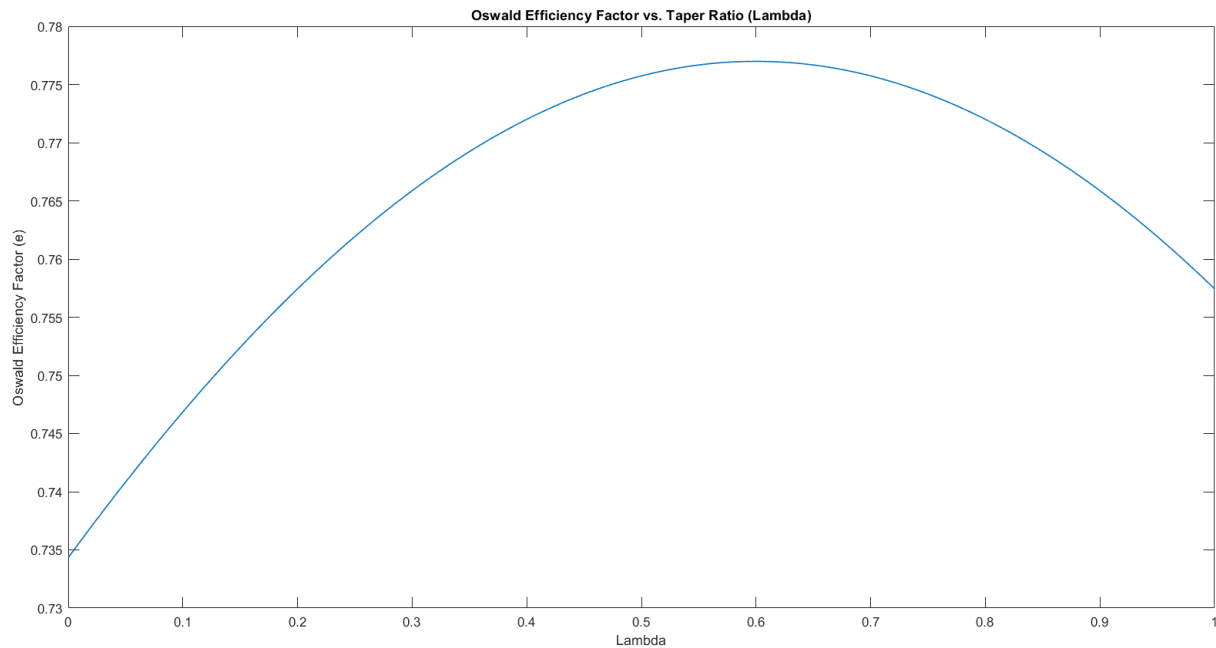


Figure 10: Oswald Efficiency factor vs taper ratio for $AR = 26$

By differentiating the Oswald efficiency factor with respect to the taper ratio, the optimal taper ratio corresponding to the highest Oswald efficiency factor was determined.

Best taper ratio = 0.6

Best Oswald Efficiency factor = 0.7770

1.3 Basic Calculations

In the basic calculation process, the chord lengths of the wing root and tip, fuselage length, required velocity, wing area, mean aerodynamic chord length, and Reynolds number are determined. To perform these calculations, the mass of the aircraft, air density, span length, and cruising lift coefficient must be provided as input parameters. For cruising flight, a wing load factor of 1 is assumed for the calculation of the required velocity.

Maximizing the range of a glider involves optimizing its aerodynamic efficiency, which is often quantified by the lift-to-drag ratio (Cl/Cd). The maximum Cl/Cd value represents the condition where the glider achieves the highest lift for the least amount of drag, thereby enabling it to glide the furthest distance for a given loss of altitude. To perform the calculations, the angle of attack (α) that yields the maximum Cl/Cd ratio must first be determined. Subsequently, the corresponding Cl/Cd value for the finite wing should be inputted as a parameter. This step can also be executed after computing all other necessary parameters.

The following parameters were taken for the calculations.

- **Mass of the aircraft = 300kg**
- **Aspect ratio = 26**
- **Cl (Finite) value at cruise flight = 1.436138**
- **Span length of the wings = 16m**

The following parameters were determined

- **Wing root chord length = 1.5385m**
- **Wing tip chord length = 0.92308m**
- **Wing Area = 9.8462m²**
- **Fuselage Length = 7.2m**
- **Mean Aerodynamic Chord (MAC) = 1.2564m**
- **Required Velocity = 17.9808ms⁻¹**
- **Re = 919148.418**

1.4 Coefficients Calculation

The distinction between 2D aerofoil curves and finite wing curves is essential in aerodynamic analysis. For a 2D aerofoil, the C_l vs. α and C_d vs. α curves represent the lift and drag characteristics as a function of the angle of attack in an idealized, infinite span condition, where there are no wingtip effects. These curves assume uniform flow around the aerofoil, leading to higher lift coefficients and lower drag coefficients compared to a finite wing.

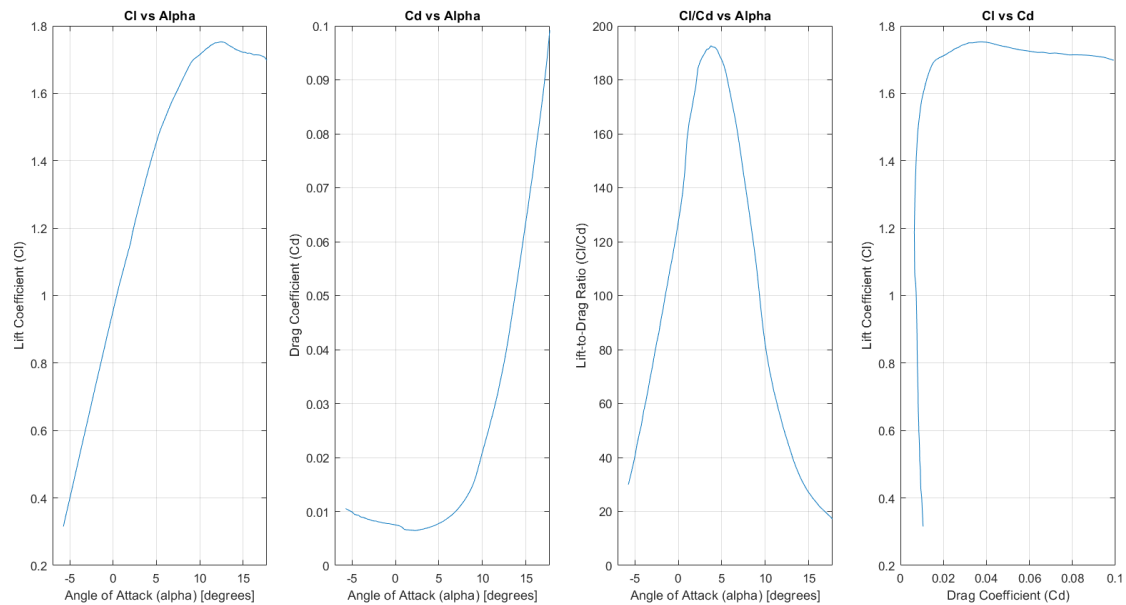


Figure 11: 2D aerofoil characteristic graphs

In contrast, finite wing curves account for the three dimensional effects, including wingtip vortices and induced drag. These effects reduce the effective angle of attack and increase the overall drag, leading to different C_l vs. α and C_d vs. α characteristics. The lift slope is typically lower, and the drag is higher for a finite wing, reflecting the real world performance more accurately. Understanding these differences is crucial for accurately predicting and optimizing the aerodynamic performance of aircraft wings.

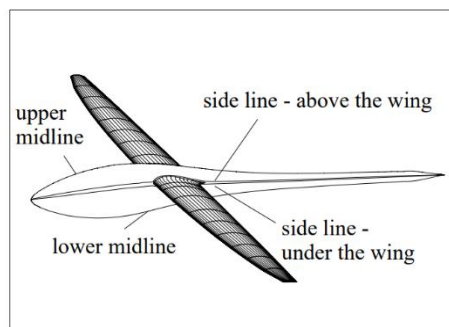


Fig. 2 : Streamlines along a fuselage

	CD num.	CD exp.[11]	x_{tr}/L num.	x_{tr}/L exp.
body 1	0.058	0.051	0.30	0.33
body 2	0.038	0.034	0.35	0.35
body 3	0.039	0.035	0.31	
body 4	0.40	0.038	0.29	0.31

Figure 12: Drag coefficients for streamline body

Using linear regression, the linear region of the infinite Cl vs. alpha curve was modelled into the form $y = mx + c$. This approach allowed for a precise determination of the slope and the y-intercept of the new infinite wing lift curve. By applying the previously mentioned equation, the modified lift curve's slope and y-intercept were accurately calculated. The results obtained from this analysis are as follows.

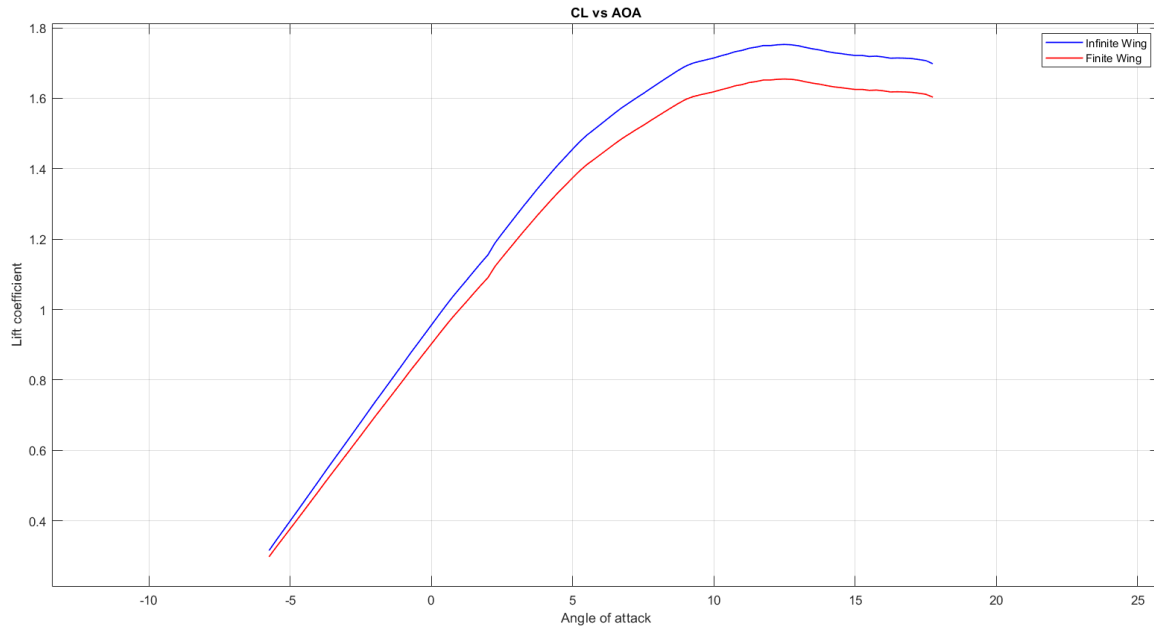


Figure 13: Cl vs alpha for finite and infinite wing

When determining the drag coefficient, only the fuselage and main wings were considered to simplify the calculations. This approach focuses on the primary contributors to drag, streamlining the analysis. For the fuselage drag coefficient at zero lift, data from published literature was used as a reference.

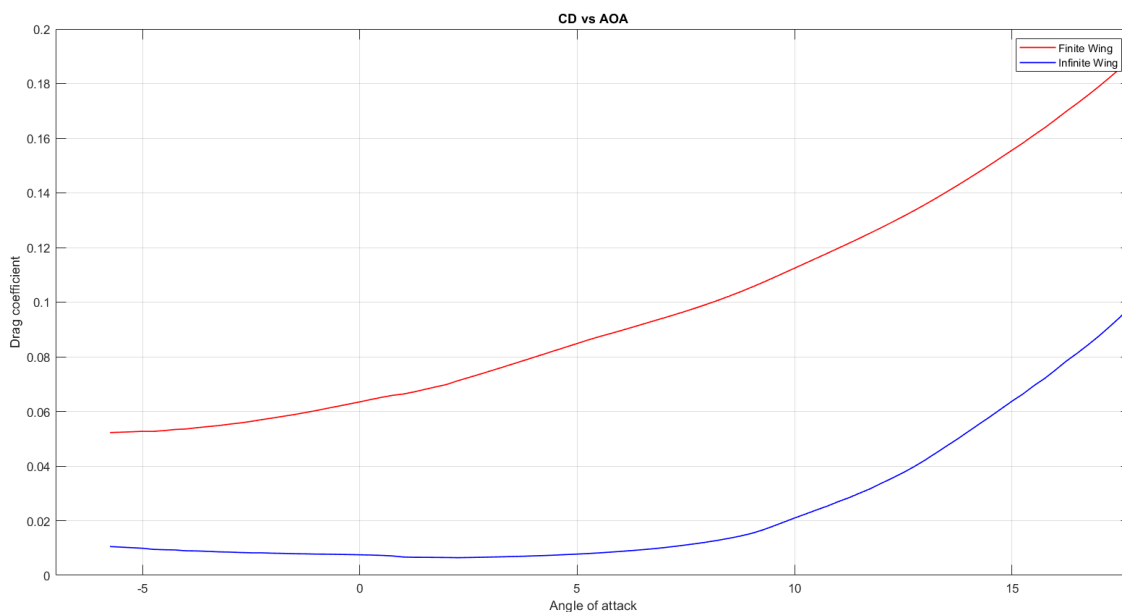


Figure 14: Cd vs alpha for finite and infinite wing

This literature provides established values and methodologies, ensuring accuracy and reliability in the calculations while reducing the complexity of accounting for additional components and interactions. So, **cd of 0.04 for fuselage** was taken as an approximation. [9]

Drag Polar Curve

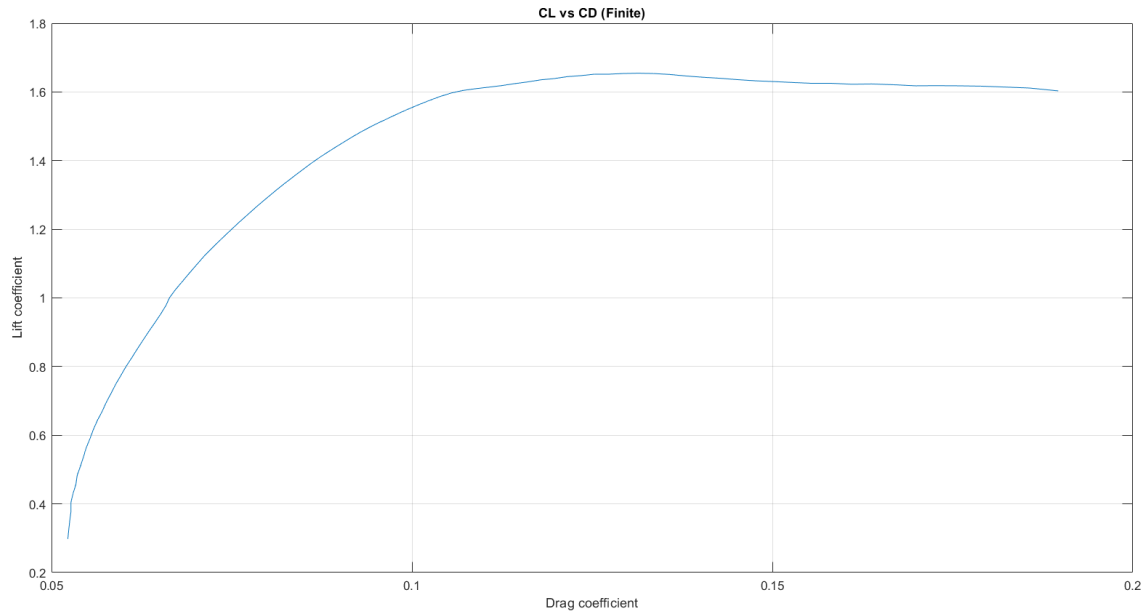


Figure 15: C_L vs C_d graph

These graphs represent the finite wing characteristics including the effect of induced drag.

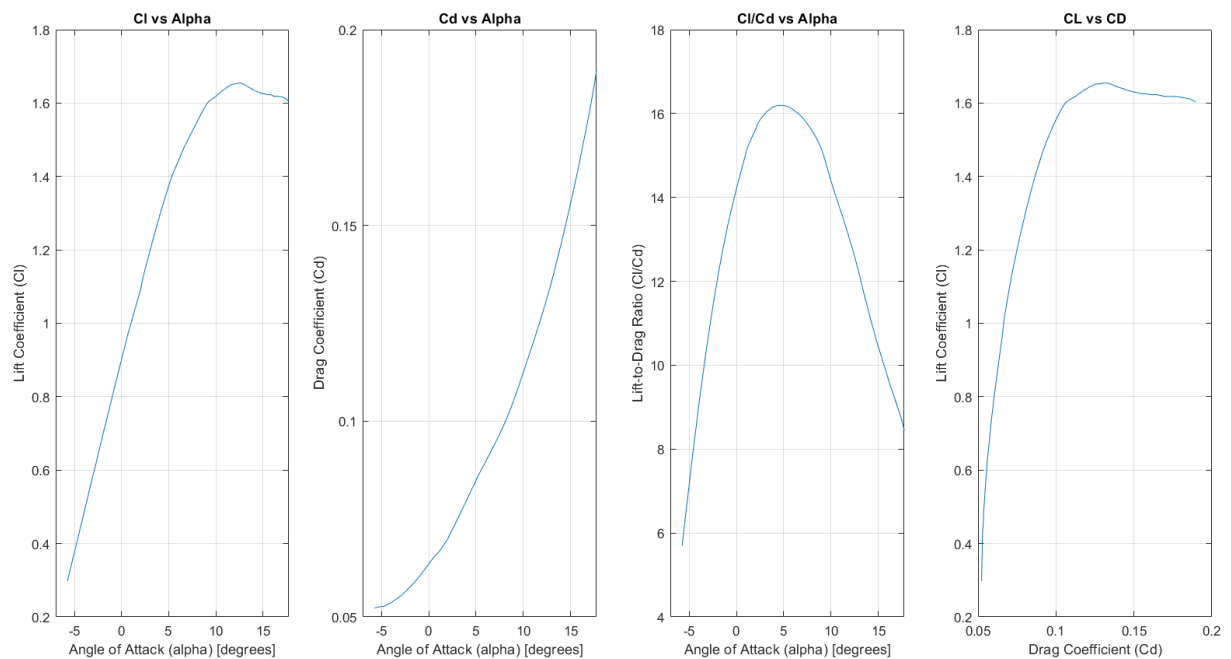
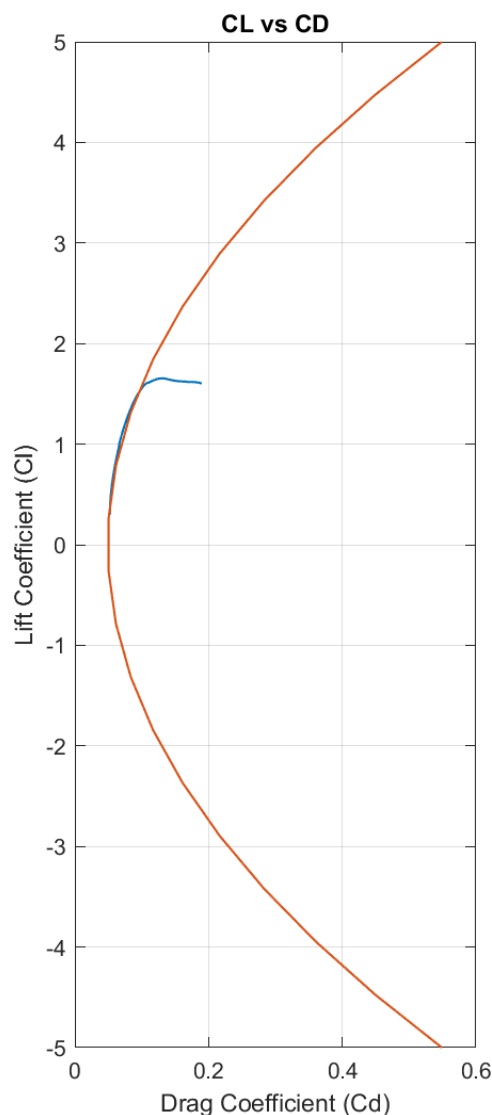


Figure 16: Finite wing characteristic graphs

Maximizing the range of a glider involves optimizing its aerodynamic efficiency by using the maximum Cl/Cd value. Operating at this optimal lift-to-drag ratio allows the glider to achieve the greatest distance for a given loss in altitude. By setting the incident angle of the wing to correspond with the angle of attack that provides the maximum Cl/Cd , the glider can minimize drag and maximize lift. This adjustment is beneficial for enhancing the overall range and performance of the glider, making it a fundamental principle in both glider design and flight strategy.

By utilizing the Cl/Cd curve obtained for the finite wing, the maximum Cl/Cd value it can attain is 18.83 at an angle of attack of 5.5 degrees. Consequently, the main wing incident angle was set to 5.5 degrees to maximize aerodynamic efficiency. At this angle of attack, the wing produces a lift coefficient Cl value of 1.43, optimizing the glider's performance and enhancing its range by achieving the most favourable lift-to-drag ratio.



For the drag polar curve of the finite wing, the following polynomial approximation can be applied, as illustrated by the orange curve.

$$C_d = 0.014 C_l^2 + 0.049$$

1.5 Glider Polar Curve

A glider polar curve is a graphical representation that illustrates the relationship between a glider's sink rate and its airspeed. This curve is fundamental in understanding the glider's performance characteristics under various flight conditions. By constructing a tangent line from the origin to the curve, pilots can determine the best gliding velocity, which is the speed at which the glider achieves the most favourable lift-to-drag ratio. This optimal airspeed is crucial for maximizing range and endurance. Analysing the polar curve provides critical insights into the most efficient flying speeds for different conditions, such as when flying into headwinds or downwind, thereby optimizing overall flight performance.

When constructing the glider polar curve for the aircraft, it is essential to analyse wing characteristics, such as the lift curve, across various gliding speeds. This requires generating different C_l curves for different Reynolds numbers. For simplicity, it is assumed that the variation in Reynolds number due to changes in velocity has a negligible effect on each C_l curve.

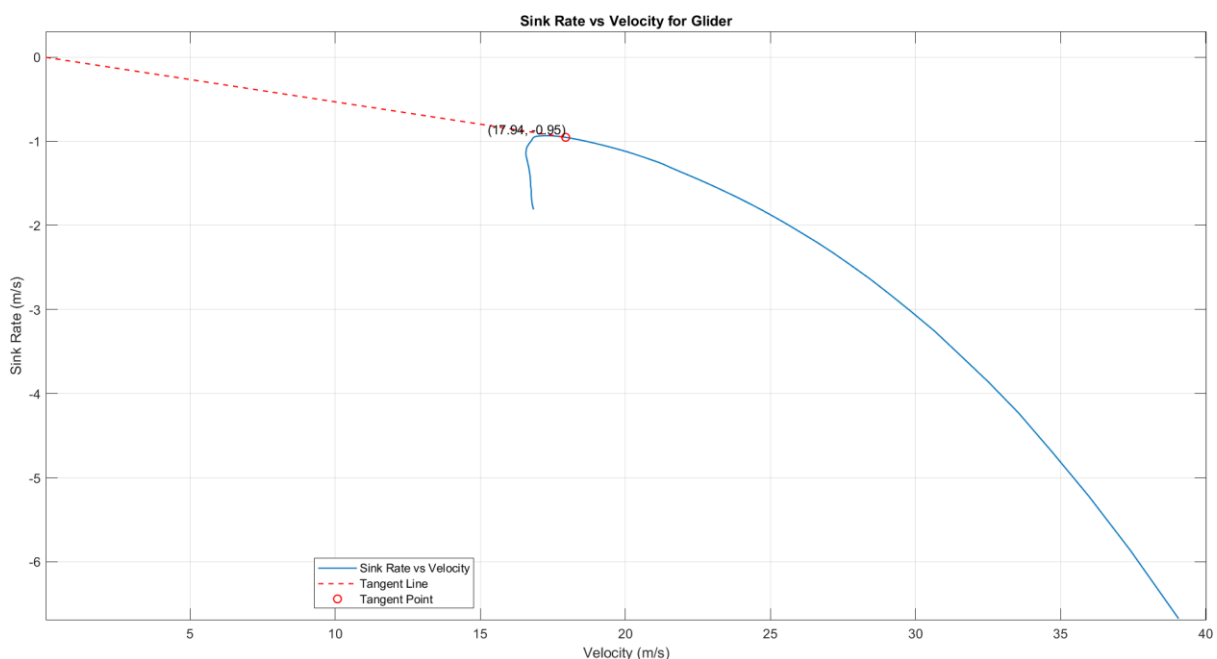


Figure 17: Glider polar curve of the aircraft

By constructing the tangent line for the glider polar curve, the tangent point was identified as (17.9424, -0.9527). This indicates that the best gliding velocity is 17.94 m/s, with a corresponding sink rate of 0.9527m/s. Furthermore, this velocity value aligns closely with the theoretical velocity determined in the basic calculation process, confirming their approximate equivalence in magnitude.

1.6 Longitudinal Stability

Longitudinal stability is a crucial aspect of an aircraft's overall stability, ensuring it maintains a steady flight path along its longitudinal axis, which extends from the nose to the tail. The need for a tail wing, or horizontal stabilizer, arises from the necessity to balance the pitching moments generated by the main wing and the fuselage. As the main wing produces lift, it also generates a nose-down pitching moment due to its aerodynamic center being ahead of the aircraft's center of gravity.

The tail wing provides a counteracting force, generating a nose-up moment to stabilize the aircraft. This is achieved by creating a downward lift force at the tail, which offsets the nose-down pitching moment. By adjusting the tail wing's angle of attack, pilots can control the aircraft's pitch attitude, ensuring smooth and stable flight. This stabilizing effect is essential for maintaining control and achieving the desired flight performance under various conditions.

For the tail wing, a symmetric aerofoil, specifically the **WORTMANN FX 76-120**, with a negative tail setting angle was utilized. Given that the tail wing is also a finite wing, it is necessary to perform the same induced drag calculations. However, since the tail wing's contribution to the overall drag of the aircraft is minimal, the infinite wing characteristics were used as an approximation for the C_d vs α graph. An initial design assumed a span efficiency factor of 0.9 and an aspect ratio of 6. Lift slope calculations accounting for downwash effects as same as for the wing were conducted to generate the finite C_l vs α curve for the tail wing, as detailed below.

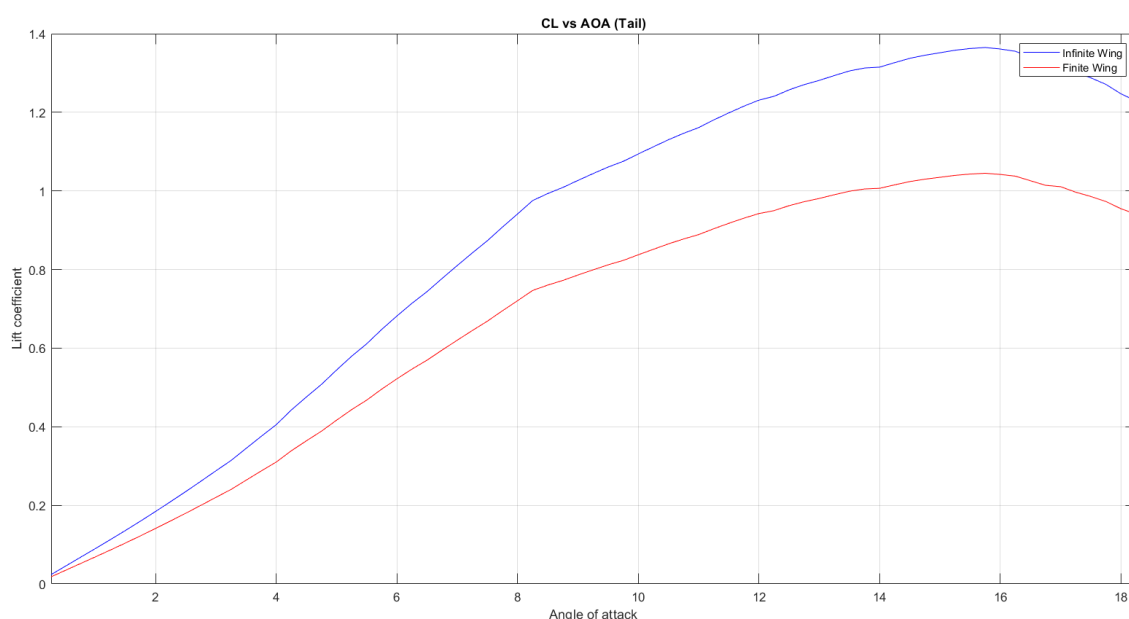


Figure 18: C_l vs α for the tail wing

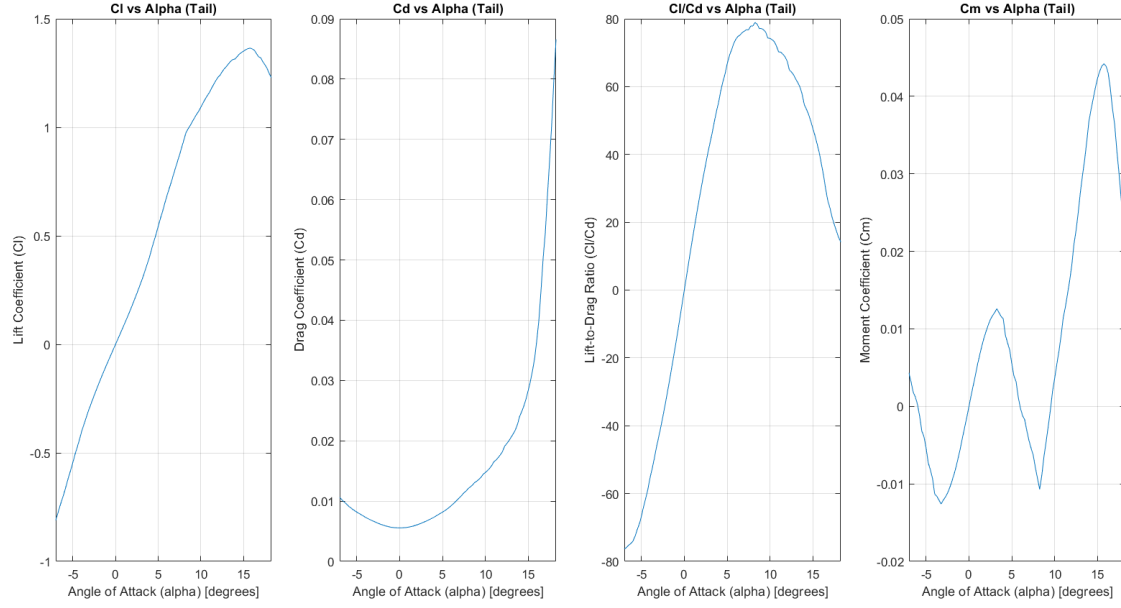


Figure 19: Characteristic graphs for the tail wing (Infinite)

Applying the same concept as before, the tail setting angle was determined using the alpha angle that yields the maximum Cl/Cd value for the tail wing, which is 8.25 degrees. Given that the tail is designed to produce negative lift, the tail setting angle was set to negative 8.25 degrees. The tail wing has a total area of 3 m², and the distance between the aerodynamic centers of the main wing and the tail wing is 4.5 meters. From the Cm vs Alpha graph, the zero-lift moment coefficient was found to be -0.21. The glider design was modelled in SolidWorks, which helped determine the center of gravity location, precisely 0.48 meters behind the leading edge. For simplicity, the downwash at zero angle of attack was calculated assuming the tail is positioned behind a main wing with an elliptical lift distribution. This assumption was also used to determine the rate of change of the downwash angle with the angle of attack. Using all these parameters, the Cm_{cg} vs Alpha graph was plotted according to the following equation, [7]

$$C_{M, cg} = C_{M, ac_{wb}} + a_{wb} \alpha_{wb} \left[h - h_{ac_{wb}} - V_H \frac{a_t}{a_{wb}} \left(1 - \frac{\partial \epsilon}{\partial \alpha} \right) \right] + V_H a_t (i_t + \epsilon_0)$$

Following equations were used for calculations. [10]

$$\varepsilon = \frac{2C_{L_w}}{\pi AR_w} \quad \text{and} \quad \frac{d\varepsilon}{d\alpha} = \frac{2C_{L_{\alpha w}}}{\pi AR_w}$$

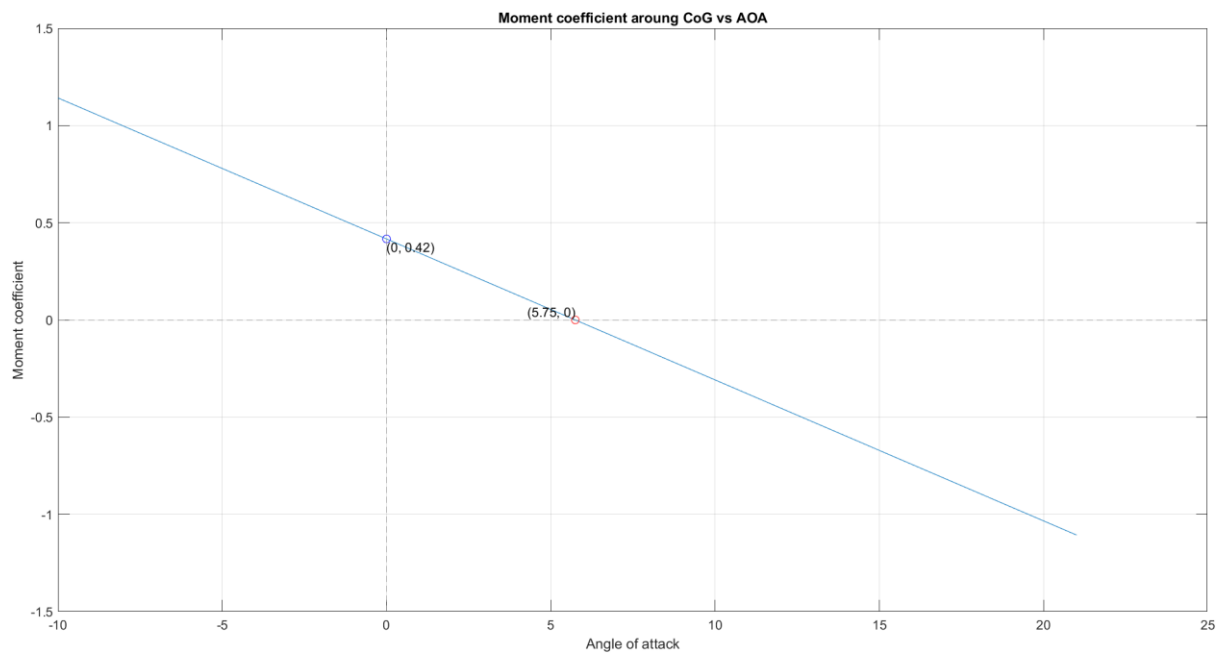


Figure 20: Moment coefficient vs alpha graph

For the aircraft to achieve longitudinal stability, the slope of the curve must be negative, and the y-intercept should be positive. The graph indicates that this design meets both of these key requirements. Additionally, the trim angle, determined from the graph, is 5.75 degrees, which closely approximates the main wing's incident angle of 5.5 degrees.

Centre of Gravity

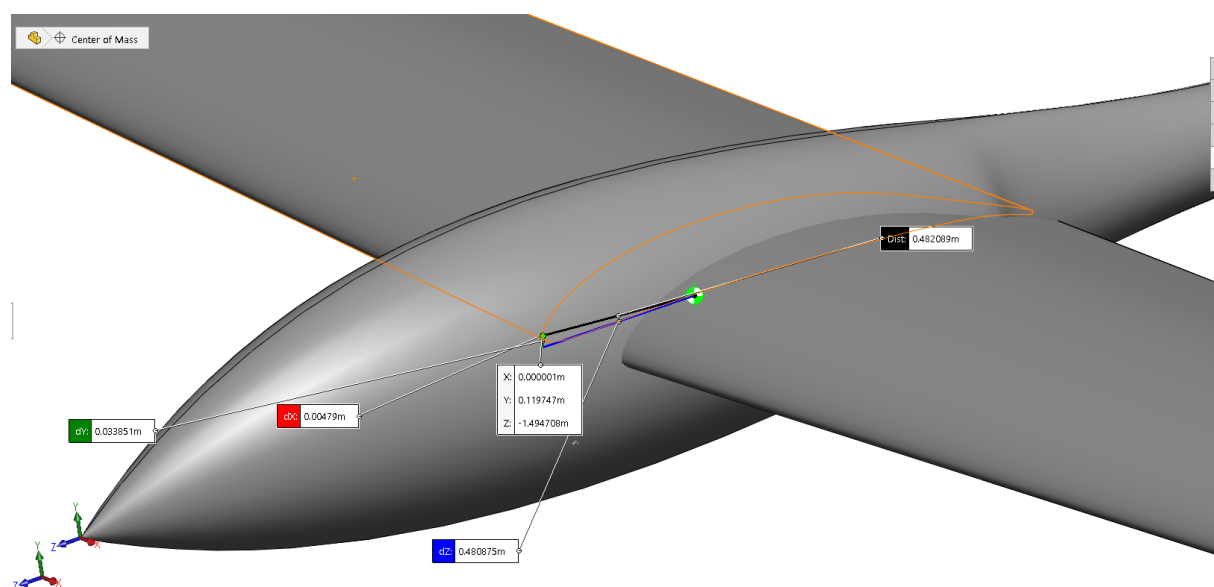


Figure 21: Center of gravity estimation using SolidWorks

1.7 CAD Model

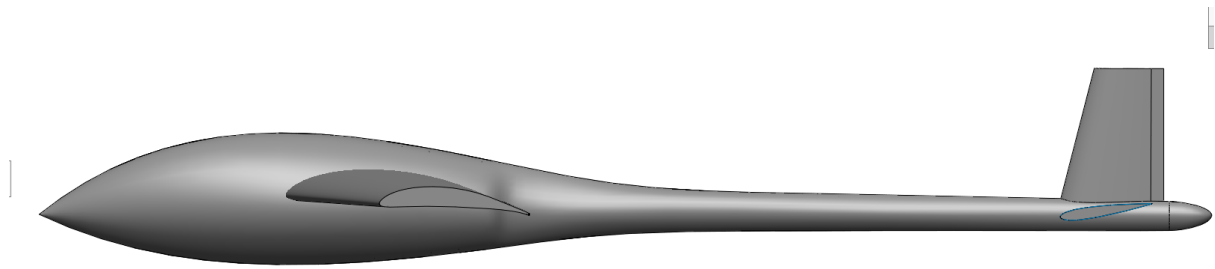


Figure 21: Side view

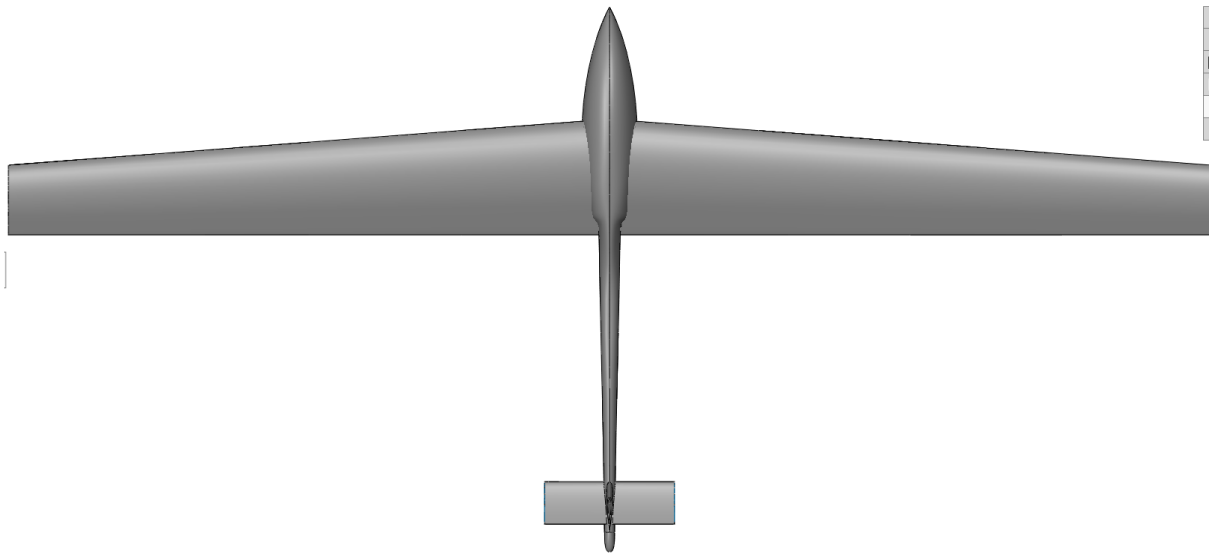


Figure 22: Top down view

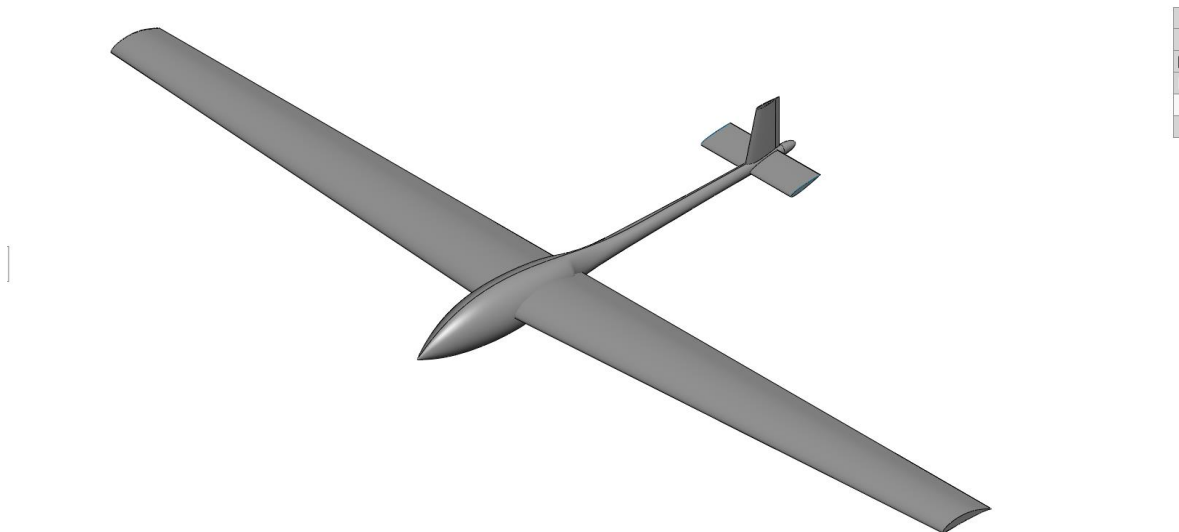


Figure 23: Isometric View

CHAPTER 02

Structural Design and Material Selection

2 Wing Structure External and Internal Loadings

The primary objective of this phase is to accurately determine the forces acting along the aircraft wing, which are critical for structural and performance analyses. To facilitate a systematic approach, the coordinate system is defined with the x-axis aligned along the chord of the wing and the y-axis aligned along the wingspan.

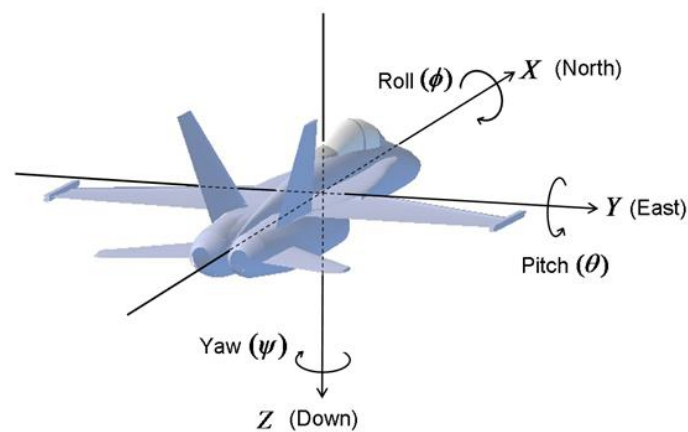


Figure 24: Aircraft Standard Coordinates System

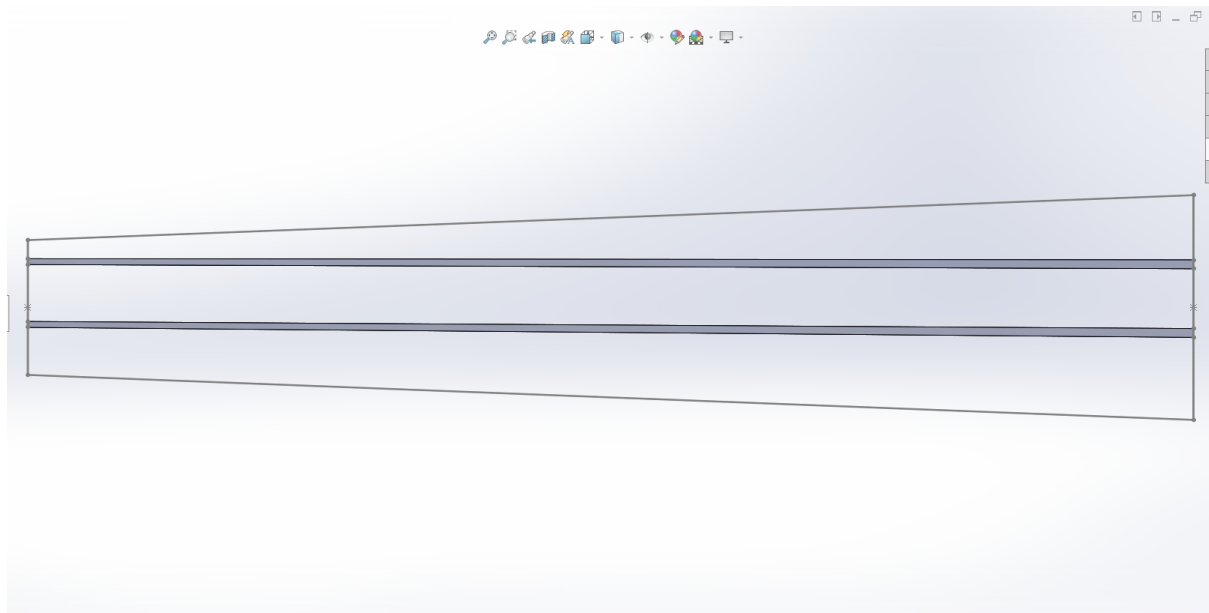


Figure 25: Wing spar configuration (Viewed from top)

In this analysis, the wing is modeled as a cantilevered beam subjected to various aerodynamic and structural loads. The mass of the fuselage is considered to exert a point load on the wing. For calculation purposes, one half of the wing is analyzed, with half of the fuselage mass represented as a point load acting at the cantilevered end of the beam. This assumption simplifies the model while adequately capturing the impact of the fuselage weight on the wing structure. **For all the calculations, Factor of Safety was taken as 1.5.**

Since the wing is tapered, a linear function of length along the spanwise direction (y) was used to model the varying chord length.

$$Wing\ Chord = \frac{-2}{Span} \times Chord\ Length \times (1 - Best_{\lambda}) \times y + Chord\ Length$$

Here, the Chord Length represents the wing chord at the root which is 1.5385m.

2.1 Lift Distribution

To simplify the force calculations, the lift distribution generated by the wing is assumed to follow an elliptical profile. This assumption is a widely accepted simplification in aerodynamics, as it reasonably approximates the actual lift distribution of an efficient wing and allows for more straightforward mathematical treatment.

The lift distribution constant was determined by integrating the elliptical distribution along the wingspan. This integral accounts for the total lift generated by the wing, ensuring that the distribution accurately reflects the aerodynamic forces acting on the wing. For the purposes of structural analysis, the maximum wing loading is considered to be 3.8 times the lift force of the aircraft. This factor represents the design load conditions that the wing must withstand, including considerations for maneuvers, turbulence, and safety margins.

$$K_{Lift} = \frac{4 \times Lift\ Needed \times 3.8}{\pi \times Span\ Length}$$

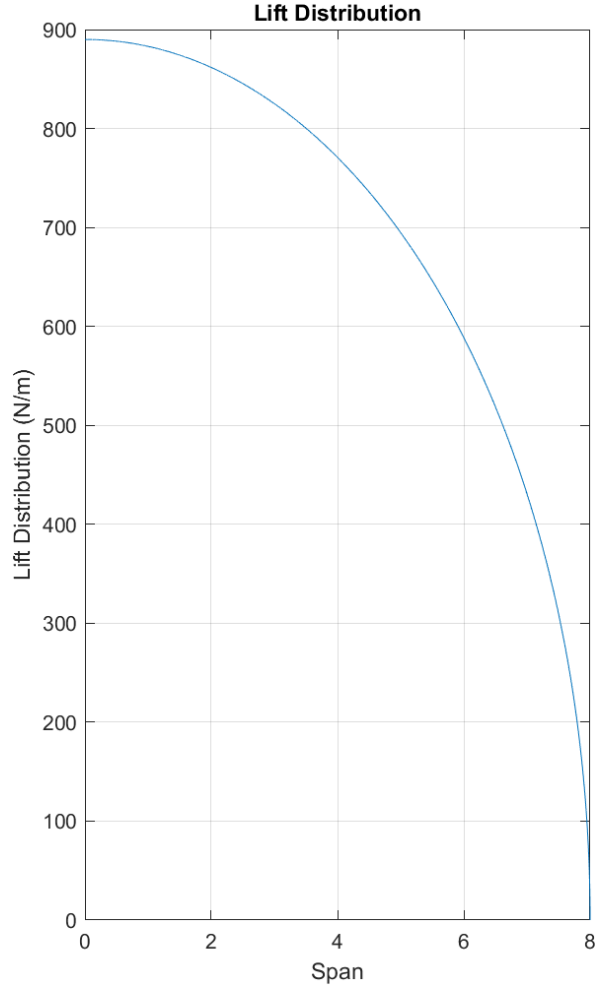


Figure 26: Lift distribution

2.2 Weight Distribution

According to the design constraints, the total weight of the wing must constitute 40% of the aircraft's total weight. Since the weight is proportional to chord length, the weight distribution along the wing is also modeled as a linear function ($y = mx + c$) of the wingspan. Constants within these equations were determined by deriving expressions for the total drag and weight, which were then integrated into the respective distributions.

$$K_{Weight} = \frac{Total\ Mass \times Wing\ Mass\ Percentage \times g}{2 \times \int_0^{\frac{Span}{2}} (Wing\ Chord) dy}$$

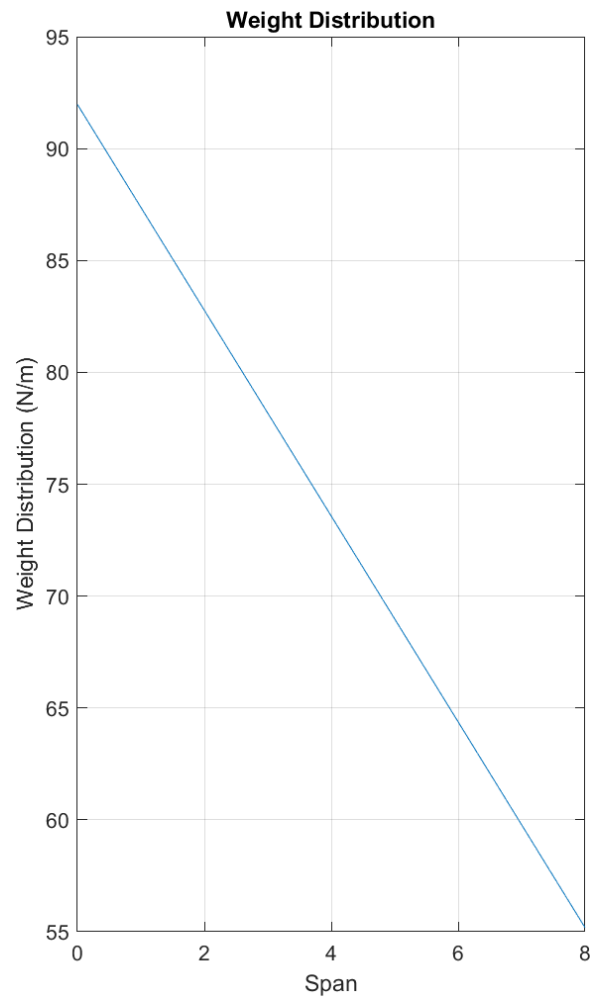


Figure 27: Weight distribution

2.3 Combined Lift and Weight Distributions

To provide a comprehensive view of the forces acting in the vertical (Z) direction, the lift distribution was combined with the weight distribution. The lift distribution, previously assumed to be elliptical for simplicity and accuracy, is contrasted with the linearly varying weight distribution to yield a net force profile along the span. This resultant graph of forces in the Z direction captures the combined effects of lift and weight, offering a clear visualization of the aerodynamic and gravitational forces the wing must support.

$$\text{Forces } Z = \text{Lift Distribution} - \text{Weight Distribution}$$

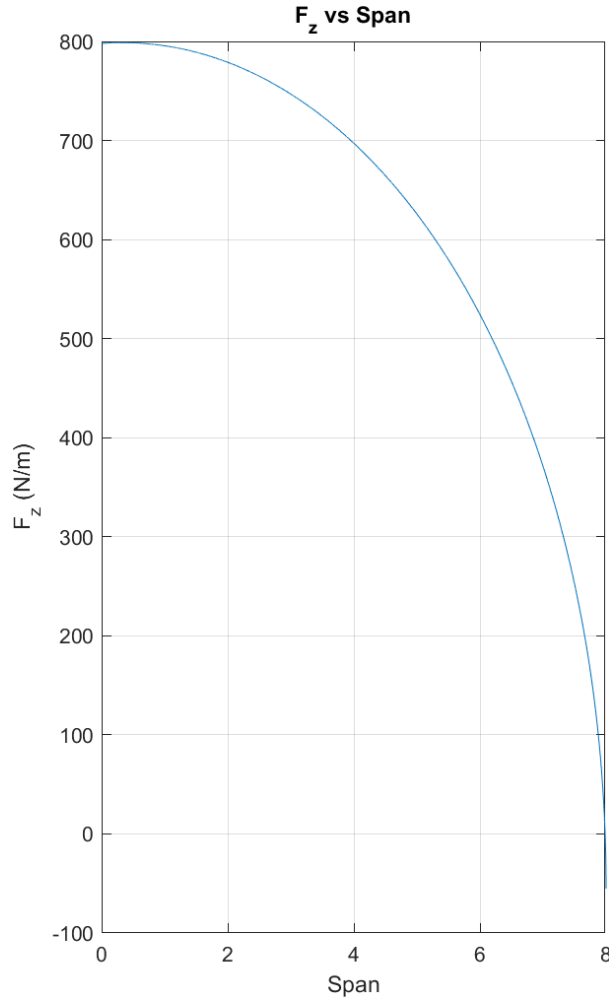


Figure 28: Combined force of lift and weight distribution

2.4 Drag Force Distribution

The total drag force is calculated at cruising conditions, under the assumption that the entire drag force acting on the aircraft is transferred to the wing.

Total drag coefficient was found out to be 0.0779. Using that, the total drag force can be calculated as follows.

$$\begin{aligned}
 Drag &= \frac{1}{2} \times \rho_{Air} \times Velocity^2 \times Wing Area \times C_d \\
 &= 160.2698 \text{ N}
 \end{aligned}$$

The drag force on each portion of the wing is assumed to be proportional to its surface area. Consequently, the drag distribution was modeled as a linear function of the chord length along the span of the wing, represented by the equation $y = mx + c$.

$$K_{Drag} = \frac{Drag \text{ (At Cruising)}}{2 \times \int_0^{\frac{Span}{2}} (Wing \ Chord) \ dy}$$

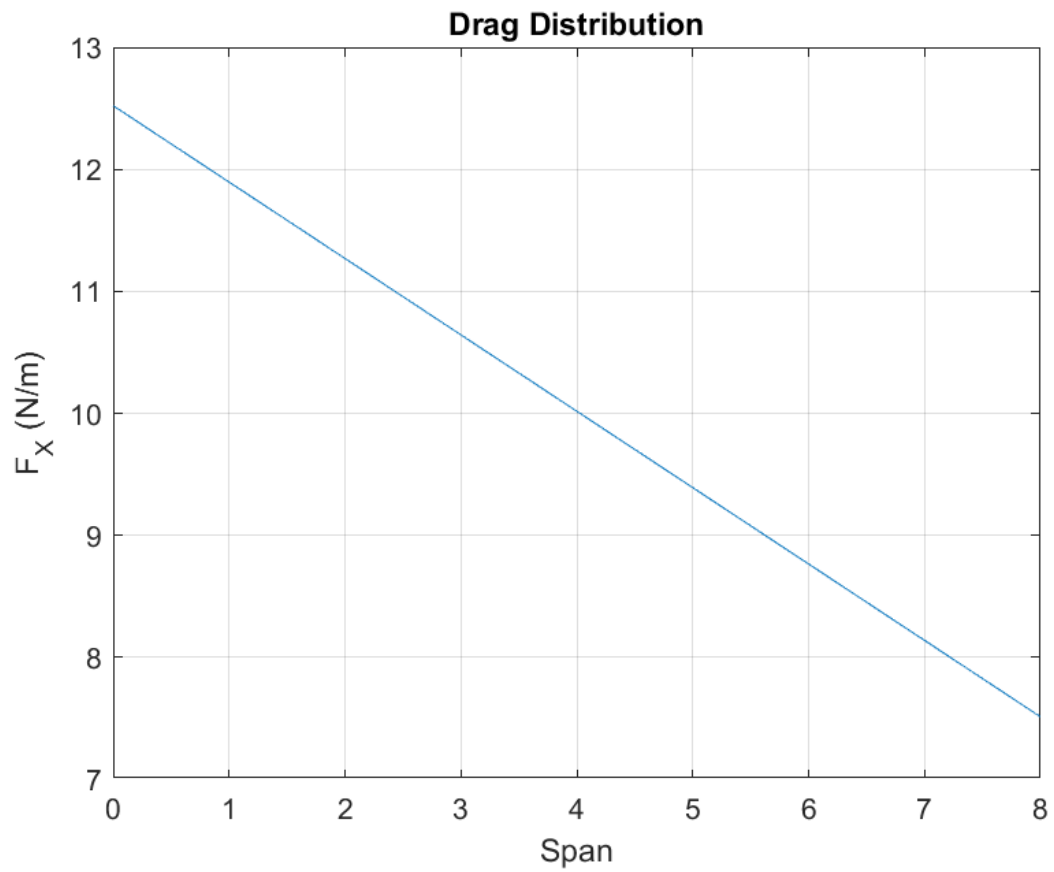


Figure 29: Drag distribution

2.5 Bending Moments and Shear Forces (Due to Lift and Weight)

The bending moments and shear forces along the span of the wing were determined using the integration method, which provides a detailed and accurate representation of the internal forces acting within the wing structure. For the sign convention, Hogging moments and clockwise shear are considered as positive. The following equations were utilized.

$$EI \frac{d^2y}{dx^2} = M(x)$$

$$EI \frac{dv}{dx} = \int M(x)dx + C_1$$

$$EIv = \int [M(x)dx]dx + C_1x + C_2$$

To derive the bending moments and shear forces, the load distribution along the span was integrated, considering the varying lift, weight, and drag distributions as previously defined. Boundary conditions were applied to solve the constants arising from the integration process. Specifically:

- The bending moment at the free end of the cantilevered wing is zero.
- The shear force at the cantilevered root (the fixed end) is equal to the total load acting at that end, which includes the aerodynamic loads and the point loads such as those from the fuselage.
- The reaction force at the root was determined by integrating the load distribution along the span, ensuring that the total shear force matches the sum of applied forces

[13]

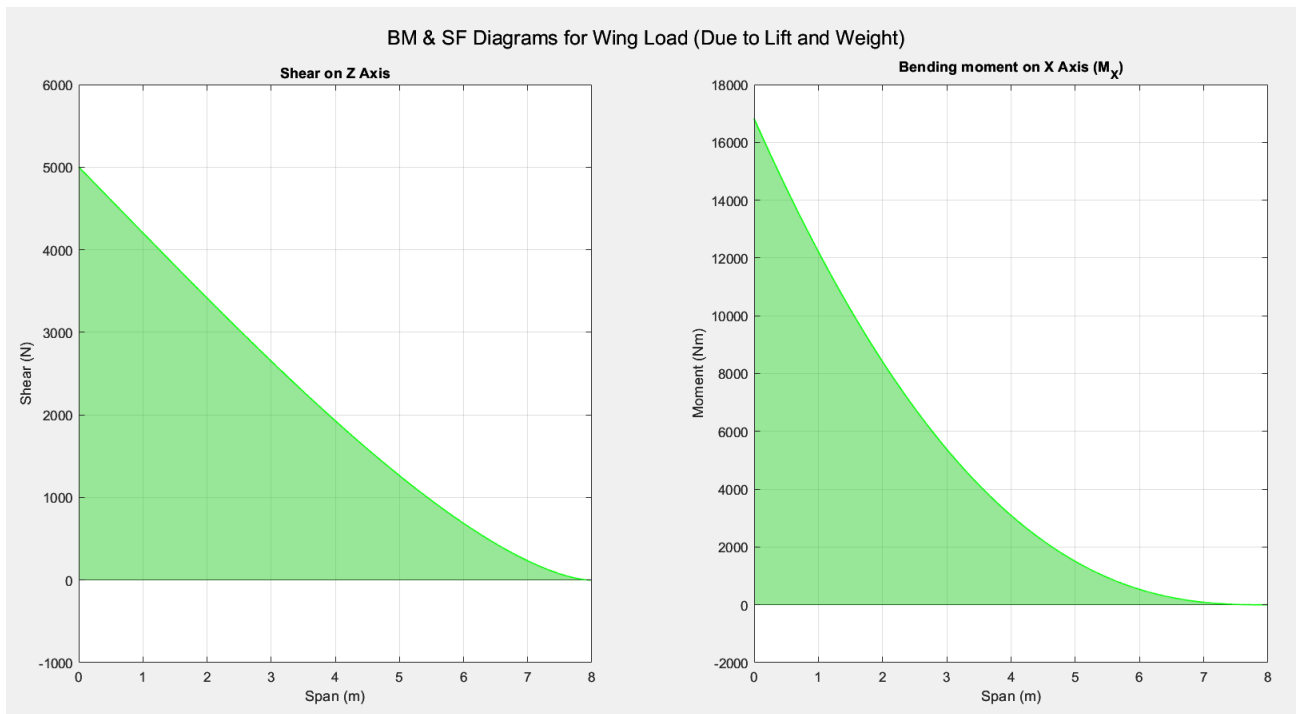


Figure 30: BM & SF Diagrams for Wing Load (Due to Lift and Weight)

2.6 Bending Moments and Shear Forces (Due to Drag)

The shear forces and bending moments due to aerodynamic drag are plotted to provide a visual representation of their distribution along the wing span. The following graph displays the shear forces and bending moments as functions of span position, reflecting the variations induced by aerodynamic drag.

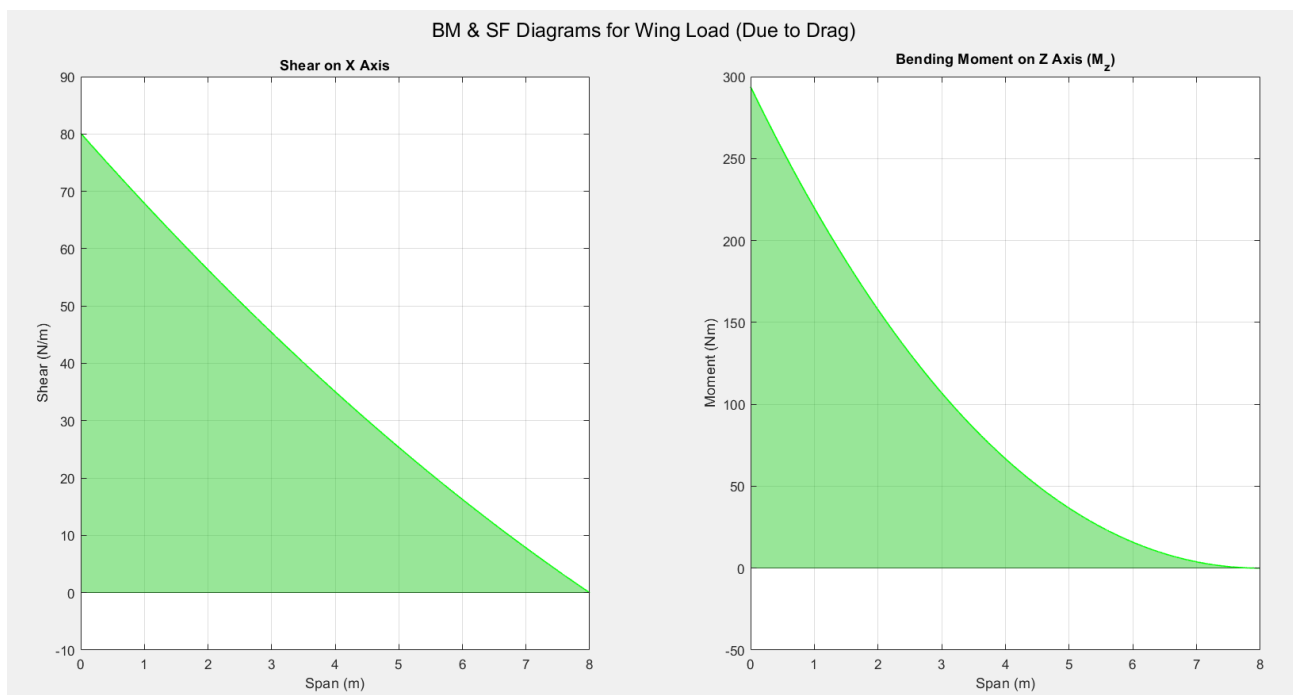


Figure 31: BM & SF Diagrams for Wing Load (Due to Drag)

In the structural analysis of the aircraft wing, it is assumed that the axial forces along the span of the wing are negligible. This assumption simplifies the analysis by focusing on the dominant forces and moments that affect the wing's performance and structural integrity.

2.7 Torsional Moments (Y Axis)

The shape of the aerofoil inherently generates a moment about the axis parallel to the aircraft's Y direction, which induces a torsional effect on the wing. This torsional moment arises due to the distribution of aerodynamic pressures along the chord length and their varying lever arms relative to the wing's shear center. As the chord length changes along the span, so does the magnitude of this moment, leading to a torsional distribution. Assuming that the torsional moment is proportional to the square of the chord length, the moment distribution along the span can be modelled using a quadratic function, represented as a parabolic curve. The moment coefficient at cruising stage was found out to be -0.2022. Using that, the proportionality constant and the torsional moment can be calculated as follows.

$$Moment = \frac{1}{2} \times \rho_{Air} \times Velocity^2 \times Wing Area \times C_m \times MAC$$

$$= -522.8376 Nm$$

$$K_{Moment} = \frac{Moment (At Cruising)}{2 \times \int_0^{\frac{Span}{2}} Wing Chord^2 dy}$$

$$M_Y = -K_{Moment} \times Wing Chord^2$$

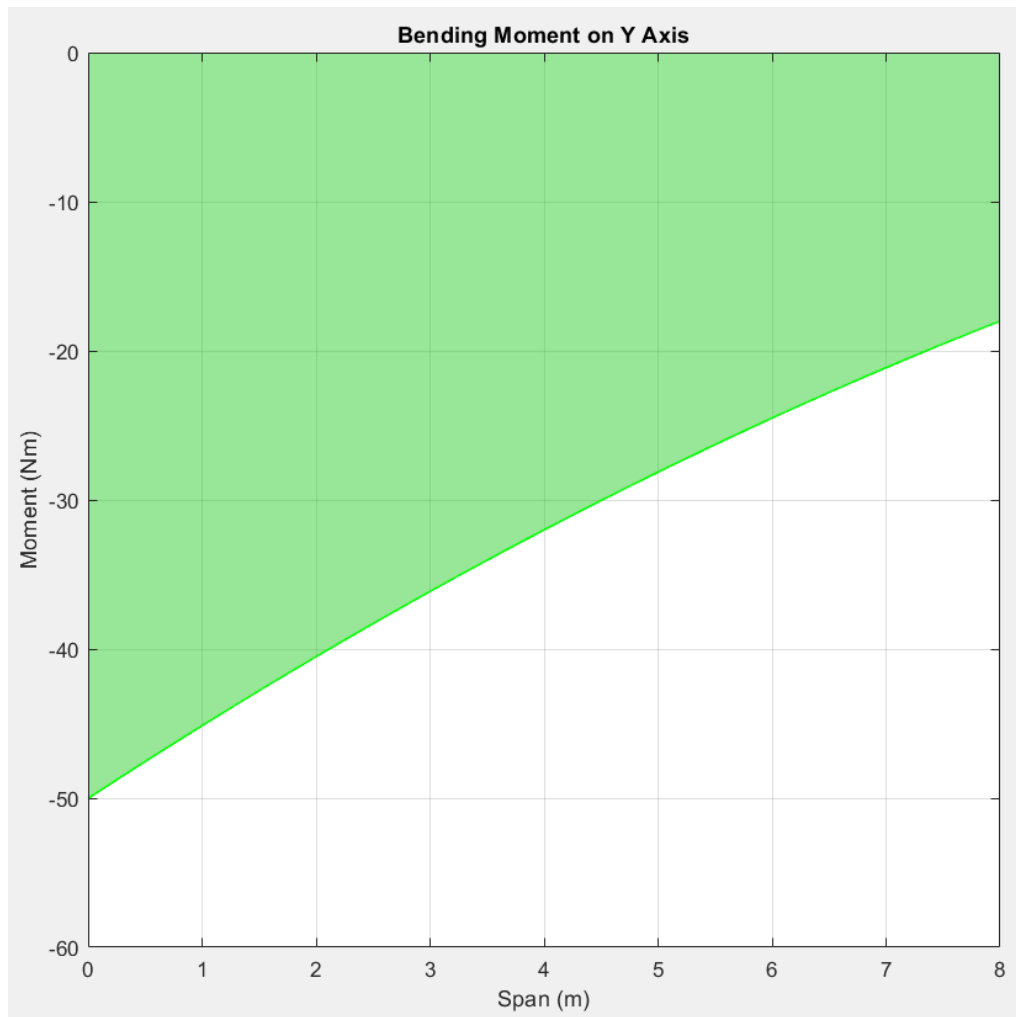


Figure 32: Bending Moment on Y Axis

2.8 Design and Analysis of the Internal Wing Structure

For the internal structure of the aircraft wing, a dual-spar design with a tapered C-shaped cross section was selected. The spars are positioned perpendicular to the fuselage to simplify the calculation of bending stresses and torsional effects. Despite the tapering of the spars, it was assumed for analytical purposes that the spars are perfectly horizontal, disregarding the minor taper angle as it has a negligible impact on the overall structural performance.

Structural Assumptions and Simplifications

To facilitate the structural analysis, the lumped area technique was employed, with an assumption that both spar cap areas and webs contribute to the bending resistance. A thin walled, single cell closed cross-section was chosen for the wing structure, which is a common approach in aircraft design due to its effectiveness in resisting torsional loads and providing adequate stiffness. [14]

Spar Design and Cap Sizing

The sizing of the spar caps, which form the flanges of the C-beam, was determined based on the available space within the wing structure and the overall weight constraints of the aircraft. The caps are designed to carry the majority of the bending loads, making their sizing critical to the structural integrity of the wing.

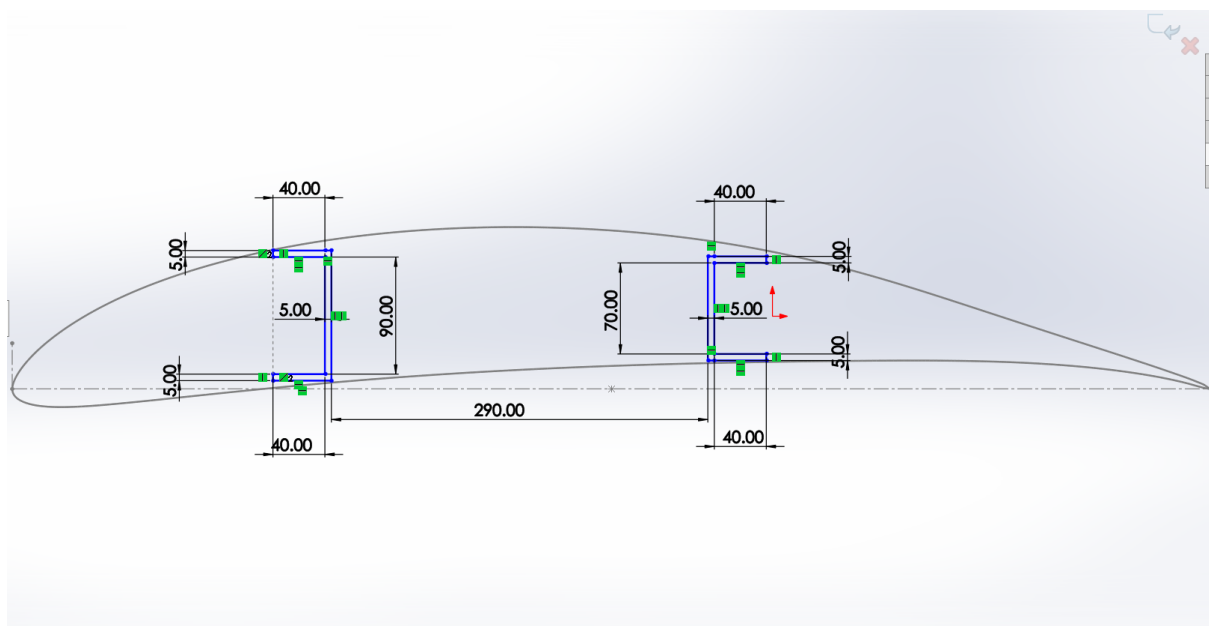


Figure 33: Wing tip spar placement

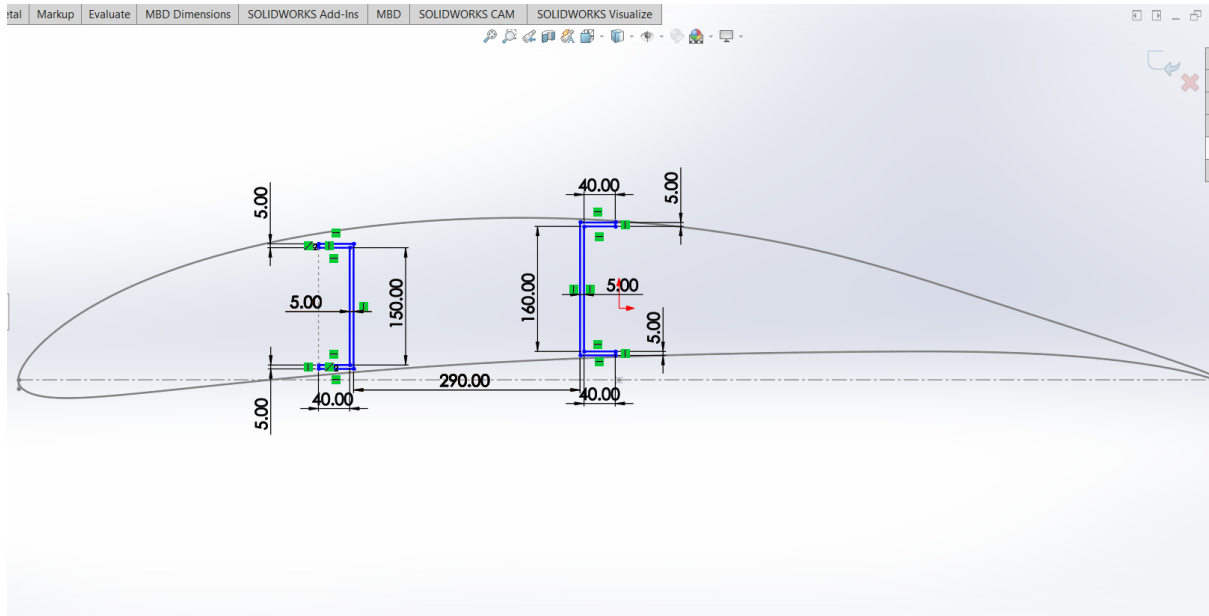


Figure 34: Wing root spar placement

For the lumped area calculations, above designs were simplified as follows. All the lumped areas marked by circle have same area of 200mm^2 .

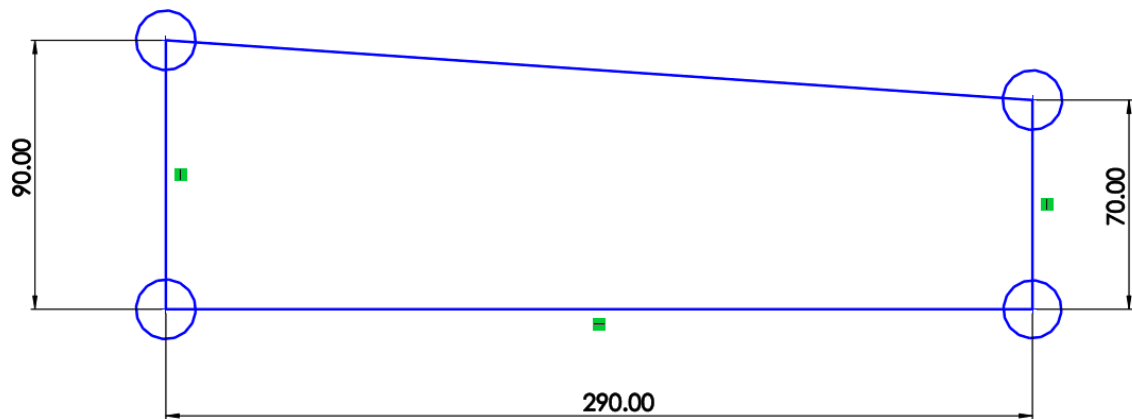


Figure 35: Wing tip lumped cap areas

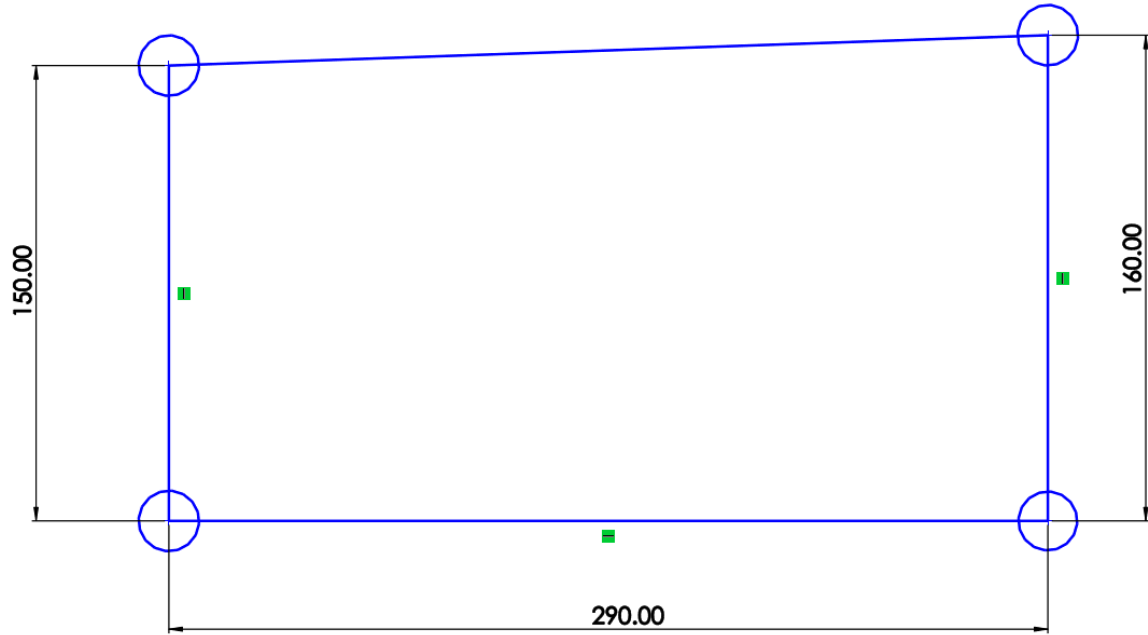


Figure 36: Wing root lumped cap areas

The dimensions of the wing box specifically the height values, change along with the span of the wing. For the simplicity of the calculations, an average value for the height was taken.

$$\text{Average Height} = \frac{(rh_1 + rh_2 + th_1 + th_2)}{4} = 117.5 \text{ mm}$$

Here,

$$rh_1 = 160 \text{ mm}$$

$$rh_2 = 150 \text{ mm}$$

$$th_1 = 70 \text{ mm}$$

$$th_2 = 90 \text{ mm}$$

Tb and Rb values represent the base length of the wing box at the tip and the root.

$$tb = 290 \text{ mm}$$

$$rb = 290 \text{ mm}$$

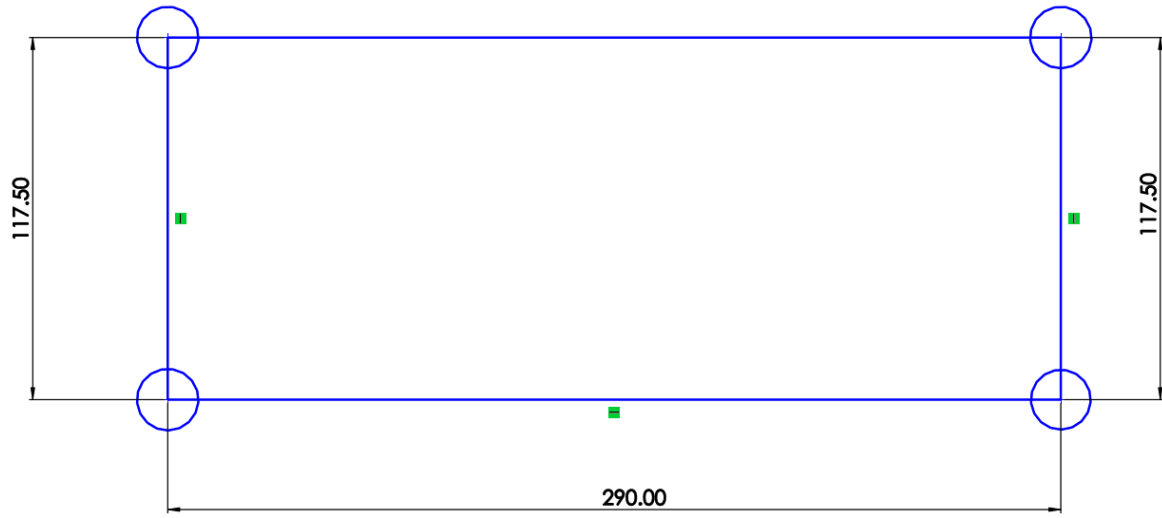


Figure 37: Simplified wing box dimensions

Analysis of Spar Cap Areas and Bending Stresses Under Unidirectional Loading

To calculate the stress values acting upon the cap areas, the neutral axis of the wing box should be located. As the wing box is simplified into a symmetrical rectangle with four lumped masses in four vertex, the neutral axis is located at the middle of the rectangle.

$$\text{Average } Z \text{ bar} = \text{Average height}/2$$

The second moments of inertia for the lumped masses can be calculated as follows: [14]

$$I_x = \sum_{i=1}^4 A_i z_i^2$$

As mentioned before, an assumption was made that both spar web sections and cap areas contribute to the bending of the whole wing. Therefore, the total second moment of inertia of both cap areas and web areas should be considered. The thickness of the web was considered as 5mm denoted as t_{spar} .

$$\text{Average } I_x = \sum_{i=1}^4 A_i z_i^2 + 2 \times t_{spar} \times \frac{\text{Average Height}^3}{12}$$

The MATLAB code was developed in such a way that, we can use any value for the cap areas and distances between them. The cap areas were considered as rectangular, where specific values for each rectangular can be given as input values namely h_{01} , b_{01} , h_{02} , b_{02} and so on. It was assumed that the cap areas and the distances between them (Both base and height values of the wing box) vary linearly along the span of the spar. This linear variation allows for the derivation of a general equation that describes the cap area sizes and their corresponding distances at any given location along the span.

The following equation can be used to determine the dimensions of the wing box at any location along the span of the wing.

$$Base = (tb - rb) \times \frac{2}{Span} \times y + rb$$

$$Height_1 = (th_1 - rh_1) \times \frac{2}{Span} \times y + rh_1$$

$$Height_2 = (th_2 - rh_2) \times \frac{2}{Span} \times y + rh_2$$

Similarly, general equations for the cap sizes at any span location are as follows,

$$h_1 = (h_{03} - h_{01}) \times \frac{2}{Span} \times y + h_{01}$$

$$h_2 = (h_{04} - h_{02}) \times \frac{2}{Span} \times y + h_{02}$$

$$b_1 = (b_{03} - b_{01}) \times \frac{2}{Span} \times y + b_{01}$$

$$b_2 = (b_{04} - b_{02}) \times \frac{2}{Span} \times y + b_{02}$$

This can be used to determine the moment of inertia of any cross section along the span of the wing so that the I_x will be a function of y (Spanwise length from the wing root).

Unidirectional Loading Considerations

For the bending analysis of the wing, unidirectional loading was considered, encompassing both lift and weight distributions which acts on the Z axis. The total drag acting on the aircraft when cruising is 160.2698 N which is significantly lower compared to the resultant force of lift and weight distribution. Therefore, the bending moment due to the drag was neglected.

The bending stresses in the spar caps were calculated using the bending stress equation for unidirectional loading.

$$\sigma = \frac{\text{Bending Moment} \times \frac{\text{Average Height}}{2}}{\text{Average } I_x}$$

Substituting values for above equation, stress values at each spar cap can be obtained.

$$\sigma_{yy_1} = - \left(M_x \times \frac{\frac{\text{Average height}}{2}}{\text{Average } I_x} \right)$$

$$\sigma_{yy_2} = - \left(M_x \times \frac{\frac{\text{Average height}}{2}}{\text{Average } I_x} \right)$$

$$\sigma_{yy_3} = \left(M_x \times \frac{\frac{\text{Average height}}{2}}{\text{Average } I_x} \right)$$

$$\sigma_{yy_4} = \left(M_x \times \frac{\frac{\text{Average height}}{2}}{\text{Average } I_x} \right)$$

The equation incorporates the second moments of area, distances from the neutral axis and the bending moment along x axis due to the lift and weight distribution, providing a comprehensive assessment of the stress distribution along the spar caps and webs

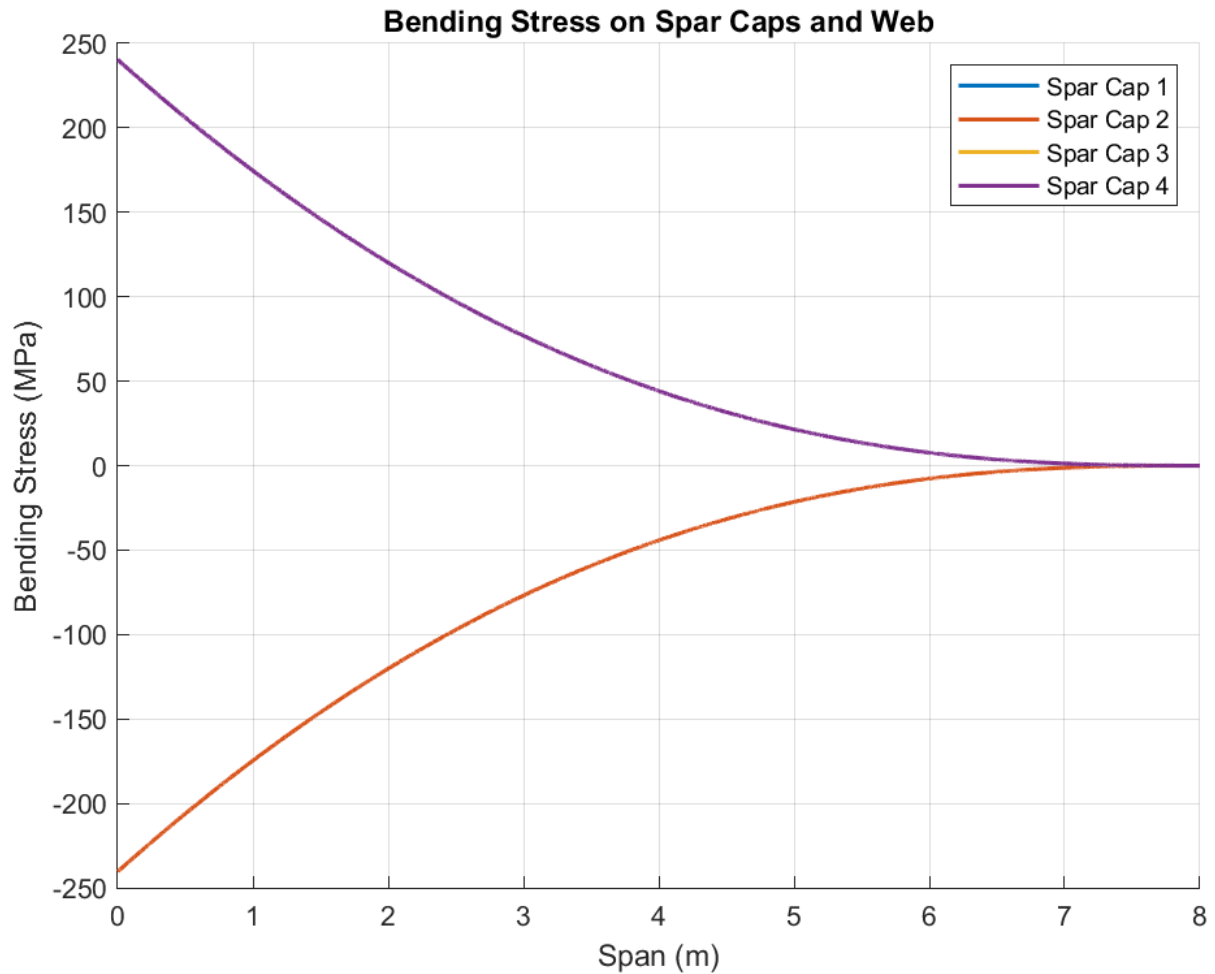


Figure 38: Simplified wing box dimensions

Based on the graph, the bending stress distribution in spar caps 1 and 2 is same and exhibits compressive stress, while spar caps 3 and 4 also have a similar stress distribution but experience tensile stress. The maximum values of both tensile and compressive stress are observed to be 240.126 MPa.

Root Stress Values at Spar Caps

- Spar Cap 1: -240.126 MPa (Compressive)
- Spar Cap 2: -240.126 MPa (Compressive)
- Spar Cap 3: 240.126 MPa (Tensile)
- Spar Cap 4: 240.126 MPa (Tensile)

Midspan Stress Values at Spar Caps

- Spar Cap 1: -44.2038MPa (Compressive)
- Spar Cap 2: -44.2038MPa (Compressive)
- Spar Cap 3: 44.2038MPa (Tensile)
- Spar Cap 4: 44.2038MPa (Tensile)

2.9 Material Selection for Spar

Given the requirement for a safety factor of 1.5, the selection of a suitable material for the spar is critical to ensure structural integrity and optimal performance. An iterative process for selecting the best material was utilized. Initially, data about mechanical properties such as yield strength, Young's modulus, shear strength and density of various commercially available Aluminium alloys were gathered and listed them with their specific alloy grade and temper.[15]

Table 02: Properties of Aluminium alloys

Name	Temper	Young's Modulus (GPa)	Shear Strength (MPa)	Yield Strength (MPa)	Density (g/cm ³)
354	T61	73.1	199	255	2.71
355	T6	70.3	235	185	2.71
355	T51	70.3	150	124	2.71
356	T6	72.4	205	152	2.68
356	T7	72.4	170	165	2.68
1100	H12	68.9	68.9	103	2.71
1100	H14	68.9	75.8	117	2.71
1100	H16	68.9	82.7	138	2.71
1100	H18	68.9	89.6	152	2.71
2014	T4	72.4	262	290	2.8
2014	T6	73.1	290	414	2.8
2017	T4	72.4	262	276	2.79
2024	T3	73.1	283	345	2.78
2024	T4	73.1	283	324	2.78
2024	T81	72.4	295	450	2.78
2025	T6	71.7	241	255	2.81
2124	T851	72	295	441	2.78
2219	T62	73.1	255	290	2.84
2618	T61	74.5	262	372	2.76
3003	H12	68.9	82.7	124	2.73
3003	H14	68.9	96.5	145	2.73
3003	H16	68.9	103	172	2.73
3003	H18	68.9	110	186	2.73
4043	H14	70	106	165	2.69
4043	H16	70	120	180	2.69
5052	H32	70.3	138	193	2.68
5052	H34	70.3	145	214	2.68
5052	H36	70.3	159	241	2.68
5052	H38	70.3	165	255	2.68
5056	H18	71	234	407	2.64

5056	H38	71	221	345	2.64
5082	H19	70	240	370	2.65
5082	H34	70.3	210	280	2.66
5652	H38	70.3	165	255	2.67
5757	H26	70.3	170	245	2.67
6061	T4	68.9	165	145	2.7
6061	T6	68.9	207	276	2.7
6063	T4	68.9	110	89.6	2.7
6063	T5	68.9	117	145	2.7
6063	T6	68.9	152	214	2.7
6063	T83	68.9	152	241	2.7
6070	T4	69	205	170	2.71
6070	T6	68.9	234	352	2.71
6151	T6	69	220	295	2.71
6262	T9	68.9	241	379	2.72
7050	T74	70	283	427	2.83
7050	T7451	71.7	303	469	2.83
7050	T7651	71.7	324	490	2.83
7075	T6	71.7	331	503	2.81
7075	T73	72	300	435	2.81
7475	T61	70.3	330	490	2.81
7475	T7651	71.7	310	462	2.81

The main objective of choosing the best alloy for spar is to minimize the deflection while keeping the stress values below the yield strength.

The deflection of the wing spar is determined by considering only the lift and weight distributions, as the impact of drag distribution is minimal and thus can be neglected. The deflection of the beam is obtained by integrating the bending moment equation twice. The constants C_1 and C_2 are determined by applying the following boundary conditions. [13]

- The slope of the beam $\left(\frac{dv}{dx}\right)$ at the cantilevered end is 0.
- The deflection of the beam (v) at the cantilevered end is 0.

$$EI \frac{d^2y}{dx^2} = M(x)$$

$$EI \frac{dv}{dx} = \int M(x)dx + C_1$$

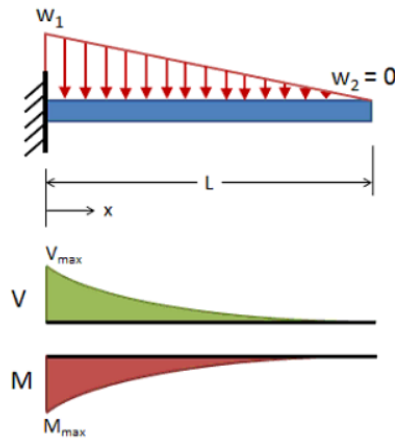
$$EIv = \int [M(x)dx]dx + C_1x + C_2$$

According to design criteria, the maximum allowable deflection of the wing is 5% of the wingspan, which is equivalent to 0.8m.

To reduce the complexity of calculations, the resultant of the lift and weight distribution was approximated to a triangularly distributed load. The W_1 value was determined by equating the area under the current Forces z distribution to the new area under the approximated triangular distribution. [16]

$$W_1 = \frac{4 \times \int_0^{\frac{Span}{2}} (Forces_z) dy}{Span}$$

Cantilever,
Triangular
Distributed Load



Deflection:

$$\delta = -\frac{w_1 x^2}{120LEI} (10L^3 - 10L^2x + 5Lx^2 - x^3)$$

$$\delta_{max} = \frac{w_1 L^4}{30EI} \quad @ x = L$$

Slope:

$$\theta = -\frac{w_1 x}{24LEI} (4L^3 - 6L^2x + 4Lx^2 - x^3)$$

$$\theta_{max} = \frac{w_1 L^3}{24EI} \quad @ x = L$$

Shear:

$$V_{max} = +w_1 L / 2 \quad @ x = 0$$

Moment:

$$M_{max} = -w_1 L^2 / 6 \quad @ x = 0$$

Iteratively maximum deflection values and approximate weight of solid spar without lightening holes were calculated for each alloy. Stress check was done by comparing the maximum stress in the spar to the yield strength of each material. The results were stored in an array.

Maximum deflection value:

$$Deflection_{Max} = \frac{w_1 \times \left(\frac{S}{2}\right)^4}{30 \times Youngs Modulus_{Current} \times Average I_x}$$

Mass was approximated as,

$$Mass_{spar} = (4 \times \text{Average Area} + 2 \times t_{spar} \times \text{Average Height}) \times \frac{\text{Span}}{2} \times \text{Density}$$

Results for each Alloy:

Alloy: 354, Mass: 42.82 kg, Max Deflection: 0.5680 m, Stress Check: Under Yield Strength
 Alloy: 355, Mass: 42.82 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength
 Alloy: 355, Mass: 42.82 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength
 Alloy: 356, Mass: 42.34 kg, Max Deflection: 0.5735 m, Stress Check: Under Yield Strength
 Alloy: 356, Mass: 42.34 kg, Max Deflection: 0.5735 m, Stress Check: Under Yield Strength
 Alloy: 1100, Mass: 42.82 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 1100, Mass: 42.82 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 1100, Mass: 42.82 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 1100, Mass: 42.82 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 2014, Mass: 44.24 kg, Max Deflection: 0.5735 m, Stress Check: Under Yield Strength
 Alloy: 2014, Mass: 44.24 kg, Max Deflection: 0.5680 m, Stress Check: Under Yield Strength
 Alloy: 2017, Mass: 44.08 kg, Max Deflection: 0.5735 m, Stress Check: Under Yield Strength
 Alloy: 2024, Mass: 43.92 kg, Max Deflection: 0.5680 m, Stress Check: Under Yield Strength
 Alloy: 2024, Mass: 43.92 kg, Max Deflection: 0.5680 m, Stress Check: Under Yield Strength
 Alloy: 2024, Mass: 43.92 kg, Max Deflection: 0.5735 m, Stress Check: Under Yield Strength
 Alloy: 2025, Mass: 44.40 kg, Max Deflection: 0.5791 m, Stress Check: Under Yield Strength
 Alloy: 2124, Mass: 43.92 kg, Max Deflection: 0.5767 m, Stress Check: Under Yield Strength
 Alloy: 2219, Mass: 44.87 kg, Max Deflection: 0.5680 m, Stress Check: Under Yield Strength
 Alloy: 2618, Mass: 43.61 kg, Max Deflection: 0.5573 m, Stress Check: Under Yield Strength
 Alloy: 3003, Mass: 43.13 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 3003, Mass: 43.13 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 3003, Mass: 43.13 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 3003, Mass: 43.13 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength
 Alloy: 4043, Mass: 42.50 kg, Max Deflection: 0.5931 m, Stress Check: Under Yield Strength
 Alloy: 4043, Mass: 42.50 kg, Max Deflection: 0.5931 m, Stress Check: Under Yield Strength

Alloy: 5052, Mass: 42.34 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5052, Mass: 42.34 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5052, Mass: 42.34 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5052, Mass: 42.34 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5056, Mass: 41.71 kg, Max Deflection: 0.5848 m, Stress Check: Under Yield Strength

Alloy: 5056, Mass: 41.71 kg, Max Deflection: 0.5848 m, Stress Check: Under Yield Strength

Alloy: 5082, Mass: 41.87 kg, Max Deflection: 0.5931 m, Stress Check: Under Yield Strength

Alloy: 5082, Mass: 42.03 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5652, Mass: 42.19 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 5757, Mass: 42.19 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 6061, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6061, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6063, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6063, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6063, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6063, Mass: 42.66 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6070, Mass: 42.82 kg, Max Deflection: 0.6017 m, Stress Check: Under Yield Strength

Alloy: 6070, Mass: 42.82 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 6151, Mass: 42.82 kg, Max Deflection: 0.6017 m, Stress Check: Under Yield Strength

Alloy: 6262, Mass: 42.98 kg, Max Deflection: 0.6026 m, Stress Check: Under Yield Strength

Alloy: 7050, Mass: 44.71 kg, Max Deflection: 0.5931 m, Stress Check: Under Yield Strength

Alloy: 7050, Mass: 44.71 kg, Max Deflection: 0.5791 m, Stress Check: Under Yield Strength

Alloy: 7050, Mass: 44.71 kg, Max Deflection: 0.5791 m, Stress Check: Under Yield Strength

Alloy: 7075, Mass: 44.40 kg, Max Deflection: 0.5791 m, Stress Check: Under Yield Strength

Alloy: 7075, Mass: 44.40 kg, Max Deflection: 0.5767 m, Stress Check: Under Yield Strength

Alloy: 7475, Mass: 44.40 kg, Max Deflection: 0.5906 m, Stress Check: Under Yield Strength

Alloy: 7475, Mass: 44.40 kg, Max Deflection: 0.5791 m, Stress Check: Under Yield Strength

By analysing the above results, **Aluminium 5056-H18** offers a good balance of strength and ductility, with a yield strength of approximately 407 MPa and Youngs modulus of 71 GPa.

This alloy has a greater resistance to fatigue which is crucial in a wing spar. This meets the required strength threshold when considering the safety factor of 1.5, ensuring that the material can withstand the stresses without excessive deformation or failure. With a density of 2.64 g/cm^3 , Aluminium 5056-H18 is lightweight, which is essential for aerospace applications. It can be easily formed into complex shapes and is readily weldable, allowing for versatile design and manufacturing options. It also exhibits corrosion resistance properties.

Approximated weight of the spars without lightening holes: 41.71 kg

Maximum wing deflection: 0.5848 m

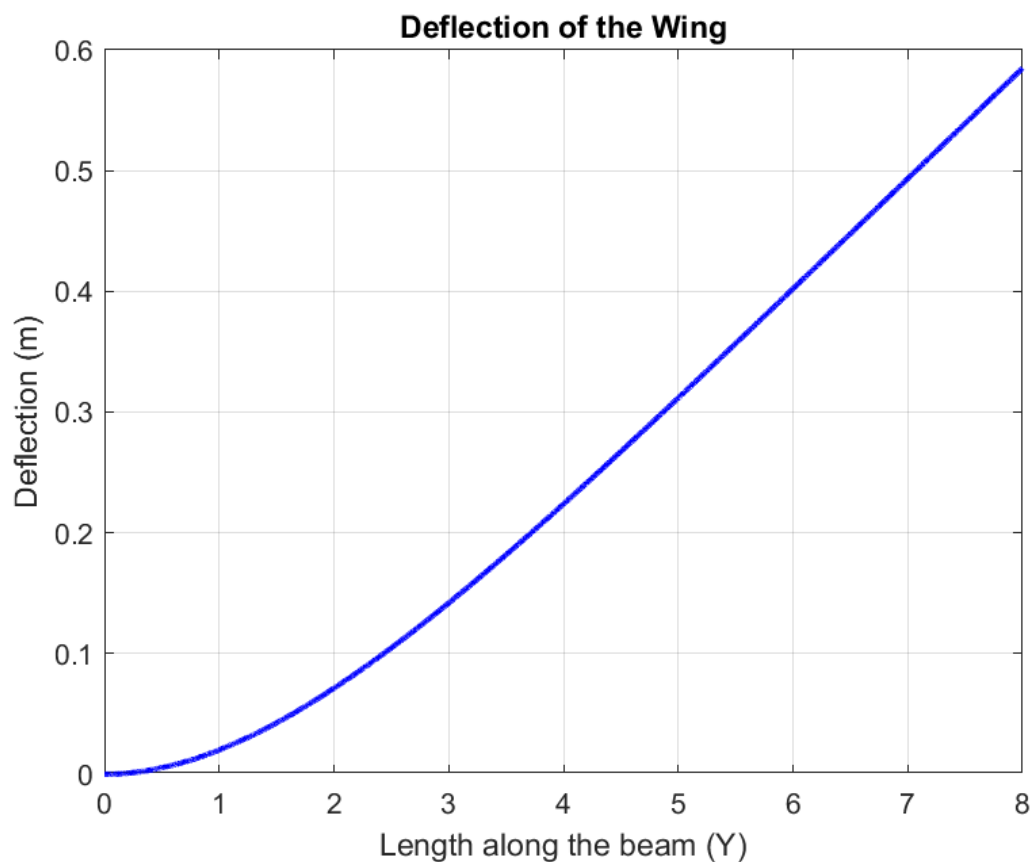


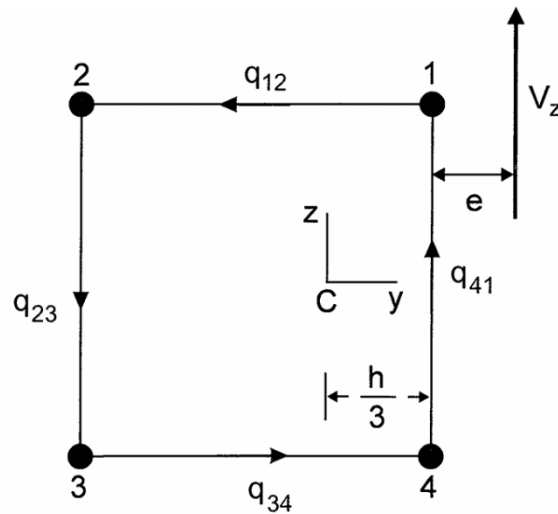
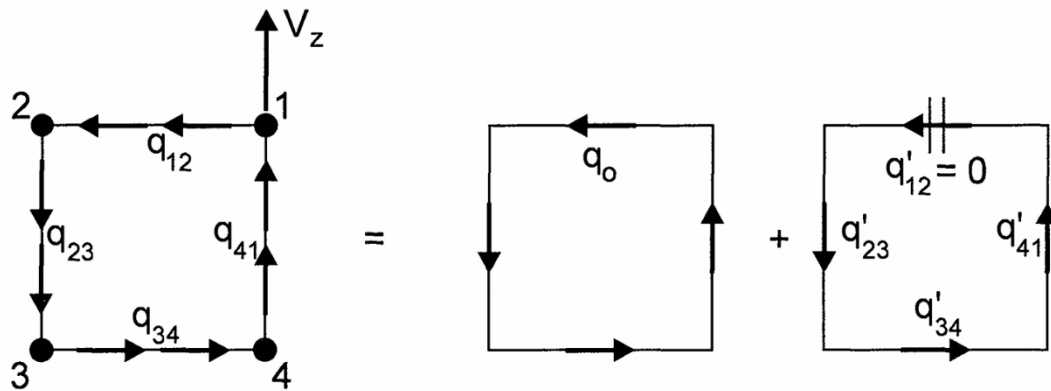
Figure 39: Deflection of the Wing

2.10 Shear Flows

The wing structure is modelled as an enclosed thin walled single cell cross section to calculate the shear flows. Only the flexural shear resulting from lift and weight is considered, as the drag forces are relatively negligible. It is assumed that both the torsional and flexural shear stresses are primarily borne by the web and skin sections of the wing. [17]

To simplify the calculations, the resultant shear force from lift and weight is assumed to pass collinearly with the rear web structure along the wingspan. To idealize the wing's single cell structure, the average height of the rectangular cross-section was considered (As mentioned in chapter 2.7) by averaging the heights of front and rear web sections of wing tip and wing root.

Using the superposition theorem, we can decompose the shear flows. [14]



V_z is the shear force along Z direction. Taking moments about spar cap 1,

$$V_z \times e + M_Y = (2 \times A_{Enclosed} \times q_0) + (q'_{23} \times \text{Average Height} \times \text{Average Base})$$

$$q_0 = \frac{V_z \times e + M_Y - q'_{23} \times \text{Average Height} \times \text{Average Base}}{2 \times A_{Enclosed}}$$

Since $e = 0$ (Assuming that shear force goes through the spar cap 1)

$$q_0 = \frac{M_Y - q'_{23} \times \text{Average Height} \times \text{Average Base}}{2 \times \text{Average Height} \times \text{Average Base}}$$

Here, M_Y represents the moment along Y axis which is a function of y. By utilizing that function, moment at any given place along spanwise y direction can be determined.

$$M_Y = -K_{\text{Moment}} \times \text{Wing Chord}^2$$

$$q'_{12} = 0$$

$$q'_{23} = -V_Z \times \text{Average Area}_2 \times \frac{\text{Average Height}}{2 \times \text{Average } I_x}$$

$$q'_{34} = 0$$

$$q'_{41} = -V_Z \times \text{Average Area}_4 \times \frac{\text{Average Height}}{2 \times \text{Average } I_x}$$

The total shear flows are,

$$q_{12} = q_0 + q'_{12}$$

$$q_{23} = q_0 + q'_{23}$$

$$q_{34} = q_0 + q'_{34}$$

$$q_{41} = q_0 + q'_{41}$$

$$\text{Thickness of the upper skin} = t_{12} = 1.5 \text{ mm}$$

$$\text{Thickness of the front web} = t_{23} = 5 \text{ mm}$$

$$\text{Thickness of the lower skin} = t_{34} = 1.5 \text{ mm}$$

$$\text{Thickness of the rear web} = t_{41} = 5 \text{ mm}$$

All total shear flow equations are function of y. So, shear flows within the thin walled section along the span can be plotted as shown below. To convert shear flows into shear stress, it should be divided by the thickness of the wall.

$$\text{Upper skin shear stress} = \frac{q_{12}}{t_{12}}$$

$$\text{Front web shear stress} = \frac{q_{23}}{t_{23}}$$

$$\text{Lower skin shear stress} = \frac{q_{34}}{t_{34}}$$

$$\text{Rear web shear stress} = \frac{q_{41}}{t_{41}}$$

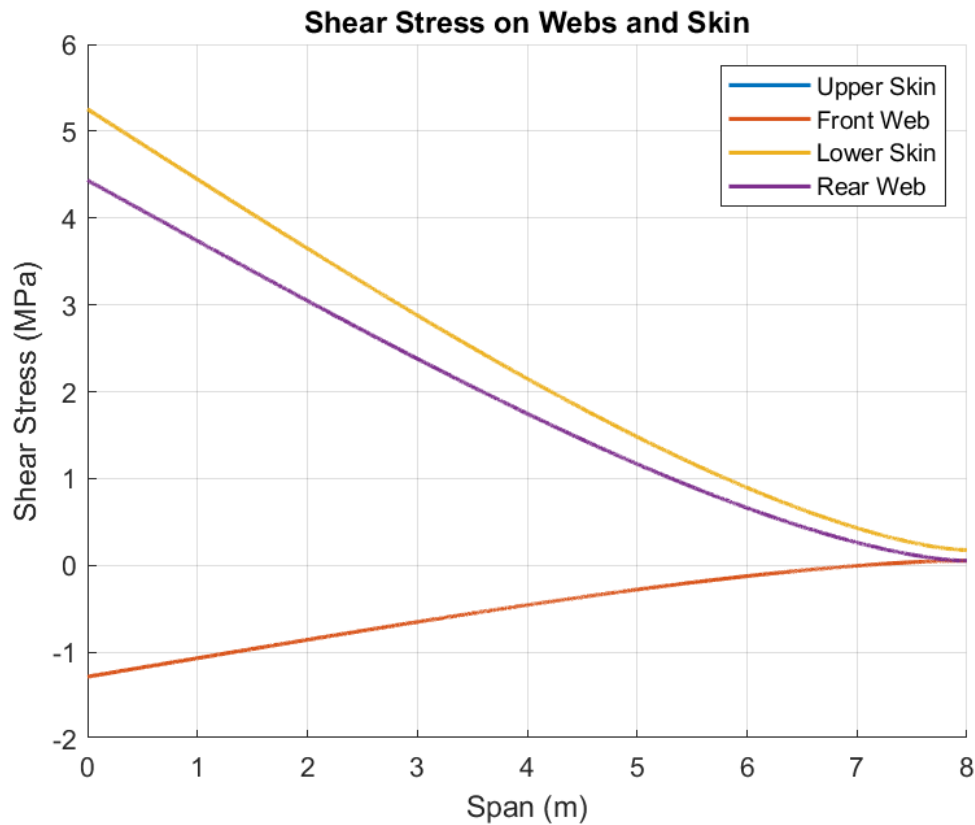


Figure 40: Shear Stress on Webs and Skin

Root Shear Stress Values

- Upper Skin: 5.25 MPa
- Front Web: -1.28 MPa
- Lower Skin: 5.25 MPa
- Rear Web: 4.43 MPa

Midspan Shear Stress Values

- Upper Skin: 2.15 MPa
- Front Web: -0.45 MPa
- Lower Skin: 2.15 MPa
- Rear Web: 1.75MPa

The shear flows in sections 12 and 34 (Upper and Lower Skin) are identical along the span, resulting in overlapping curves on the graph. The graph indicates that the maximum shear stress occurs in the upper and lower skin, reaching approximately 8 MPa. Considering a safety factor

of 1.5, an appropriate material should be selected for constructing the skin section to ensure it meets the required strength criteria.

Same iterative process was used for choosing a material for the skin section. Stress check was done by comparing the maximum shear stress in the above graph with the shear strength of each alloy. The mass of the skin surface was approximated as,

Mass was approximated as,

$$Mass_{Skin} = 2 \times t_{12} \times Chord\ length \times (1 + Best\ Lambda) \times \frac{Span}{4} \times Density$$

Results for each alloy:

Alloy: 354, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 355, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 355, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 356, Mass: 39.58 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 356, Mass: 39.58 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 1100, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 1100, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 1100, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 1100, Mass: 40.02 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2014, Mass: 41.35 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2014, Mass: 41.35 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2017, Mass: 41.21 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2024, Mass: 41.06 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2024, Mass: 41.06 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2024, Mass: 41.06 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2025, Mass: 41.50 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2124, Mass: 41.06 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2219, Mass: 41.94 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 2618, Mass: 40.76 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

55

Alloy: 7475, Mass: 41.50 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Alloy: 7475, Mass: 41.50 kg, Max Shear: 5.2534 Mpa, Stress Check: Under Shear Strength Limit

Having a low density and a higher Young's modulus were considered as main factors when selecting a suitable material. Therefore, **Aluminium Alloy 2024-T3** was selected. 2024-T3 offers excellent mechanical properties with high yield strength and Young's modulus of 73.1 GPa, making it suitable for structural applications like aircraft skins where strength is critical. With a typical shear strength of approximately 283 MPa, Aluminum 2024-T3 easily exceeds the 5.25 MPa requirement, providing a robust safety margin even when considering a safety factor. It can be easily formed, machined, and treated, making it a versatile option for complex aircraft wing geometries.

Approximated weight of the skin: 41.06 kg

2.11 Buckling of Skin Plates

In aircraft wing design, maintaining structural integrity under various load conditions is essential to ensure safety and performance. One critical aspect is preventing the thin wing skin from buckling under compressive and shear loads encountered during flight. Wing skins are subject to complex aerodynamic forces that induce compression, tension, and shear stresses

To counteract these stresses, ribs and stringers are incorporated as internal reinforcements within the wing. Ribs provide shape and structural support across the chord of the wing, while stringers run longitudinally along the span to add stiffness and increase resistance to buckling. Determining the optimal number and placement of these reinforcements is essential for weight efficiency and structural stability.

Using plate buckling equations, critical stress levels at which buckling occurs can be determined. It can be utilized to determine the minimum number of ribs and stringers required to prevent skin buckling under operational loads.

In a wing structure, the primary loading comes from bending forces caused by wing deflection under lift and the wing's own weight. This upward deflection creates compressive forces on the upper surface of the skin. As a result, for buckling analysis, only compression buckling was considered, while neglecting the buckling due to shear loads. [18]

Compression buckling of flat rectangular plates

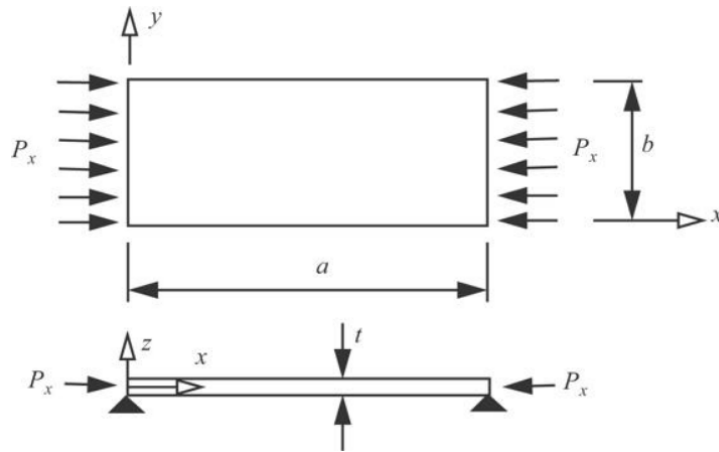


Figure 41: Forces applied to the rectangular plate

$$\sigma_{cr} = k_c \pi^2 \frac{E}{12(1 - \nu^2)} \left(\frac{t}{b}\right)^2$$

The above equation was used to calculate the length b for the skin plate to prevent buckling. This length b helps to determine the required number of stringers for the wing. Here, σ_{cr} represents the compressive (or crippling) stress at which the plate begins to buckle. The goal is to select b value so that the stresses acting on the skin plate remain below this crippling stress, ensuring stability. An assumption was made that the compressive stress acted upon the skin is uniformly distributed along the opposite longitudinal edges

First, the maximum stress experienced by the skin plate needs to be calculated. In an aircraft wing, the skin plates are secured to the stringers and ribs with rivets. To ensure that the skin remains attached to the wing, the deflection of the skin should be equal to the deflection of the wing spar. Knowing the deflection of the main spars allows to determine the stress developed in the skin.

When calculating the second moment of area along the x axis, the skin was treated as two long rectangular beams considering the thickness of the skin as height and the mean aerodynamic chord length (MAC) as the width of the cross section. Both upper and lower skin are at $\frac{\text{Average Height}}{2}$ distance from the neutral axis of bending. Parallel axis theorem was used to calculate the second moment of area

$$Skin_{Ix} = 2 \times \left[\frac{MAC \times (t_{12})^2}{12} + \left(\frac{Average\ Height}{2} \right)^2 \times t_{12} \times MAC \right]$$

Here, t_{12} represents the thickness of the skin

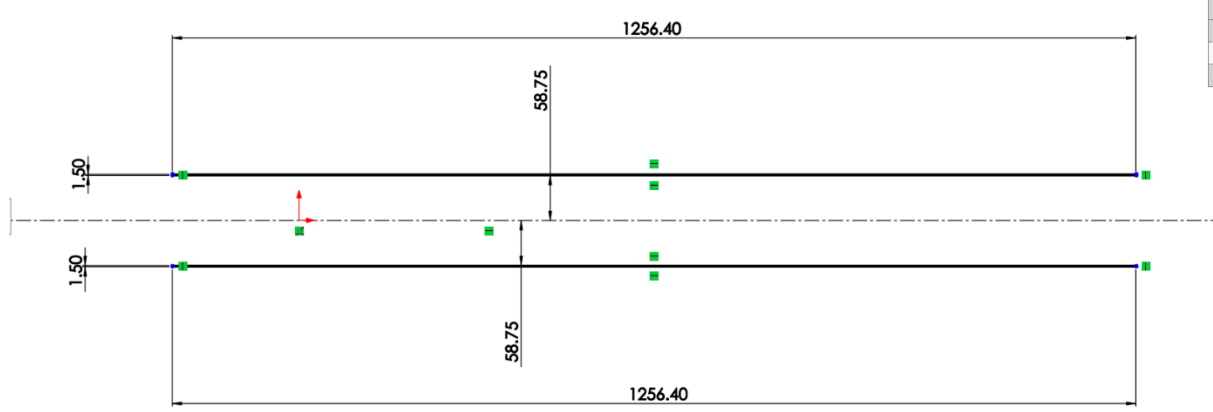


Figure 42: 2D cross section of the wing modeled for calculations

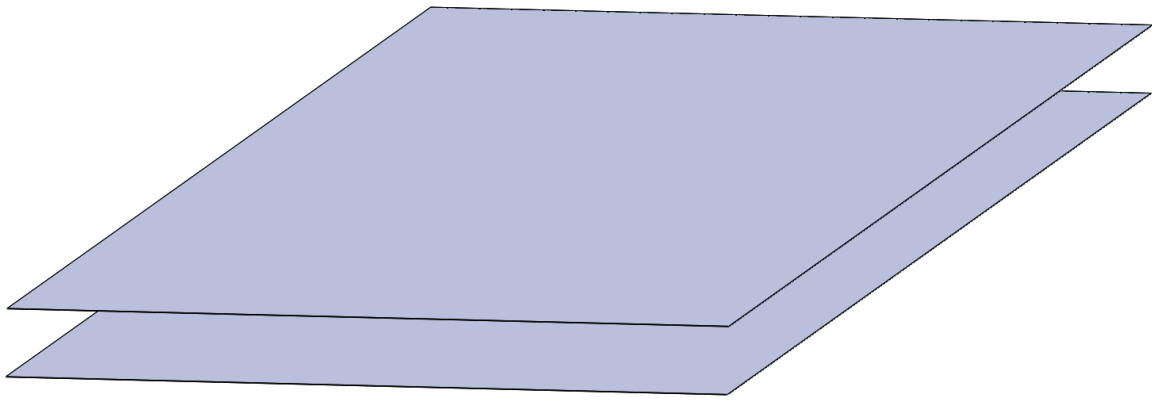


Figure 43: 3D view of modeled skin for calculations

K_C is the non dimensional compression buckling coefficient. It depends on the boundary conditions of both loaded edges and free edges of the rectangular plate. Since the skin plates are rivetted to the stringers and ribs, both loaded edges and free edges were considered as clamped. The following chart shows how the compression buckling coefficient varies according to the boundary conditions.

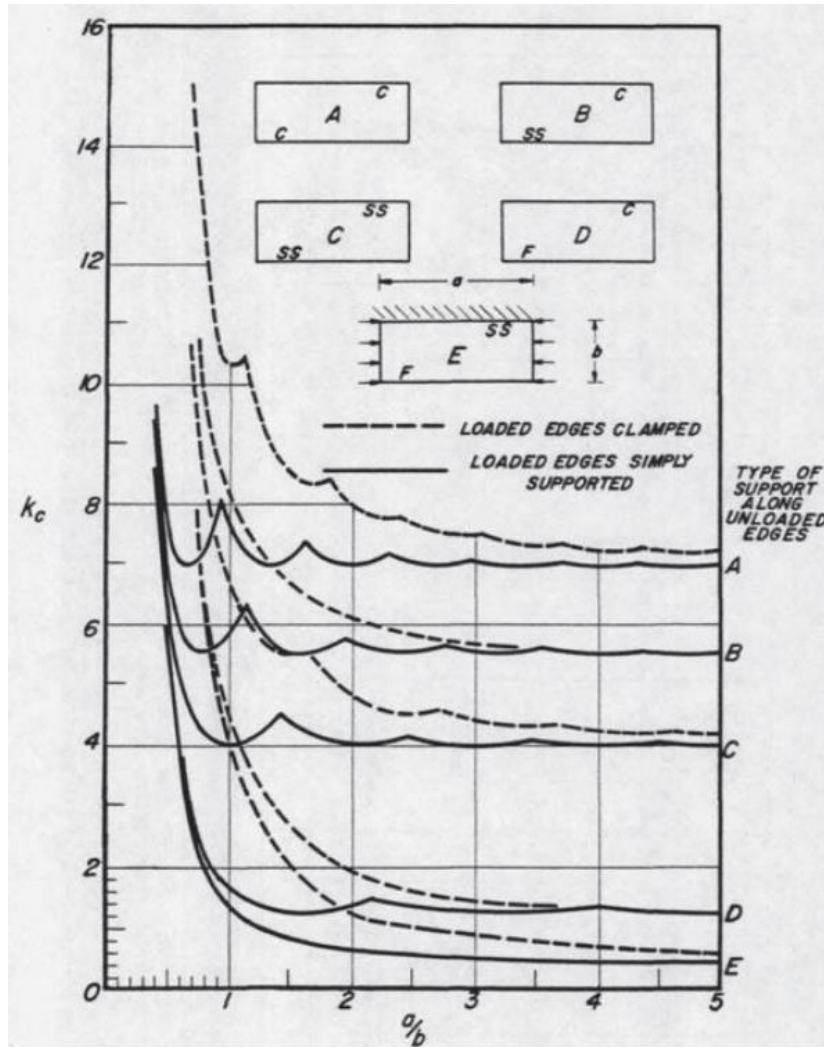


Figure 44: Compression buckling coefficients for flat rectangular plates

According to the above chart, for a selected boundary condition, the compression buckling coefficient varies with the aspect ratio of the rectangular plate.

For the buckling calculations, a/b ratio was assumed to be 3.5, which gives the K_c value of approximately 7.5. Poisson ratio was taken as 0.33 for the alloy.

Considering the deflection of the spar, the bending moment at any location can be determined using the following formula. [19]

$$M_x \text{ Skin} = \frac{W_1 \times \left(\frac{\text{Span}}{2} - y \right)^3}{6 \times \frac{\text{Span}}{2}}$$

Using the moment at any location along the span, we can determine the compressive stress built up on the skin using the Euler Bernoulli bending formula.

$$\sigma_{cr} = \frac{M_x \times z}{I_x}$$

$$\sigma_{cr} = M_x \text{ Skin} \times \frac{\text{Average height}}{2 \times \text{Skin}_{I_x}}$$

Substituting this into the compression buckling equation, b value can be found.

$$b \text{ Length} = \sqrt{\frac{(K_c \times \pi^2 \times E_{skin} \times (t_{12})^2)}{\sigma_{cr} \times 12(1 - \nu^2)}}$$

Since $M_x \text{ Skin}$ is a function of y , it varies along the spanwise direction. This approach enables the stringer count to be determined at any spanwise location along the wing. The maximum bending moment acts on the cantilevered edge of the wing, which refers to the root of the wing. When the moment increases, the stress increases.

As a result of that, the b length at the root of the wing is small. Small b value results in higher number of stringers to be used at the root of the wing. On behalf of that, the chord length also varies along the span. By taking both of these considerations into account, an equation for the number of stringers can be formed.

$$\text{Number of Stringers} = \frac{\text{Wing Chord}}{b \text{ Length}}$$

Both Wing chord and b length are functions of y . At the root, 12 stringers are needed to prevent the skin plates from buckling. As we go further towards the tip of the wing, this bending moment reduces, causing the stringer count to decrease. By utilizing that effect, variable number of stringers at each part of the wing can be determined rather than adding 12 stringers all throughout the wing. This will be beneficial for reducing the weight of excess stringers. Using all above equations, a graph can be plotted for the stringer count along the spanwise length of the wing.

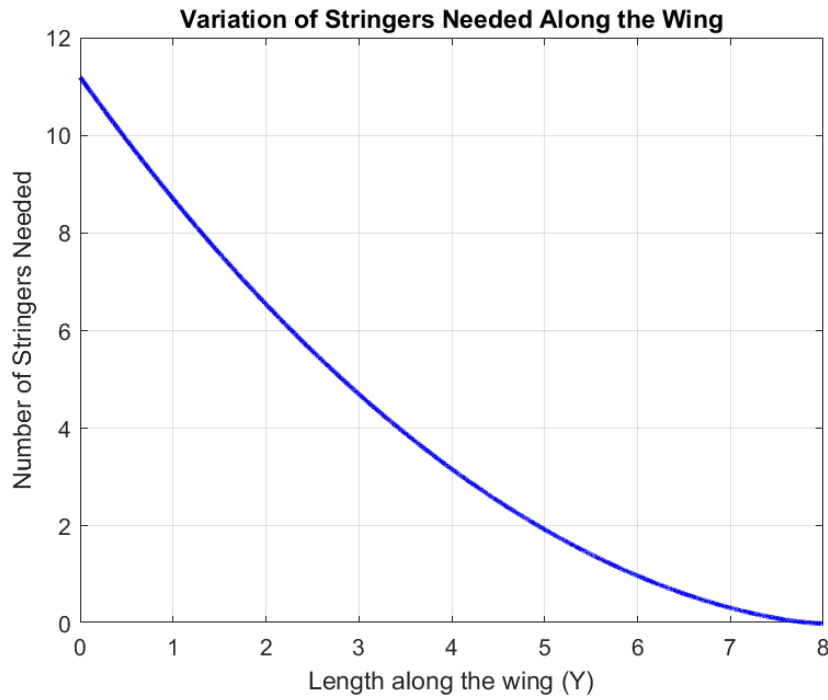


Figure 45: Variation of stringer count along the wing

How ever, the wing structure will be more complex when using variable stringer count for each section of the wing. So, for the simplicity of the design, 12 stringers are considered for the whole wing. This also helps to enhance the FOS of the wing design.

As per the assumptions, the aspect ratio of the rectangular plate was taken as 3.5. By substituting the b value, a value can be calculated which refers to the number of ribs.

$$a \text{ Length} = \frac{a}{b} \text{ Ratio} \times b \text{ length}$$

$$\text{Number of Ribs} = \frac{\frac{\text{Span}}{2}}{a \text{ Length}}$$

Here, **a length** refers to the ribs spacing of the wing which is equal to 0.4812 m. Using that spacing, total of 17 ribs are needed for the wing.

CHAPTER 03

Design for X

3 Design for X

3.1 Design Parameters and Fabricating Procedures

Spars

Spars are fabricated from 5mm Al 5056 H18 alloy sheets, which are cut to the required dimensions using a waterjet cutter. This cutting method offers several advantages, including high precision, minimal material wastage, no thermal distortion, and the ability to cut complex shapes efficiently.

The initial length of the spar is 8 meters. However, this can vary based on available resources, such as the maximum cutting bed size and the capabilities of the bending machine. If necessary, a single spar can be made by joining two 4-meter sheets using a trap joint and riveting them together. Each spar features two bending lines.

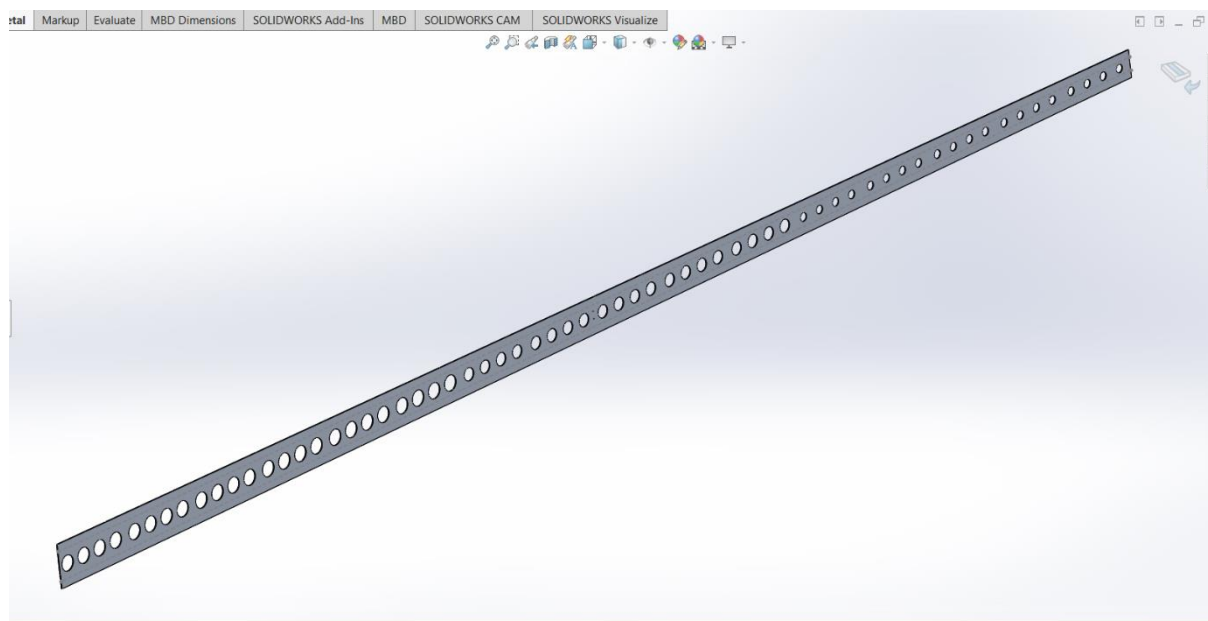


Figure 46: Flatten spar section

After cutting the sheet to the desired shape, including the lightening holes, it is bent at 90 degrees using a compatible bending machine to form the cap areas of the spar. During the bending process, the springback effect must be accounted for. To minimize this effect, the spar is bent slightly beyond 90 degrees to achieve the desired final angle. [20]

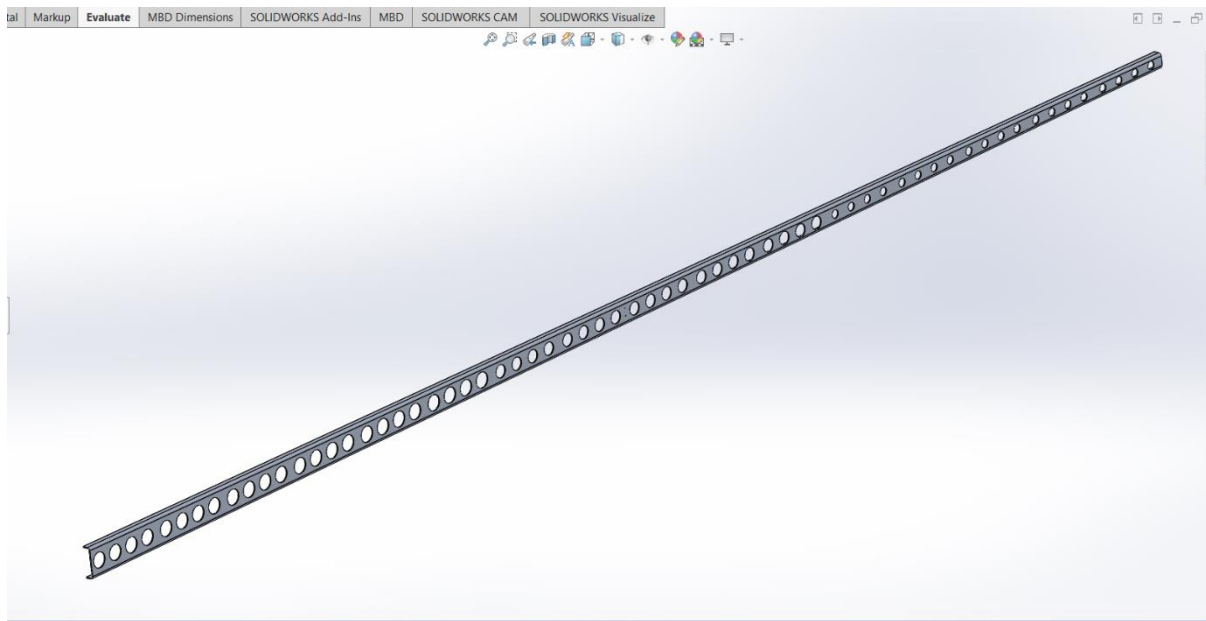


Figure 47: Bent spar section

Wing ribs

The ribs are fabricated using Al 5056 H18 alloy sheets by utilizing CNC milling. It starts with selecting an aluminium billet with required dimensions. The flange of the rib is 20mm, so that the billet thickness should be greater than or equal to 20mm. The developed 3D CAD model with lightening holes, cutouts and flanges should be imported to the CAM software. Having mounted the workpiece in the bed, the process begins with roughing to remove bulk material, followed by finishing passes to ensure high dimensional accuracy and a smooth surface finish. Post machining steps include deburring to eliminate sharp edges, surface treatments like anodizing for enhanced corrosion resistance, and a thorough inspection to ensure dimensional accuracy and structural integrity.

When it comes to the pros and cons of CNC milling, it offers exceptional accuracy, ensuring the parts meet the exact design parameters. It also allows to create intricate shapes, including lightening holes, slots for the stringers and flanges in a single step. It also gives highly consistent results across multiple ribs which is critical for mass production. Since the process is automated, it reduces the labor intensity and human errors, increasing the overall efficiency.

One of the major disadvantages of CNC milling is its high cost. Using CNC machines and related tooling setups for low volume production can be very expensive. Since milling is a

subtractive manufacturing method, it produces waste material. Extremely thin sections such as the 2mm thickness might be prone to deflections or deformations during machining. [21]

The rib sections were specifically designed for CNC milling. Design parameter changes along with modifications should be done if using an another fabricating method. As an example, the same rib can be fabricated using thin metal sheets. But the flanges are created using a hydraulic press. The basic dimensions including the ripping edges, lightening holes should be cut using laser cutter or a waterjet cutter. Then, it can be hydraulic pressed and formed into the final rib design.

Stringers

Stringers cross section was chosen as L section. It can be fabricated using 1mm thickness Al 5056 H18 alloy sheets. The sheets are cut into the required dimensions including the riveting holes using LaserJet cutters. Then using a bending machine, the L shape is formed. The angle of the L cross section varies along the wingspan.

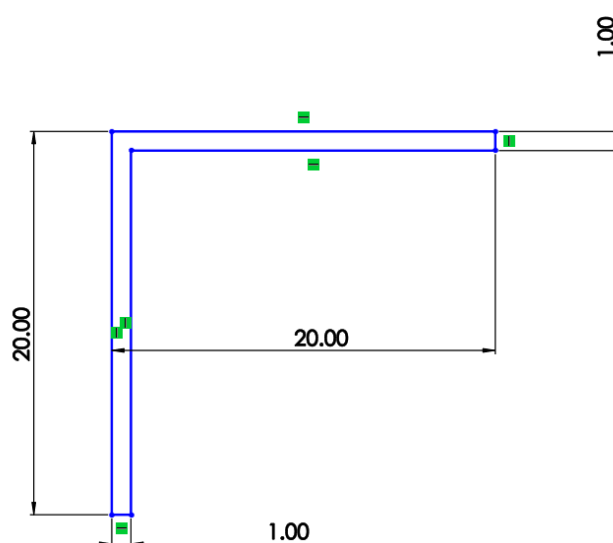


Figure 48: Stringer Cross Section

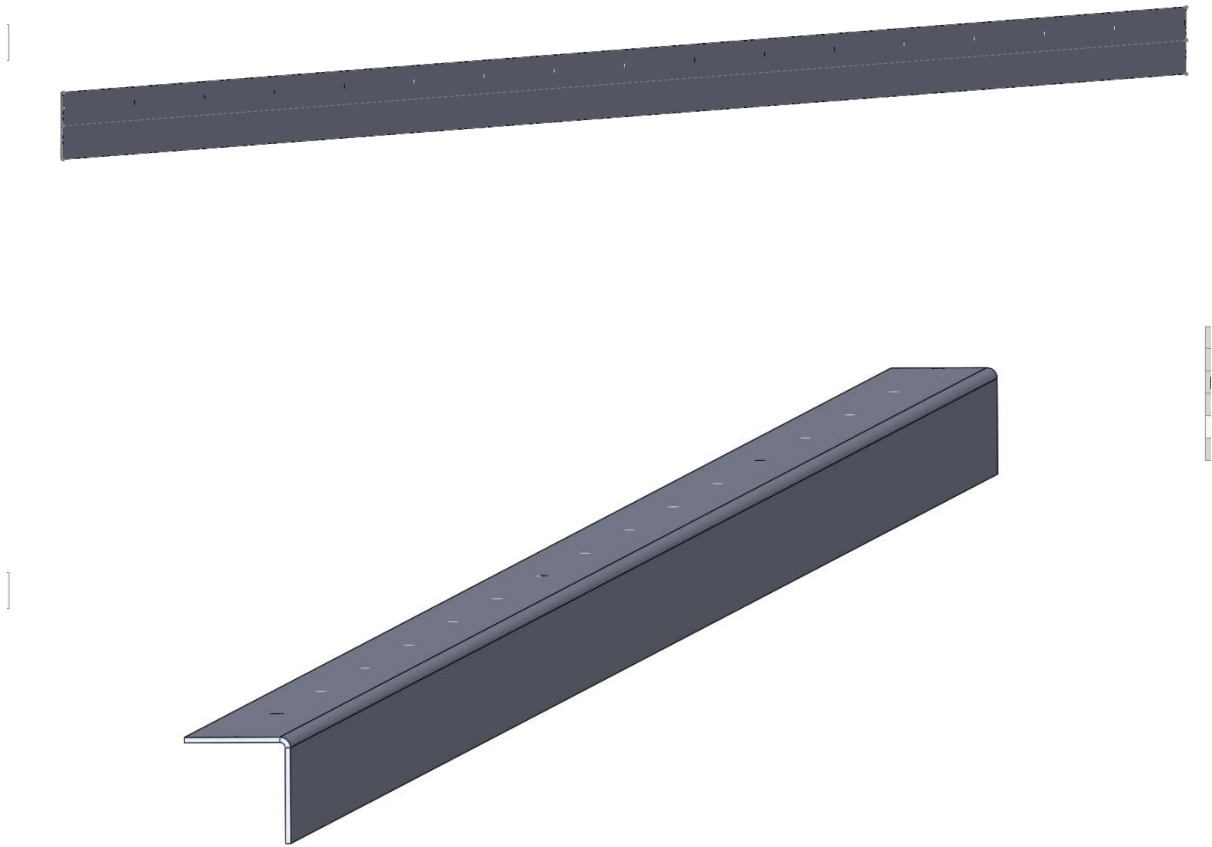


Figure 49: The flattened and bent sheet metal, used for the stringers

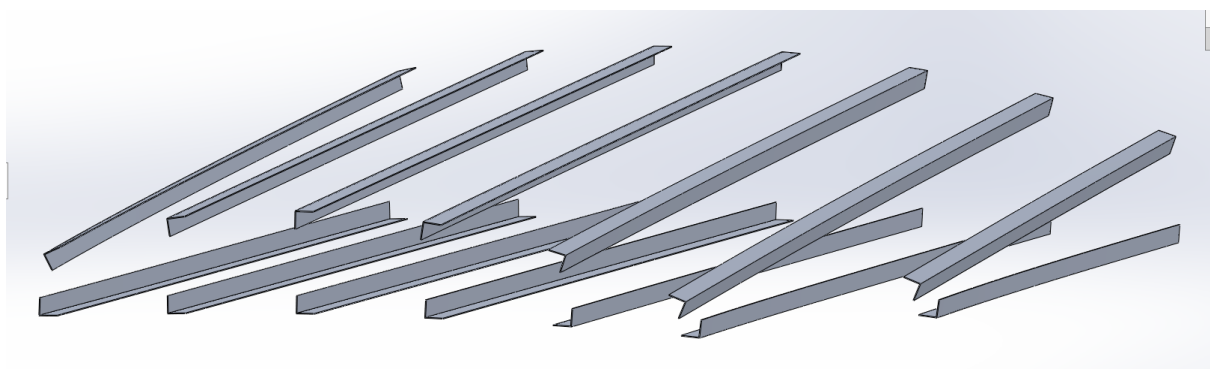


Figure 50: Stringer placement in the wing

Wing Skin

The skin is made up with 1.5mm thickness Aluminium 2024 T3 sheets. The 2d sketch with riveting holes is cut into the sheet using water jet cutter. [22]

ALUMINUM AND ALUMINUM ALLOY SHEET AND PLATE

Standard Specification/ GB/T 3880, ASTM B209, EN 485

Alloy and Temper:

Alloy	Temper
1xxx: 1050, 1050A, 1060, 1100	O, H112, H12, H14, H16, H18, H22, H24, H26
3xxx: 3003, 3004, 3005, 3105	
5xxx: 5005, 5052, 5754, 5083	O, H111, H112, H22, H24, H26, H28, H32, H34, H36, H38
6xxx: 6061, 6082	T4, T451, T6, T651
2xxx: 2014, 2017, 2024	T351, T451
7xxx: 7075	T651

*H116 and H321 for alloy 5083 are provided as per Mill's Standard or by agreement

> AVAILABLE SIZE RANGE

Dimension	Range
Thickness:	0.5 ~ 6.0mm for sheet, 6.0 ~ 120mm for plate
Width:	900 ~ 2,200mm
Length:	2,000 ~ 10,000mm



> STANDARD WIDTH AND LENGTH

1000x2000mm, 1250x2500mm, 1500x3000mm,
1219x2438mm, 1524x3048mm



> SURFACE FINISH

Mill Finish, unless otherwise specified

> SURFACE PROTECTION

Paper interleaved, PE/PVC filming (if specified)



Figure 51: Available Aluminium alloy sheets and plates

3.2 Assembly Process

The assembly process starts with positioning the spars. Two spars are securely held in place using jigs to maintain alignment. Each rib is made up of three separate parts. Since the rivet holes are pre-drilled in the correct locations on the ribs and spars, the front rib section, the front spar, and the middle rib section can be precisely aligned by matching their holes.

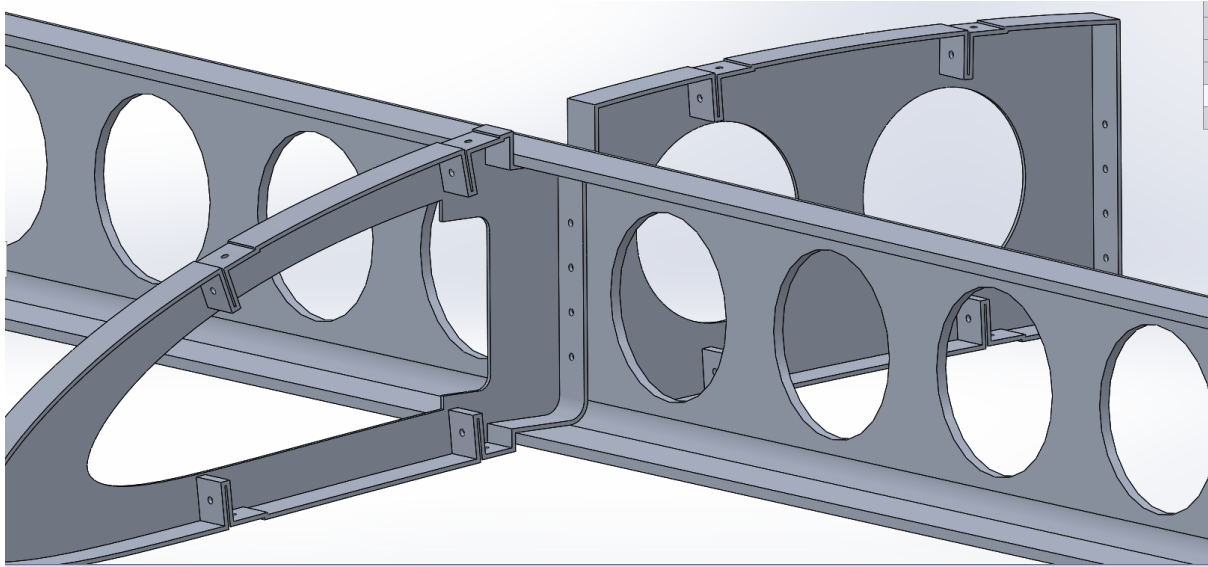


Figure 52: Aligning the front rib and the middle rib part with the front spar

Similarly, the middle rib section, the rear spar, and the rear rib section are aligned in the same way. Once the holes are aligned, rivets are inserted, and the components are fastened using a rivet gun and a bucking bar to ensure a strong and secure connection.

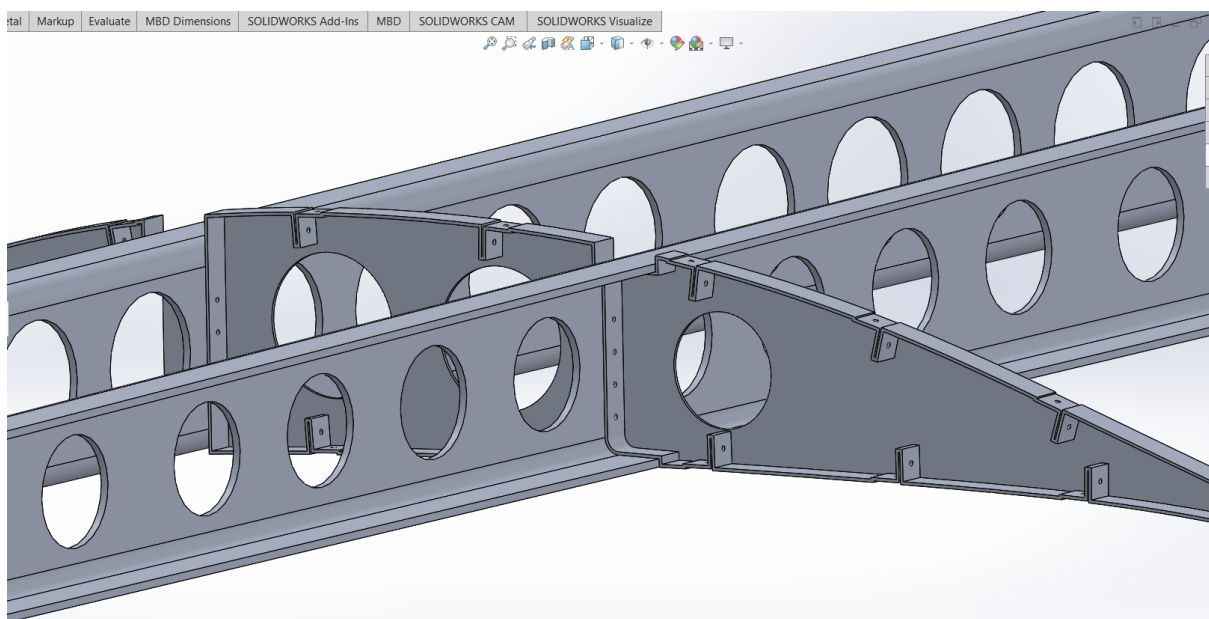


Figure 53: Aligning the middle rib and the rear rib part with the rear spar

Next, the stringers are placed into the precut slots on each rib. It is essential to ensure that the rivet holes in the rib flanges align with the corresponding holes in the stringers. The precut slot features a flange, which can be used to securely fix the stringer to the rib as shown in the figure below.

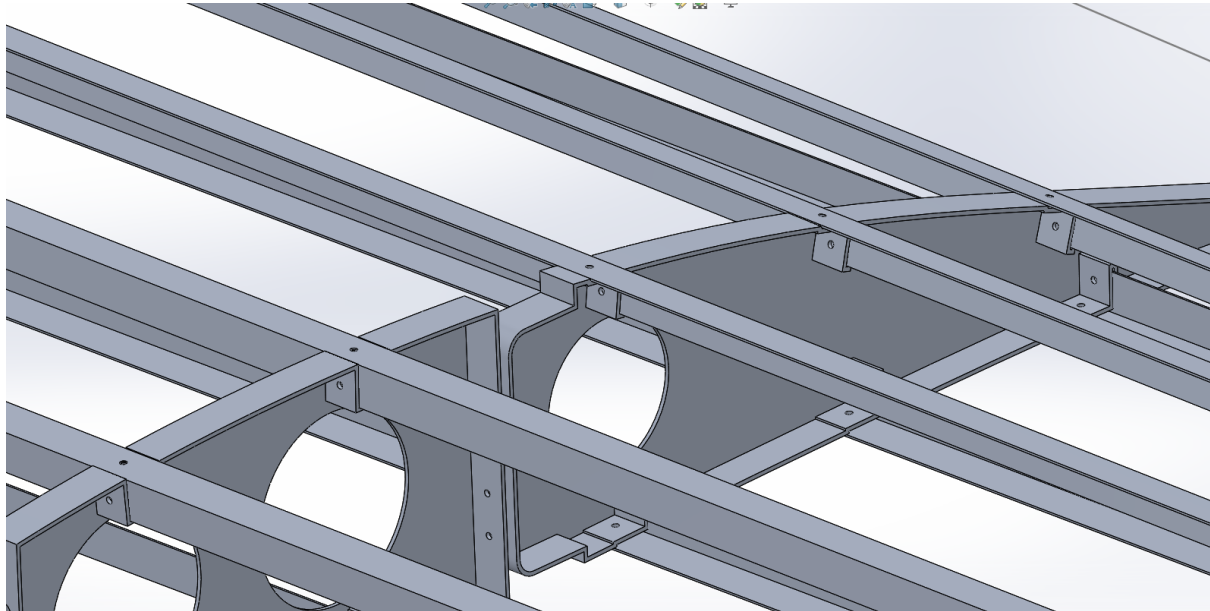
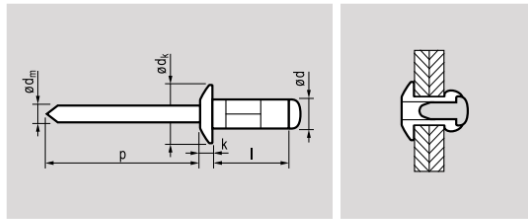


Figure 54: Aligning the stringers with the rivet holes of the rib

At this stage, the spars, ribs, and stringers have been assembled. Next, the skin plates are attached to the stringers. The rivet holes in the skin plates must be aligned with those in the top part of stringers, allowing the rivet to pass through the skin plate, stringer, and rib simultaneously. For the rib's leading edge, the skin plate is carefully wrapped around the rib and secured in place with rivets, ensuring a tight and uniform fit.

For all the rivets in the wing structure, **3.2mm Diameter Dome Head Aluminium rivets** are used. The length of the rivet varies with the attaching location. As an example, to connect the ribs and spars together, 12.7mm length rivets will be used, while 6mm length rivets are used for fixing the skin onto the ribs and stringers. [23]

MFX 1171



PLIA I multigrip I dome head white

$\varnothing d$	l		Item nr.	$\varnothing d_k$	k	$\varnothing d_m$	p		
[mm]	[mm]	[mm]		[mm]	[mm]	[mm]	[mm]	[N]	[N]
3,2	6,0	0,5-3,0	11713206	6,0 [+/-0,24]	$\leq 1,4$	$\sim 1,78$	≥ 27	980	680
[+0,05/-0,13]	8,0	0,5-5,0	3208						
	9,5	2,0-6,5	3209						
$\varnothing 3,3$ [3,5 max]	10,0	2,5-7,0	3210						
	11,1	3,5-8,0	3211						
	12,0	4,5-9,0	3212						
	12,7	5,5-9,5	3213						
	14,0	6,5-11,0	3214						
	16,0	8,5-13,0	3216						

Figure 55: Rivet catalogue

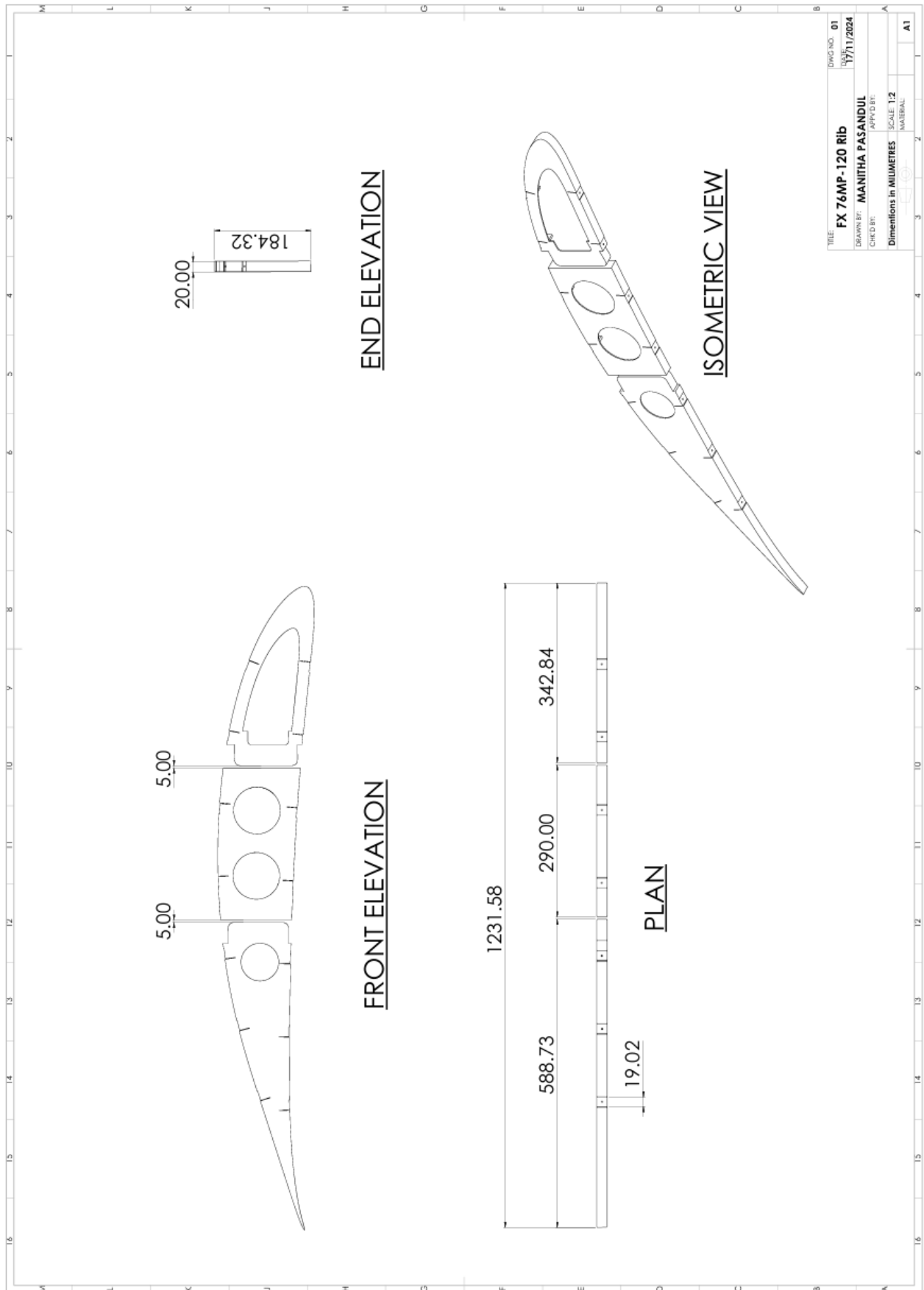
3.3 Material Selection for Components

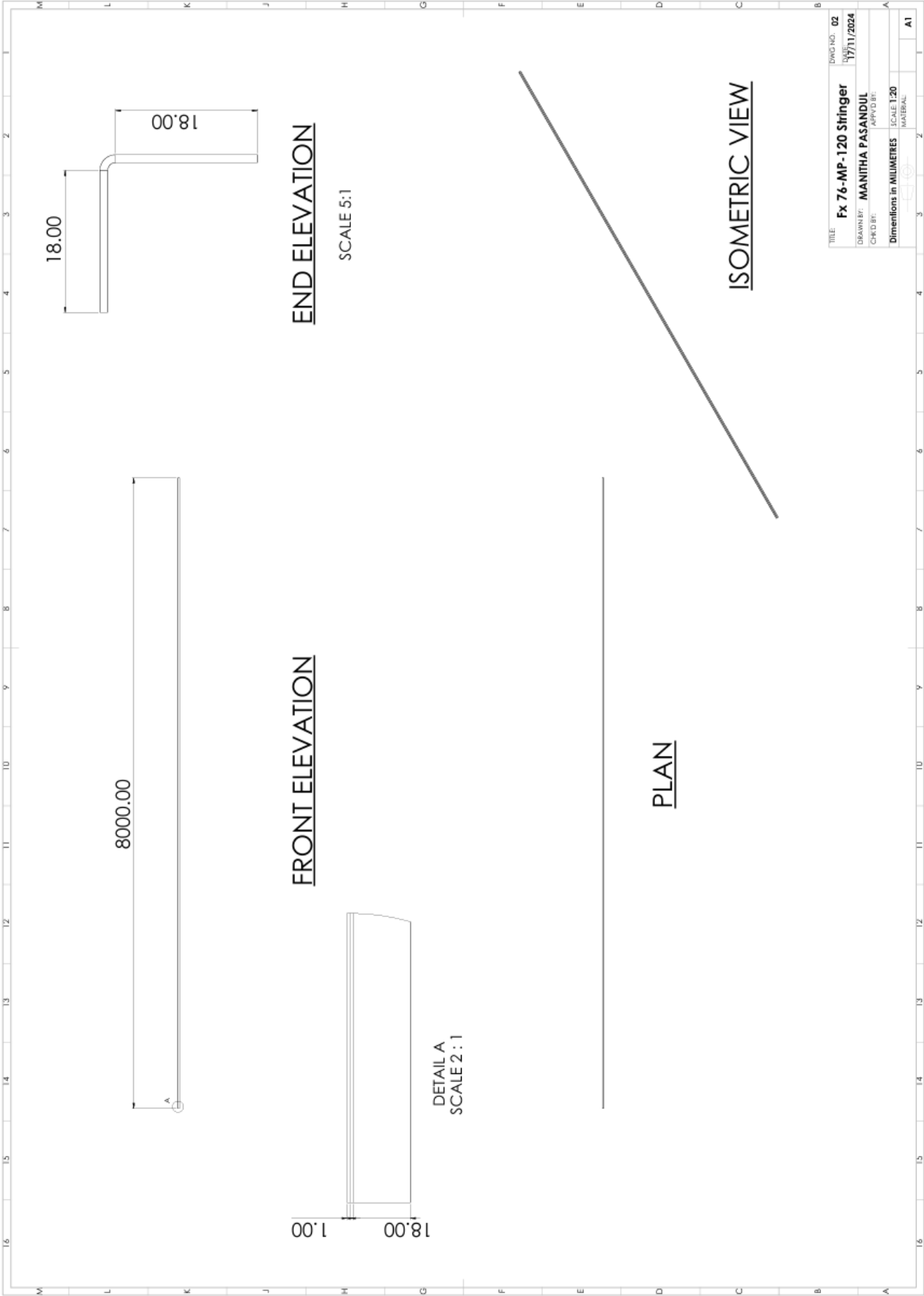
Table 03: Material Selection

Wing Component	Selected Material	Availability
Front and rear spar	Al 5056 H18 Alloy Sheets	5mm Thickness Sheets
Ribs	Al 5056 H18 Alloy Billets	1000 X 200 X 25 mm
Skin	Al 2024 T3 Alloy Sheets	1.5mm Thickness Sheets
Stringers	Al 5056 H18 Alloy Sheets	1mm Thickness Sheets
Rivets	Aluminum	AlMg2,5 Material Rivets

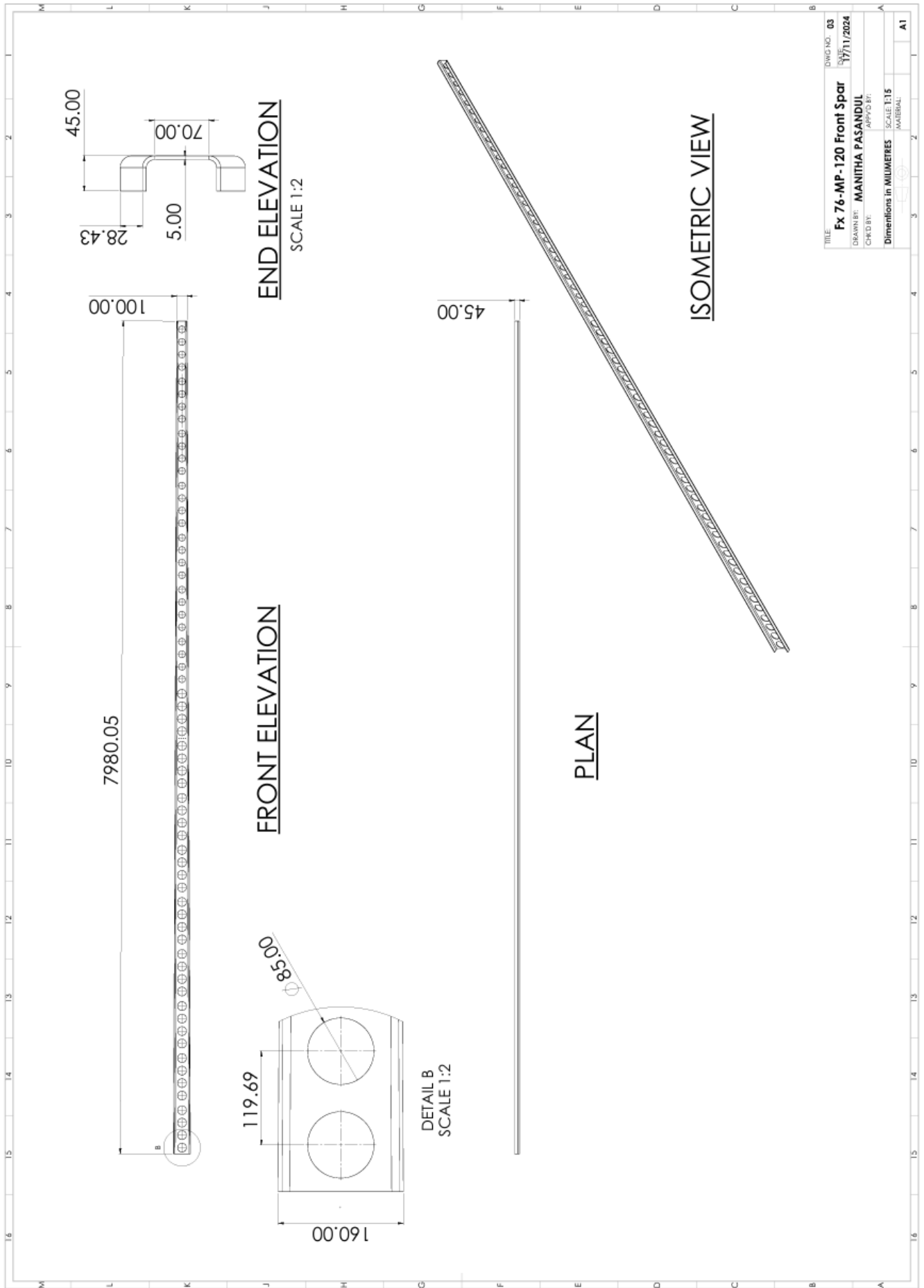
NOTE: A detailed model of a single rib was modelled, incorporating features such as flanges, rivet holes, and slots for stringers. As the remaining ribs are identical in design, the assembly drawing omits the inclusion of these detailed rib components for demonstration purposes, representing only the basic rib geometry instead.

4 Annex 1

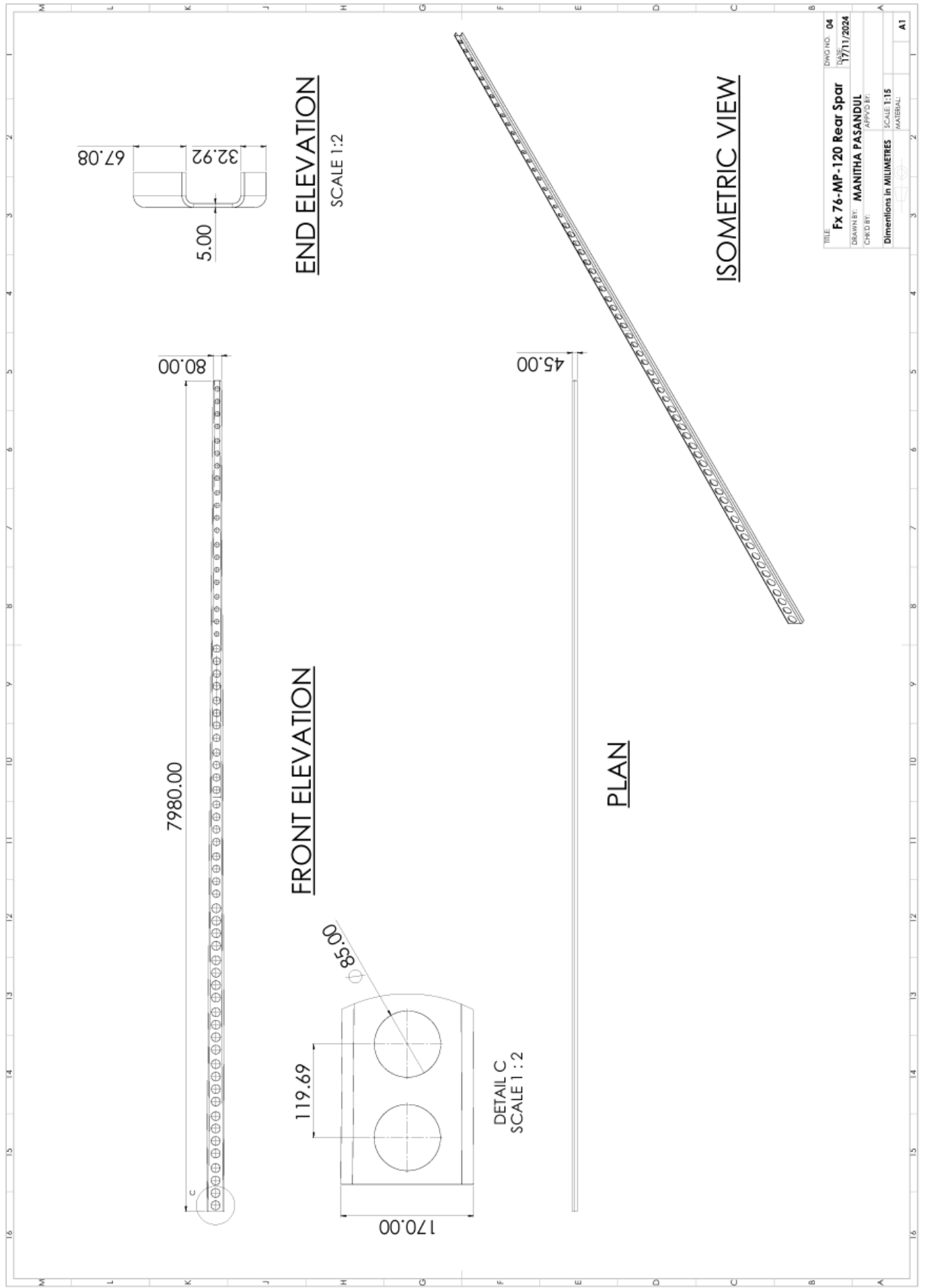


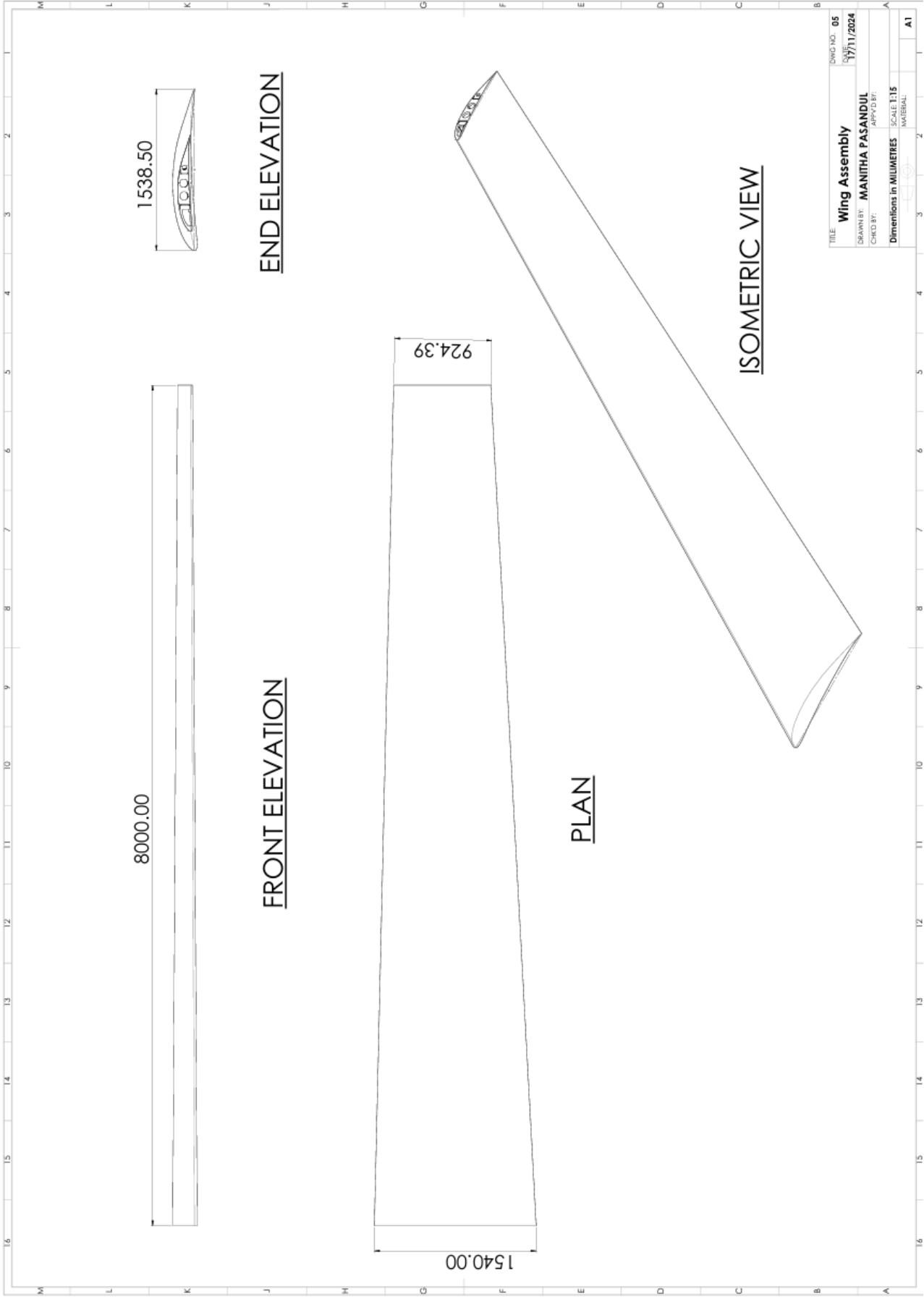


TITLE	DWG NO.	02
	DATE	17/11/2024
Fx 76-MP-120 Stringer		
DRAWN BY:	MANITHA PASANDUL	
CHECKED BY:	ARTHA BIL	
Dimensions in MILLIMETRES	SCALE	1:20
	MATERIAL	A1



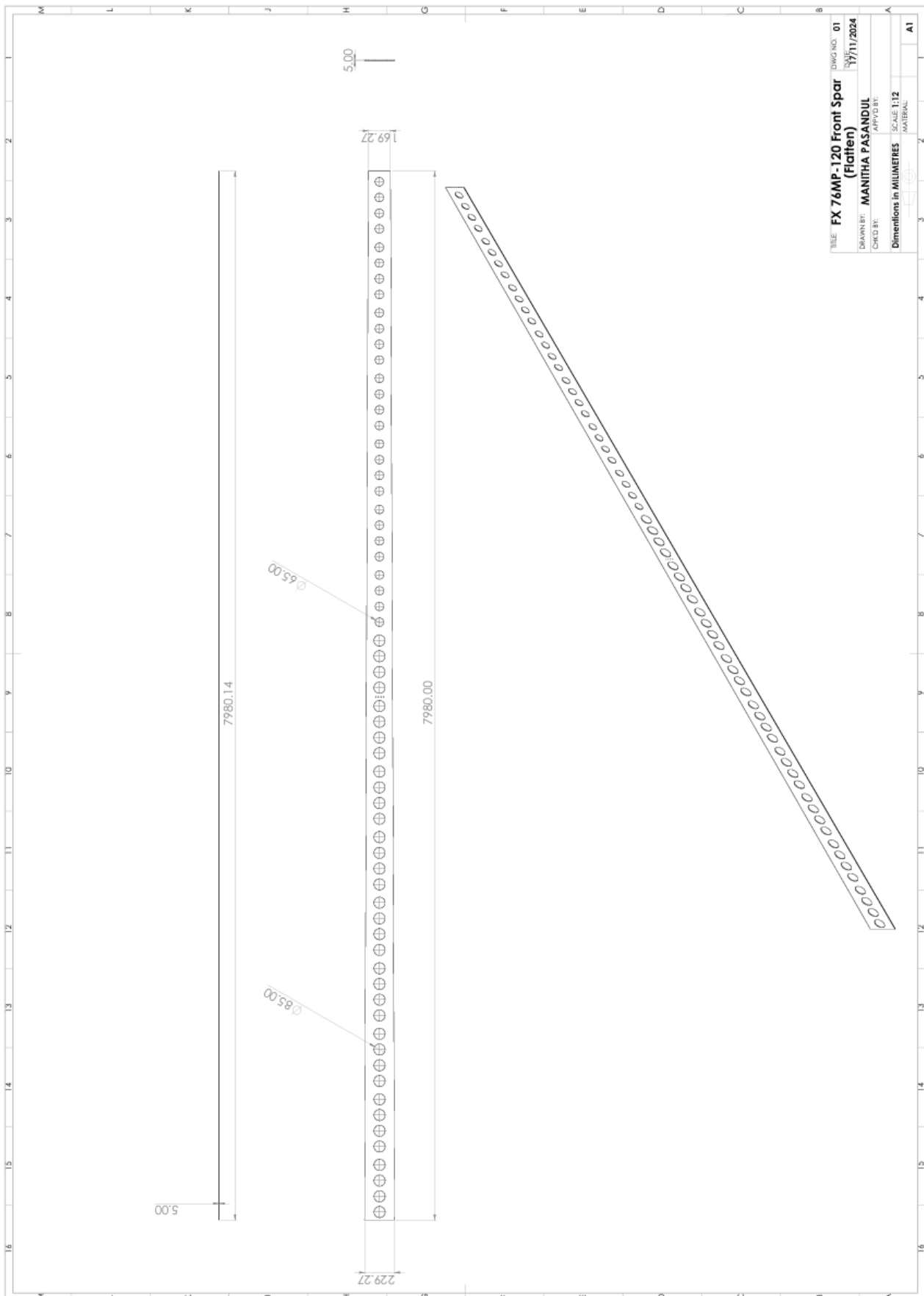
TITLE	DWG NO.	03
Fx 76-MP-120 Front Spar	DATE	17/11/2024
DRAWN BY: MANITHA PASANDUL	APP'D BY:	
CHECKED BY:	SCALE	1:15
Dimensions in MILLIMETRES	MATERIAL	A1



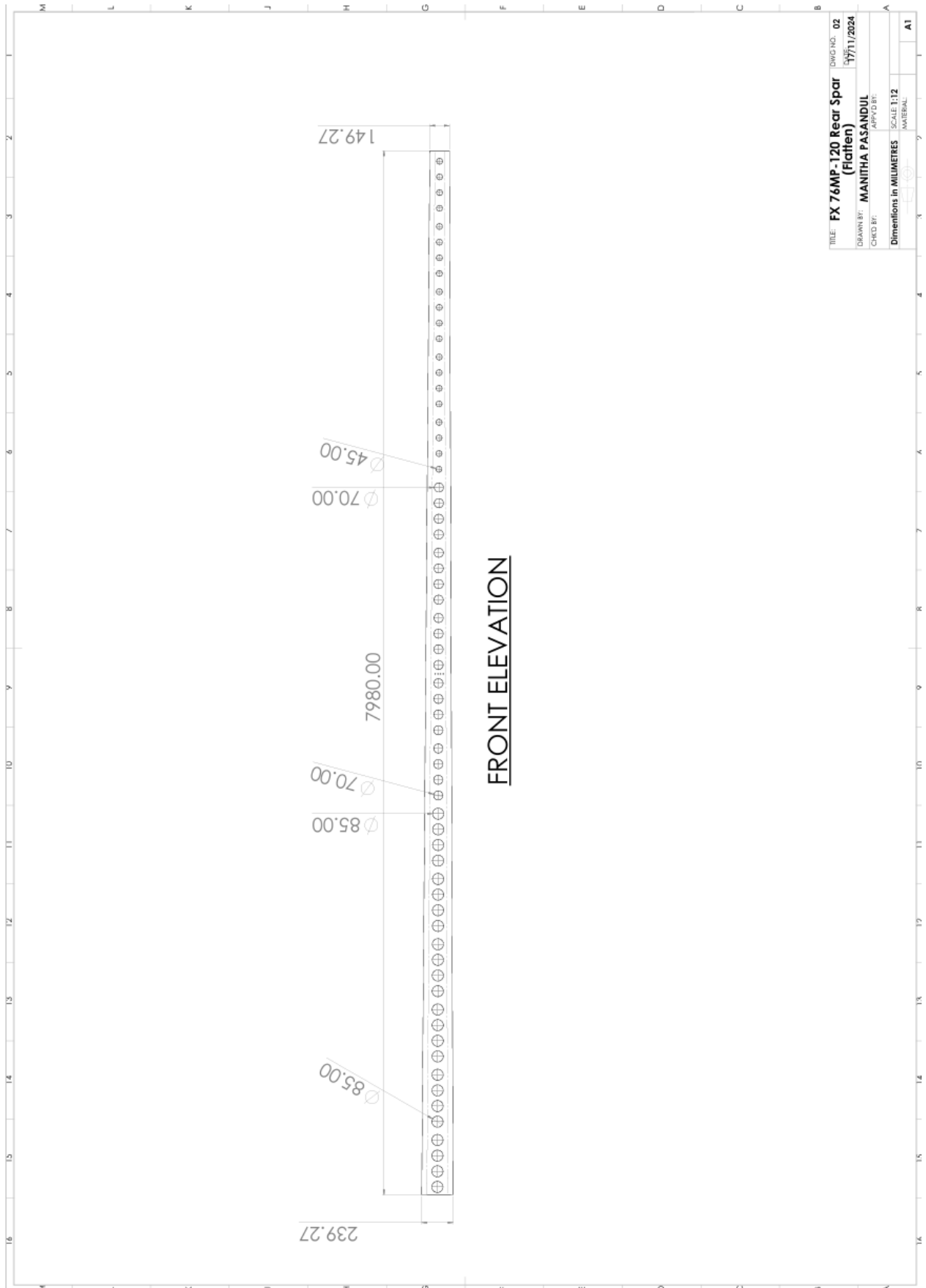


TITLE	Wing Assembly	DWG NO.	05
DRAWN BY	MANITHA PASANDUI	DATE	17/11/2024
CHECK BY		APPROVED BY	
Dimensions in MILLIMETRES	SCALE 1:15	MATERIAL	A1

Fabrication Drawing of Front Spar



Fabrication Drawing of Rear Spar



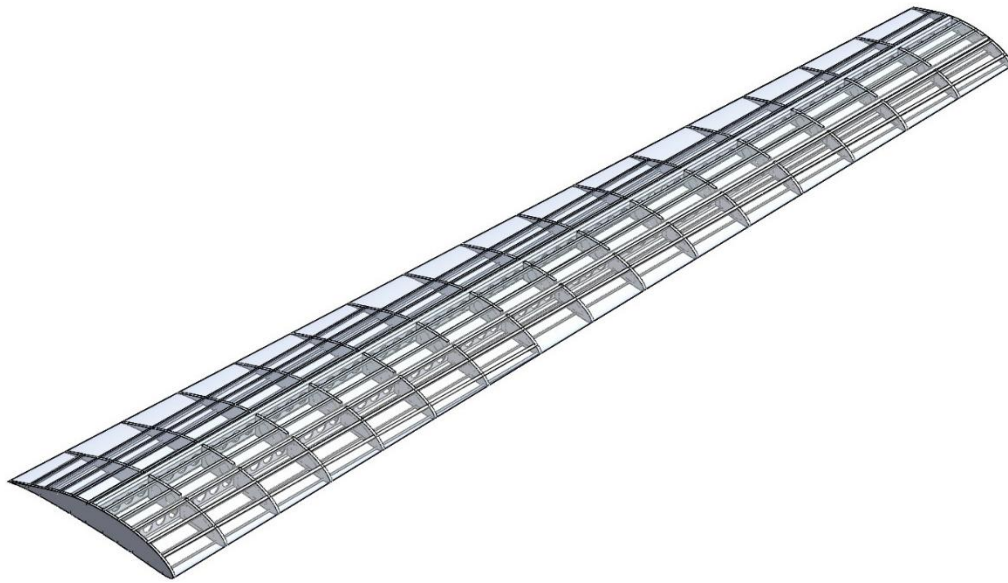


Figure 56: Complete assembly view 01

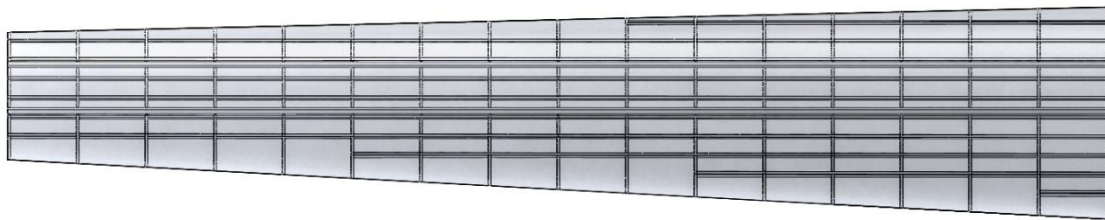


Figure 57: Complete assembly view 02

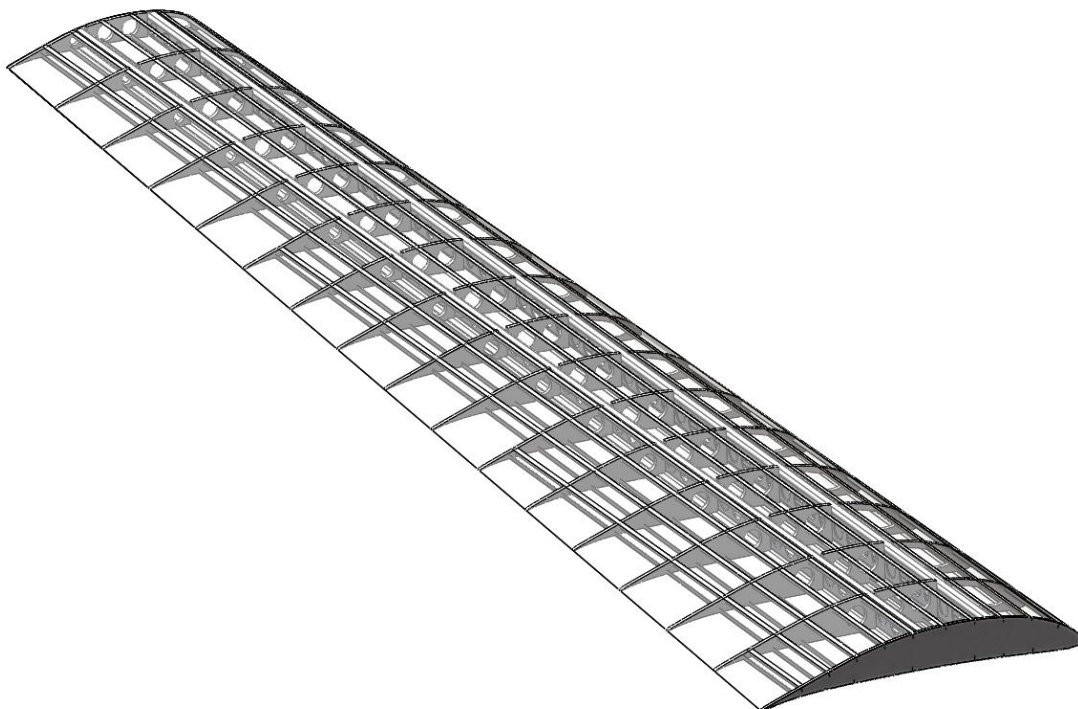


Figure 58: Complete assembly view 03

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